



PROJECT REPORT

HCL Electronic Shop

Semester: Programming Fundamentals

Class: PF1124

Group: HCL

Instructor Nguyen Dinh Cuong

Team members: Nguyen Hai Long

Pham Nam Cuong

Pham Ngoc Huy

Index

HCL Electronic Shop	Error! Bookmark not defined.
Index.....	2
I. Project introduction	3
II. Analyze System Requirements	4
III. Design Details	53
IV. Test.....	86
V. Task Assign (to each team member)	99
VI. Installation Instructions	100
Appendix	103

I. Project introduction

The system will provide these following features:

- Register/Login system:
 - + Multiple user cases (Customer, Admin, Super Admin which each will have their own option menus).
- General product & functions:
 - + View Product/Category information list function
 - + Search for Product/Category functions.
- Shopping cart & placing order functions
 - + Adding & removing Products to or from Shopping Cart to place Order
 - + View Shopping Cart functions for all roles/user cases.
 - + Complete & place order, cancelling order for all user cases.
 - + Picking payment methods & checking out.
 - + Checking Order details & status function for every user cases.
- Management functions:
 - + Adding/removing/editing Product functions for Admin & Super Admin.
 - + Adding/removing/editing Category functions for Admin & Super Admin.
 - + Adding/removing/editing Customer functions for Admin & Super Admin.
 - + Adding/removing/editing Admin functions for Super Admin.

1. Proposed System

- To create a fully functional console app with database for digital marketing.
- Have multiple user cases and different permissions for each user roles: customers, staffs and managers.

2. The scope of the project to be applied

- Made to be implemented into any kind of shops and stores.

3. System Name

Electronic Shop

4. Deployment environment

Window, Mac, Linux

5. Development Tools

- Visual Studio Code
- Draw.io Diagram
- My SQL Workbench

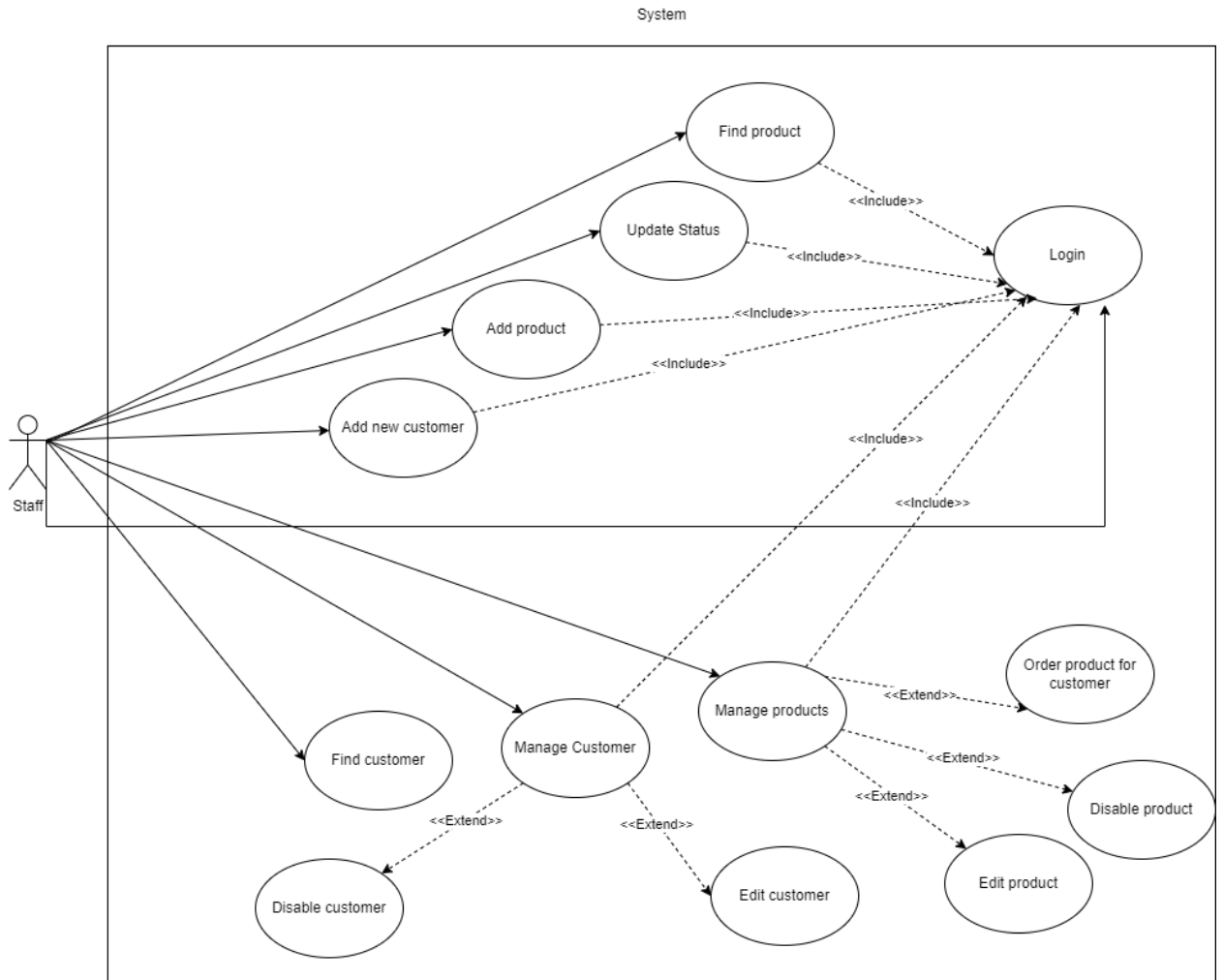
6. Customer Requirements (System features)

- You must login to use the system.
- You must have your account role set to be Admin or Super Admin to use their menu functions.
- You must have your address, phone number and payment method information to place an order.
- You must contact Admins or Super Admins in order to change your Customer information details.

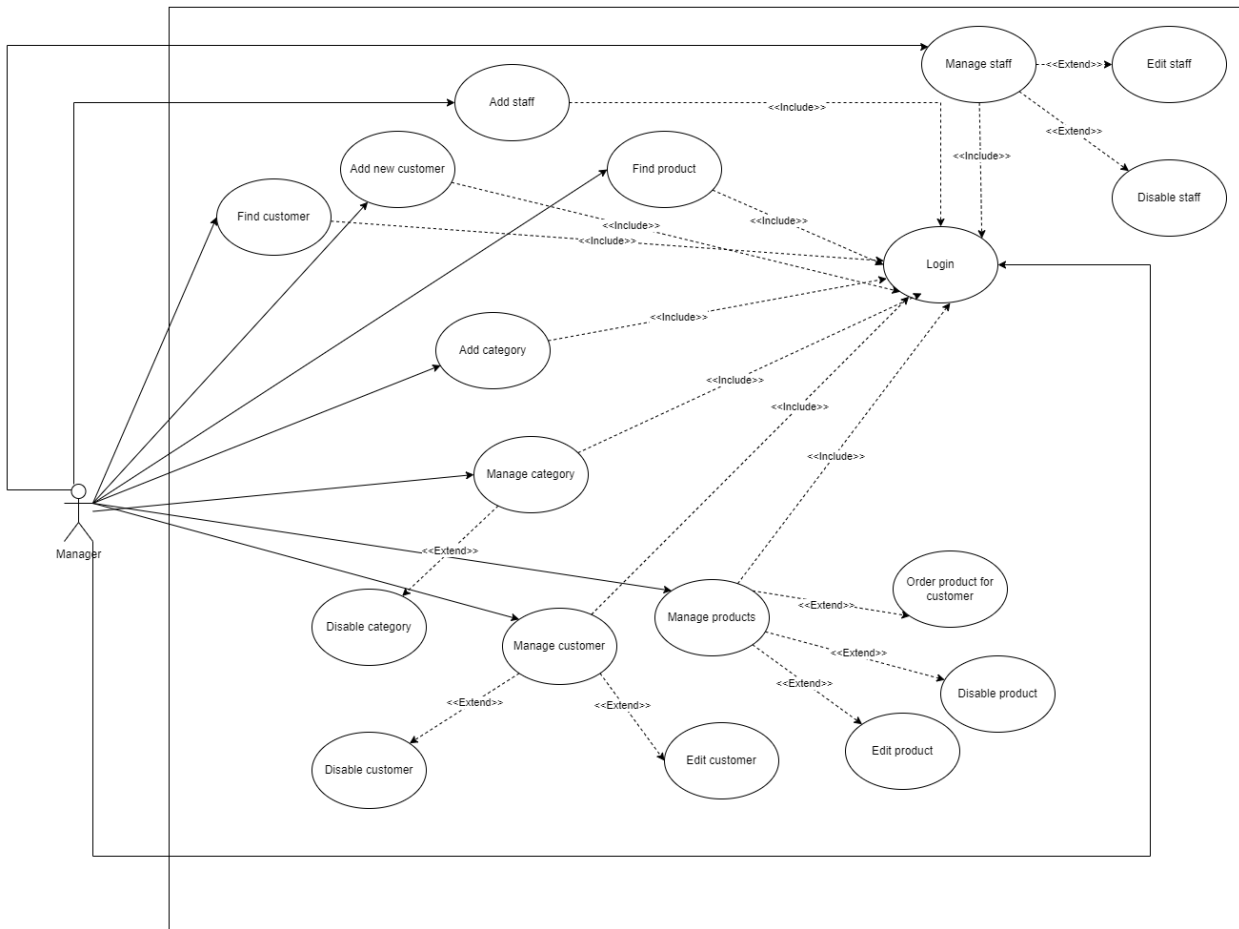
II. Analyze System Requirements

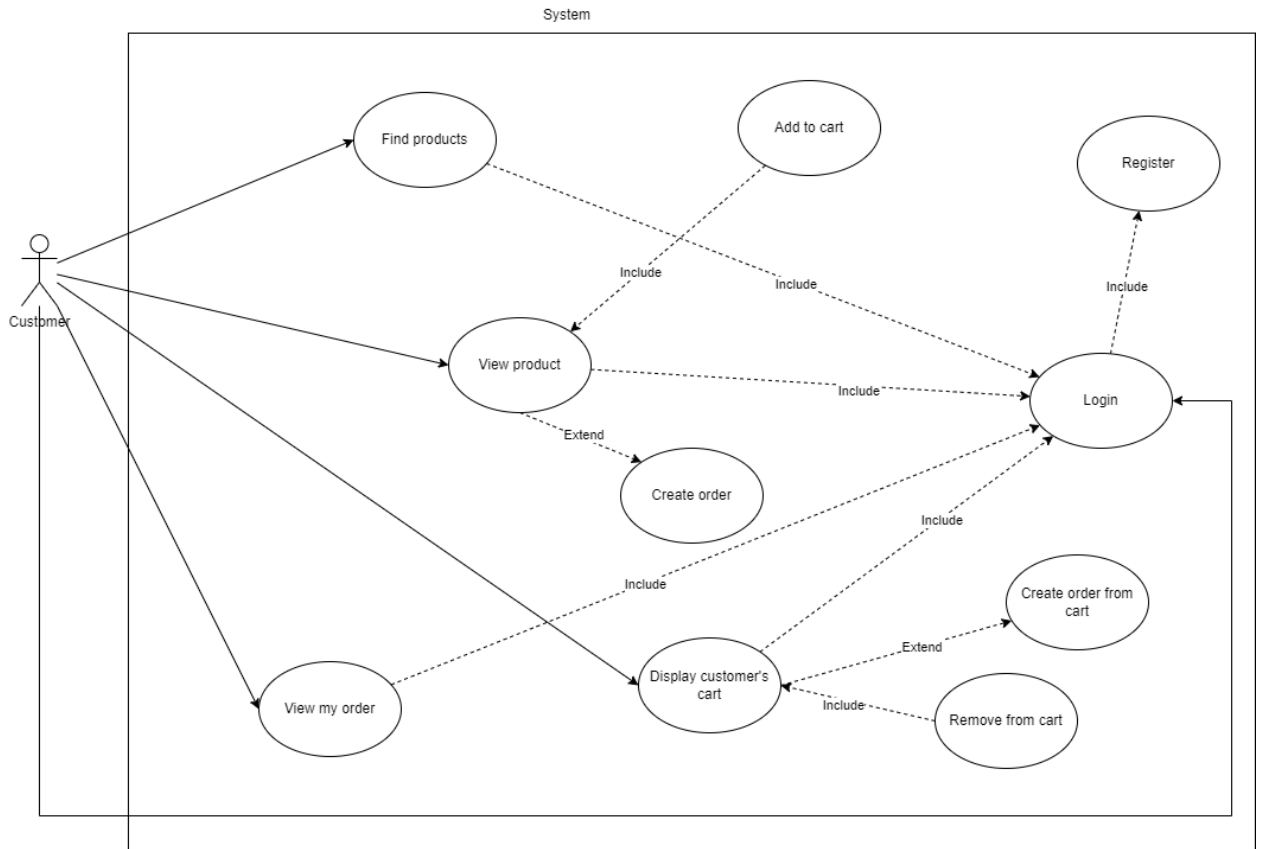
This application project was created to help with managing digital marketing business models, it helps to improve management efficiency. Staffs and customers have easy access to products informations, which include price, name, product details or even it's status, they can also easily place in any order with the shopping cart function. The main function of this system is for management, it helps the business with managing their products, orders as well as customers or even employees. Staffs can also create new user account for their customers or place in any order according to the customer's request, allow them to view the information of unpaid orders and confirm successful payment if enough money has been received from the customers as well as canceling the order if necessary.

1 Use Case



System





1.1.Login/Register

Use Case Name	Login
Use Case ID	UC_Login
Description	Users log in to the system to use functions
Actor	Customer, Staff, Manger
Organizational Benefits	This feature is used to define the roles and data users can access in the program.
Triggers	The user initiates the login process by entering their username and password and clicking the "Login" button
Preconditions	The user must be registered in the system. The application is running and the login window is displayed.
Postconditions	Success: - A success message is displayed according to the user's role.

	- The user is redirected to their respective menu based on their role (user, admin, superadmin). Fail: - An error message is displayed: "Invalid username or password!" - The login window remains open for the user to retry.
Main Course	- Step 1: Initialize and Display Login Window: - Step 2: Handle Login Button Click: - Step 3: Redirect Based on Role
Alternate Courses	• Invalid Username or Password: - Step 1: User enters incorrect username or password. - Step 2: Clicks "Login" and receives an error message: "Invalid username or password!" • User Clicks Register Button: - Step 1: User clicks the "Register" button. - Step 2: The login window is closed and the registration process is initiated.
Exceptions	Database Connection Error: - Step 1: Unable to connect to the database during authentication. - Step 2: Display an error message: "An error occurred during the login process. Please try again." • SQL Command Execution Error: - Step 1: An error occurs during the SQL query execution. - Step 2: Display an error message: "An error occurred during the login process. Please try again." • Unexpected Error: - Step 1: An unexpected error occurs. - Step 2: Display an error message: "An error occurred during the login process. Please try again."

Use Case Name	Register
Use Case ID	UC_Register
Description	Customer create their new user and information.
Actor	Customers
Organizational Benefits	Create new information
Triggers	Customer click register button
Preconditions	The application is running and the registration window is displayed. The username must be unique and not already in use.
Postconditions	Success: - A success message is displayed: "Registration successful!" - The registration window is closed. - The user is redirected to the login page. Failure: - An error message is displayed indicating the problem (e.g., duplicate username, invalid data). - The registration window remains open for the user to correct the errors.
Main Course	- Step 1: nitialize and Display Registration Window: - Step 2: Handle Register Button Click:
Alternate Courses	Duplicate Username: - Step 1: User enters a username that already exists. - Step 2: User clicks "Register" and receives an error message: "Username is duplicated. Please choose a different username." Invalid Input Data: - Step 1: User enters invalid data (e.g., incorrect date format, invalid email).

	- Step 2: User clicks "Register" and receives an appropriate error message indicating the issue.
Exceptions	• Database Connection Error: - Step 1: Unable to connect to the database during registration. - Step 2: Display an error message: "An error occurred during registration. Please try again." • SQL Command Execution Error: - Step 1: An error occurs during SQL query execution (e.g., insert operations). - Step 2: Rollback the transaction and display an error message: "An error occurred during registration. Please try again." • Unexpected Error: - Step 1: An unexpected error occurs. - Step 2: Rollback the transaction and display an error message: "An error occurred during registration. Please try again."

1.2. Display Product

Use Case Name	Display Products
Use Case ID	UC_DisplayProduct
Description	Customers want to go through all of the available products. Staffs want to view all of the products list.
Actor	Customer, staff, Manager
Organizational Benefits	View and know about products information Make orders or add to the customer's cart
Triggers	The function will be activated when the user choose the display product option in their menu.
Preconditions	- The application is running and the user has selected to view products. - User authentication has determined their role.

Postconditions	Success: - A list of products with action buttons is shown. - The user can interact with the products based on their role. Failure: - An error message is shown and the user is returned to the previous menu..
Main Course	Step 1: Initialize and Display Product Window Step 2: Retrieve and Display Product Data
Alternate Courses	AC 1: Role-Based Visibility: - Step 1: User role is "user", "admin", or "superadmin". - Step 2: Display buttons with different visibility based on the role: - Users see "Add to Cart" and "Order" buttons. - Admins and Superadmin see "Edit", "Disable", and "Order for Customer" buttons. AC 2: No Products Available: - Step 1: The query returns no results. - Step 2: Display a message indicating no products are available.
Exceptions	EX 1: Database Connection Error: - Step 1: Unable to connect to the database. - Step 2: Display an error message: "An error occurred while retrieving product data. Please try again." EX 2: SQL Command Execution Error: - Step 1: An error occurs during SQL query execution. - Step 2: Display an error message: "An error occurred while retrieving product data. Please try again." EX 3: Unexpected Error: - Step 1: An unexpected error occurs.

	- Step 2: Display an error message: "An unexpected error occurred. Please try again."
--	---

1.3. Order for customer

Use Case Name	Order For Customer
Use Case ID	UC_OrderForCustomer
Description	Staff make order for customer
Actor	Staff
Organizational Benefits	Customers can order directly through staff members or managers
Triggers	Staffs can pick the "order for customer" option
Preconditions	- Customer must be registered in the system. - Product must be available.
Postconditions	Successfully: - Staff successfully put in a new order for the customer. Fail: - Staff failed to create a new order for the customer.
Main Course	- Step 1: Chose the Order button next to the desired product. - Step 2: Enter the ID of the customer that's making the order through Staff. - Step 3: Fill the order information and submit.
Alternate Courses	Continue to make another order for the customer. After submitting the order for the customer. Redo the steps to make another order.
Exceptions	Ex 1: The quantity of products in stock is insufficient - Step 1: Display message: "The quantity of products in stock is insufficient for the order".

	- Step 2: Re-enter the sufficient amount to place the order or choose another product to order for the customer. EX 2: Invalid data type filled in information - Step 1: Display error message. - Step 2: Re-fill the correct information. EX 3: Database connection error - Step 1: Crash. - Step 2: Turn the program back on.
--	---

1.4. Display Cart to Order or Remove

Use Case Name	Display Cart to order, edit or remove items from the cart
Use Case ID	UC_DisplayCartToOrderOrRemove
Description	Customer can see their shopping cart to place the order, remove items from cart or delete the items from the shopping cart.
Actor	Customer
Organizational Benefits	Customers can place order/edit or delete the item from the cart
Triggers	Customer click the view cart button
Preconditions	Customer must be logged in and have items in their cart.
Postconditions	Success: - Cart items are displayed, and actions can be performed on them. Fail: - Customer cannot make any order or edit/removing items from the cart.
Main Course	- Step 1: Customer opens the display cart window. - Step 2: System retrieves cart items from the database. - Step 3: System displays cart items in a scrollable window.

	Case 1: Customer want to make an order - Step 1: Choose products to add into the shopping cart. - Step 2: Fill in order information from the cart. EX 2: Invalid data type filled in information - Step 1: Display error message. - Step 2: Re-fill the correct information. - Step 3: Confirm order. Case 2: Customer want to remove items from the shopping cart - Step 1: Click the remove button next to product's information in the cart. - Step 2: Remove successfully.
Alternate Courses	Case 1: Customer continue to make another order or remove more product from the cart. Case 2: Customer navigates back to the main menu.
Exceptions	Ex 1: The quantity of products in stock is insufficient - Step 1: Display message: "The quantity of products in stock is insufficient for the order". - Step 2: Re-enter the sufficient amount to place the order or choose another product to order for the customer. EX 2: Invalid data type filled in information - Step 1: Display error message. - Step 2: Re-fill the correct information. EX 3: Database connection error - If the system fails to connect to the database, an error message will displayed.

1.5. Add new Customer

Use Case Name	Add new Customer
Use Case ID	UC_AddCustomer
Description	Staff create a new user account for the customer
Actor	Staff
Organizational Benefits	Allows any staff to register a new customer by entering user and customer details.
Triggers	Staff choose add customer option
Preconditions	Customer's information doesn't already exist in database
Postconditions	Success: Register successfully Failure: Failed to register new customer
Main Course	- Step 1: Choose Add Customer option in Staff's menu screen. - Step 2: Fill in all the necessary information to register. - Step 3: Click Register button.
Alternate Courses	Case 1: Staff clicks the "Close" button without registering: - Staff decides not to register a new customer and clicks the "Close" button. - System closes the registration window and returns to the admin menu. Case 2: Staff returns to Admin Menu from Registration window: - Staff navigates back to the admin menu from the registration window without registering a new customer.
Exceptions	EX 1: Username already taken - Step 1: Display error message - Step 2: Re-enter new username EX 2: Invalid data type filled in information - Step 1: Display error message - Step 2: Re-fill the correct information

	EX 3: Database connection error - If the system fails to connect to the database, an error message is displayed, and the registration process cannot proceed.
--	--

1.6. Add new Staff

Use Case Name	Add new Staff
Use Case ID	UC_AddStaff
Description	Manager create a new Admin user for new Staff member
Actor	Manager
Organizational Benefits	Create new Admin user for staff member
Triggers	Manager choose create new Admin option
Preconditions	New user information isn't already exist in database
Postconditions	Success: Register successfully Failure: Failed to create new user
Main Course	- Step 1: Choose add staff option in Super Admin's menu screen. - Step 2: Fill in all the necessary information to create the new user. - Step 3: Chose the Register option.
Alternate Courses	Case 1: Manager clicks "Close" button without registering: - If the system fails to connect to the database, an error message is displayed, and the registration process cannot proceed. Case 2: Manager returns to Super Admin Menu from Registration window:

	- Manager navigates back to the super admin menu from the registration window without registering a new staff member.
Exceptions	EX 1: Username already exist - Step 1: Display error message - Step 2: Re-enter new username EX 2: Invalid data type filled in information - Step 1: Display error message - Step 2: Re-fill the correct information EX 3: Database connection error - If the system fails to connect to the database, an error message is displayed, and the registration process cannot proceed.

1.7. Add new Product

Use Case Name	Add new Product
Use Case ID	UC_AddProduct
Description	Staff add new information about newly imported products
Actor	Staff
Organizational Benefits	Add new information about newly imported products
Triggers	Staff choose add product option
Preconditions	Staff login
Postconditions	Success: - The new product is added to the database. - The new product appears in the static list of products. - A success message is displayed. - The user is returned to the admin menu.

	Fail: - An error message is displayed if any required fields are empty, contain invalid data, or if there is an issue inserting the product into the database. The product is not added to the database or the static list.
Main Course	- Step 1: Staff open Add Product window. - Step 2: Staff fills required fields. - Step 3: Staff click the save button. - Step 4: Display success message. - Step 5: Staff can either return to menu or add another product.
Alternate Courses	Alternate Course 1: Field(s) Left Blank - Step 1: The user leaves one or more fields empty. - Step 2: When clicking "Save": - o Display an error message: "All fields must be filled." - o The add product window remains open for user correction. - o Alternate Course 2: Invalid Stock Quantity - Step 1: The user enters a non-numeric value in the Stock Quantity field. - Step 2: When clicking "Save": - o Display an error message: "Stock Quantity must be a valid number." - o The add product window remains open for user correction. Alternate Course 3: Invalid Category ID - Step 1: The user enters a non-numeric value in the Category ID field. - Step 2: When clicking "Save": - o Display an error message: "Category ID must be a valid number." - o The add product window remains open for user correction.

	○ Alternate Course 4: Invalid Price - Step 1: The user enters a non-decimal value in the Price field. - Step 2: When clicking "Save": ○ Display an error message: "Price must be a valid decimal number." ○ The add product window remains open for user correction.
Exceptions	EX 1: Database Connection Error: - Step 1: Display error message: "An error occurred while adding the new product. Please try again." - Step 2: Return to the input form without closing the window EX 2: Database Insertion Error - Step 1: Display error message: "An error occurred while adding the new product. Please try again." - Step 2: Return to the input form without closing the window.

1.8. Add new Category

Use Case Name	Add new Category
Use Case ID	UC_AddCategory
Description	Manager will add a new product category
Actor	Manager
Organizational Benefits	Add a new product category
Triggers	Manager click add category in Super Admin Menu
Preconditions	Manager login on Super Admin account
Postconditions	Success: Create a new category successfully

	Fail: Fail to create a new category
Main Course	- Step 1: Click Add category button in Super Admin's menu screen - Step 2: Fill all necessary information for the new category - Step 3: Click Add button
Alternate Courses	Case 1: Manager click "Close" button without adding new category. - Admin decides not to add a new category and clicks the "Close" button. - System prompts confirmation before closing the window. - If confirmed, system closes the category addition window and returns to the super admin menu. Case 2: Admin returns to Super Admin Menu from Add Category window. - Admin navigates back to the super admin menu from the category addition window without adding a new category.
Exceptions	EX 1: Invalid data type filled in information - Step 1: Display error message - Step 2: Re-fill the correct information EX 2: Database connection error - If the system fails to connect to the database, an error message is displayed, and the addition process cannot proceed.

1.9. Edit Product

Use Case Name	Edit Product
Use Case ID	UC_EditProduct
Description	Staffs can edit product information
Actor	Staff

Organizational Benefits	Edit product information
Triggers	Staff click edit product button
Preconditions	Product must exist in database
Postconditions	Success: Updated product information is saved to the system. Fail: Can't edit product.
Main Course	- Step 1: Click Display Product to open the list of products. - Step 2: Click Edit product option. - Step 3: Edit new product information and then choose save. - Step 4: Edit successfully.
Alternate Courses	Case 1: Staff clicks "Close" button without saving changes: - Staff decides not to save the changes and clicks the "Close" button. - System prompts confirmation before closing the window. - If confirmed, system closes the edit product information window and returns to the product display menu. Case 2: Staff returns to Product Display Menu from Edit Product Information window: - Staff go back to the product display menu from the edit product information window without saving changes.
Exceptions	EX 1: Can't find product - Step 1: Display error: "Can't find product". - Step 2: Re-enter valid product id to find. EX 2: Invalid data type filled in information - Step 1: Display error message - Step 2: Re-fill the correct information EX 3: Database connection error

	- If the system fails to connect to the database, an error message is displayed, and the update process cannot proceed.
--	---

1.10. Edit Customer

Use Case Name	Edit Customer
Use Case ID	UC_EditCustomer
Description	Staff help customer edit their user information
Actor	Staff
Organizational Benefits	Staff helps keeping customer information up-to-date and accurate.
Triggers	Staff click edit customer button
Preconditions	Customer account must already exist within the database
Postconditions	Success: Edit customer successfully Fail: Can't edit customer's user information
Main Course	- Step 1: Click the Display Customer list option. - Step 2: Click Edit customer option. - Step 3: Edit new Customer's user information and then choose save. - Step 5: Edit successfully.
Alternate Courses	Case 1: Staff clicks "Close" button without saving changes: - Staff decides not to save the changes and clicks the "Close" button. - System prompts confirmation before closing the window. - If confirmed, system closes the edit customer information window and returns to the customer display menu. Case 2: Staff returns to Customer Display Menu from Edit Customer Information window: - Staff navigates back to the customer display menu from the edit customer information window without saving changes.

Exceptions	EX 1: Can't find customer - Step 1: Display error: "Can't find customer". - Step 2: Re-enter the valid customer id to find their user. EX 2: Invalid data type filled in information - Step 1: Display error message. - Step 2: Re-fill the correct information. EX 3: Database connection error - If the system fails to connect to the database, an error message is displayed, and the update process cannot proceed.
------------	---

1.11. Edit Staff

Use Case Name	Edit Staff
Use Case ID	UC_EditStaff
Description	Manager find staff and edit their user information
Actor	Manager
Organizational Benefits	Edit staff information
Triggers	Manager click edit staff button
Preconditions	Staff account must exist in database
Postconditions	Success: Edit Staff successfully Failure: Can't edit Staff
Main Course	Step 1: Click Display Staff list in Super Admin menu. Step 2: Find the staff's user you want to edit and click Edit. Step 3: Edit new staff's user information and then choose save. Step 4: Edit successfully.
Alternate Courses	Case 1: Manager clicks "Close" button without saving changes: - Superadmin decides not to save the changes and clicks the "Close" button.

	- System prompts confirmation before closing the window. - If confirmed, system closes the edit staff information window and returns to the staff display menu. Case 2: Manager returns to Staff Display Menu from Edit Staff Information window: - Superadmin navigates back to the staff display menu from the edit staff information window without saving changes.
Exceptions	EX 1: Can't find Staff - Step 1: Display error: "Can't find Staff" - Step 2: Re-enter Staff id to find EX 2: Invalid data type filled in information - Step 1: Display message error. - Step 2: Re-fill the correct information. EX 3: Database connection error - If the system fails to connect to the database, an error message is displayed, and the update process cannot proceed.

1.12. Find Product

Use Case Name	Find Product
Use Case ID	UC_FindProduct
Description	Customer or Staff can find product information
Actor	Customer or Staff
Organizational Benefits	Customer or Staff can easily find the desired product
Triggers	Customer or Staff click find product button in their menu
Preconditions	Product must already exists within the database
Postconditions	Success: Displaying the product you want to find Fail: List null

Main Course	- Step 1: Click find product button in menu. - Step 2: Enter the name of the product you want to find. - Step 3: The screen will display all the products that have similar names like the one you've entered.
Alternate Courses	
Exceptions	EX 1: Database connection error - If the system fails to connect to the database, an error message is displayed, and the search process cannot proceed

1.13. Find Customer

Use Case Name	Find Customer
Use Case ID	UC_FindCustomer
Description	Staff can find customer information
Actor	Staff
Organizational Benefits	Staff can easily find customer's information
Triggers	Staff click find Customer button in staff menu
Preconditions	Customer must already have an account in the database
Postconditions	Success: Display the information of the Customer you want to find Failure: List null
Main Course	- Step 1: Click find button in menu. - Step 2: Enter the name of the customer you want to find. - Step 3: The screen will display all the customers that have similar names like the one you've entered.
Alternate Courses	Case 1: Staff clicks "Back" button without saving changes: - Staff decides not to search for a customer and clicks the "Back" button.

	- System clears the product list and returns to the respective menu (admin menu or superadmin menu). Case 2: Staff returns to respective menu: - Staff navigates back to their respective menu without performing a search.
Exceptions	EX 1: Database connection error - If the system fails to connect to the database, an error message is displayed, and the search process cannot proceed. EX 2: No customers match the search criteria: - If no customers match the search criteria, the system displays an error message indicating that no customer were found.

1.14. Find staff

Use Case Name	Find Staff
Use Case ID	UC_FindStaff
Description	Manager can find staff information
Actor	Manager
Organizational Benefits	Staff can easily find staff's information
Triggers	Manger click find staff button in manager menu
Preconditions	Staff must already have an account in the database
Postconditions	Success: Display staffs that matched with the search criteria. Failure: List null.
Main Course	- Step 1: Click find button in menu. - Step 2: Enter the name of the staff you want to find. - Step 3: The screen will display all the staffs that have similar names like the one you've entered.

Alternate Courses	Case 1: Manager Clicks "Close" Button Without Searching: - Manager decides not to search for a staff member and clicks the "Close" button. - System clears the staff list and returns to the super admin menu. Case 2: Manager Returns to Super Admin Menu: - Manager navigates back to the super admin menu without performing a search.
Exceptions	EX 1: Database connection error - If the system fails to connect to the database, an error message is displayed, and the search process cannot proceed. EX 2: No Staff matches the search criteria: - If no staff matches the search criteria, the system displays an error message indicating that no staff members were found.

1.15. Display customer

Use Case Name	DisplayCustomer
Use Case ID	UC_DisplayCustomer
Description	Staff can view all customer's information
Actor	Staff
Organizational Benefits	Can view all of customer's information
Triggers	Staff clicks display customer in staff menu
Preconditions	Customer must already have an account in the database
Postconditions	Success: Display a list of all the current customer's information Failure: List empty
Main Course	- Step 1: Staff clicks display customers in staff menu. - Step 2: Menu display all of the customer's information.

Alternate Courses	Case 1: Staff clicks "Back" button without any action: - Staff decides not to perform any actions and clicks the "Back" button. - System clears the customer list and returns to the main menu.
Exceptions	EX 1: Database connection error - If the system fails to connect to the database, an error message is displayed, and the customer list cannot be retrieved. EX 2: Error occurs during data retrieval: - If an error occurs during data retrieval, the system displays a generic error message.

1.16. Display staff

Use Case Name	DisplayStaff
Use Case ID	UC_DisplayStaff
Description	Manager can view all staff's information
Actor	Manager
Organizational Benefits	Can view all staff's information
Triggers	Staff clicks display staffs in Super Admin menu
Preconditions	Staff must already have an account in the database
Postconditions	Success: Display a list of all the current staff's information Failure: List empty
Main Course	Step 1: Manager clicks display staffs in Super Admin menu Step 2: Menu display all of the staff's information
Alternate Courses	Manager clicks "Back" button without any action: - Manager decides not to perform any actions and clicks the "Back" button. - System clears the staff list and returns to the main menu.

Exceptions	EX 1: Database connection error - If the system fails to connect to the database, an error message is displayed, and the staff list cannot be retrieved. EX 2: Error occurs during data retrieval: - If an error occurs during data retrieval, the system displays a generic error message.
------------	---

1.17. Delete product

Use Case Name	DeleteProduct
Use Case ID	UC_DeleteProduct
Description	Staff can remove product from the database
Actor	Staff
Organizational Benefits	Remove product from the database
Triggers	Staff clicks delete product in staff menu
Preconditions	Product must already exist in the database
Postconditions	Success: Remove product successfully Fail: Can't remove product
Main Course	Step 1: Staff click delete product in staff menu. Step 2: Enter ID of the product you wish to remove. Step 3: Remove successfully.
Alternate Courses	Staff cancels the deletion: - Staff decides not to proceed with the deletion and cancels the operation. - System does not perform any update operation on the database.
Exceptions	EX 1: Can't find product - Step 1: Display error message

	- Step 2: Re-enter valid product ID EX 2: Invalid data type filled in information - Step 1: Display error message - Step 2: Re-fill the correct information EX 3: Database connection error - If the system fails to establish a connection to the database, an error message is displayed indicating the failure.
--	---

1.18. Delete customer

Use Case Name	DeleteCustomer
Use Case ID	UC_DeleteCustomer
Description	Marks a customer and their corresponding user account as inactive in the database
Actor	Staff
Organizational Benefits	Ensures customer records are up-to-date.
Triggers	Staff clicks delete customer in staff menu
Preconditions	Customer must already have an account in the database
Postconditions	Success: Remove customer successfully Fail: Can't remove customer
Main Course	- Step 1: Staff click delete customer in the Admin menu. - Step 2: Enter ID of the customer you want to remove. - Step 3: Remove successfully.
Alternate Courses	Staff cancels the deletion: - Staff decides not to proceed with the deletion and cancels the operation. - System does not perform any update operation on the database.
Exceptions	EX 1: Can't find customer

	- Step 1: Display error message - Step 2: Re-enter valid customer ID EX 2: Invalid data type filled in information - Step 1: Display error message - Step 2: Re-fill the correct information EX 3: Database connection error - If the system fails to establish a connection to the database, an error message is displayed indicating the failure.
--	---

1.19. Delete staff

Use Case Name	DeleteStaff
Use Case ID	UC_DeleteStaff
Description	Marks a staff member and their corresponding user account as inactive in the database using a transaction.
Actor	Manager
Organizational Benefits	Ensures staff records are up-to-date.
Triggers	Manager clicks delete staff in super admin menu
Preconditions	Staff must already have an account in the database
Postconditions	Success: Remove staff successfully Fail: Can't remove staff
Main Course	- Step 1: Manager click delete staff in Super Admin menu. - Step 2: Select the staff that need to be remove. - Step 3: Remove successfully.
Alternate Courses	Manager cancels the deletion: - Manager decides not to proceed with the deletion and cancels the operation. - System does not perform any update operation on the database.

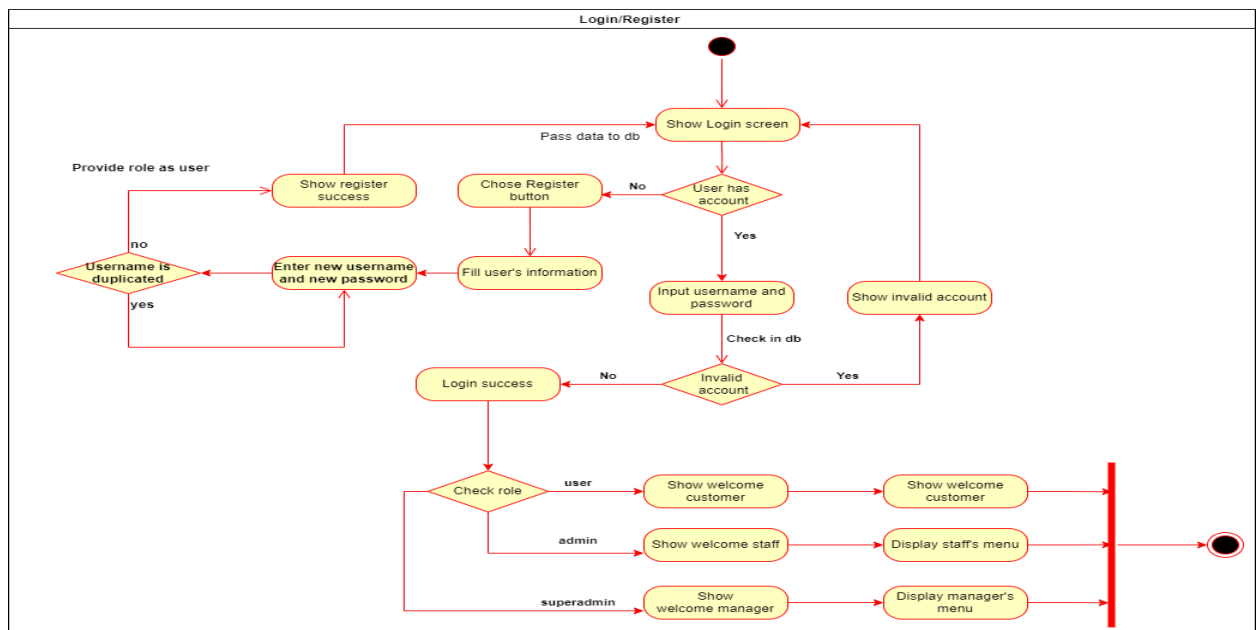
Exceptions	EX 1: Can't find staff - Step 1: Display error message. - Step 2: Re-enter staff ID. EX 2: Invalid data type filled in information - Step 1: Display error message. - Step 2: Re-fill the correct information. EX 3: Database connection error - If the system fails to establish a connection to the database, an error message is displayed indicating the failure..
------------	---

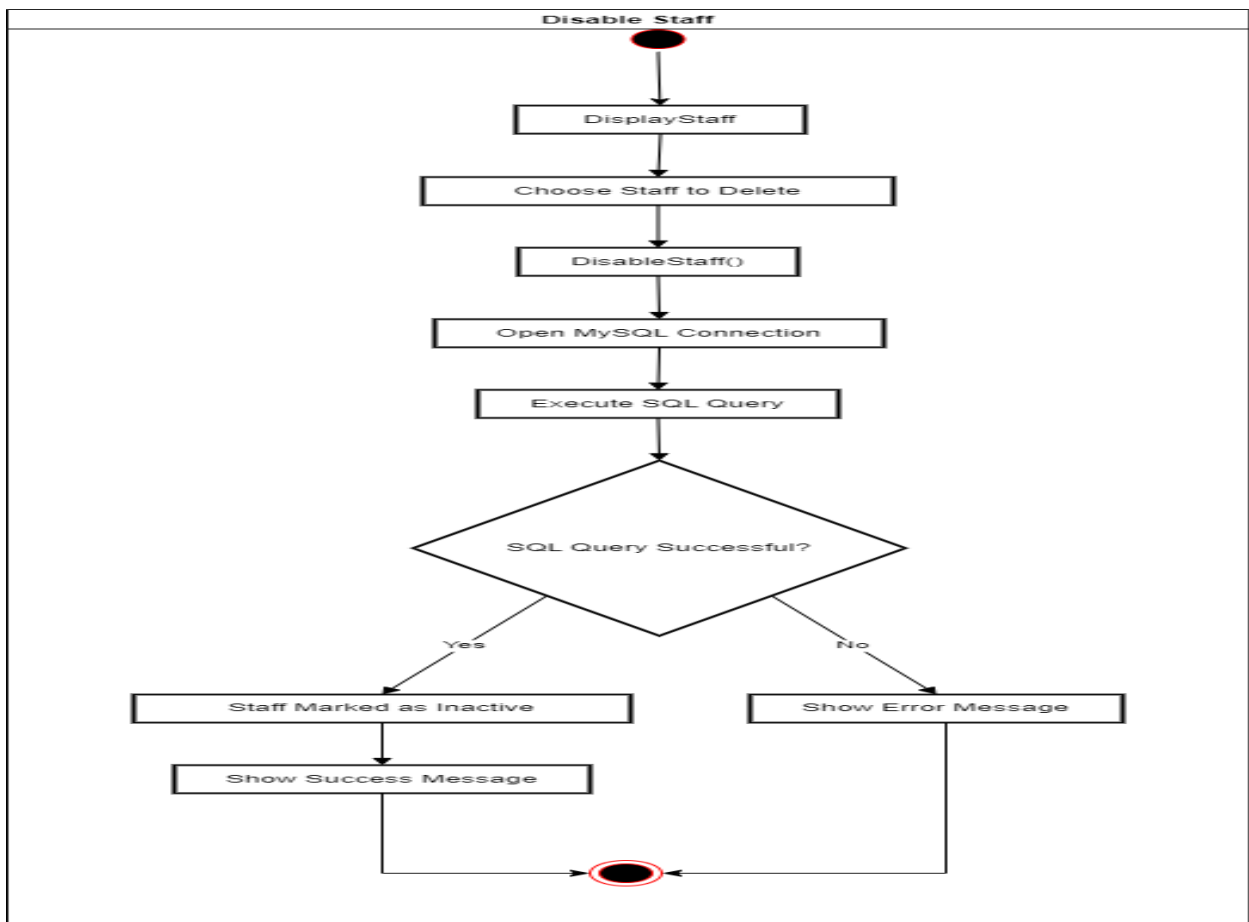
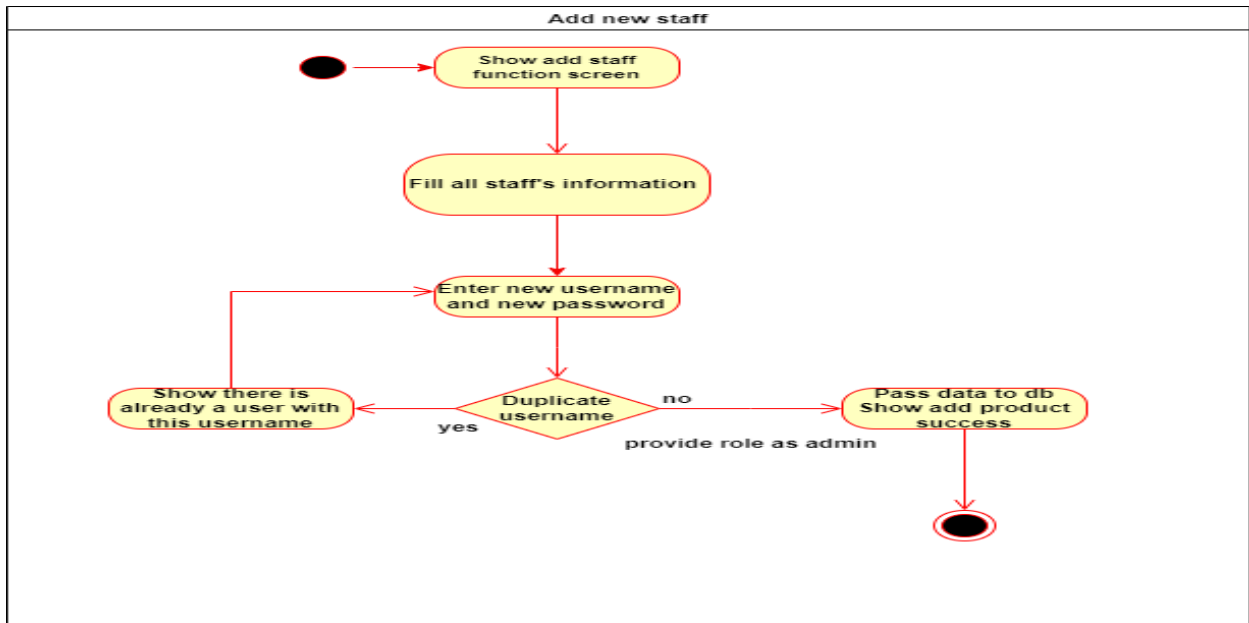
1.20. Delete category

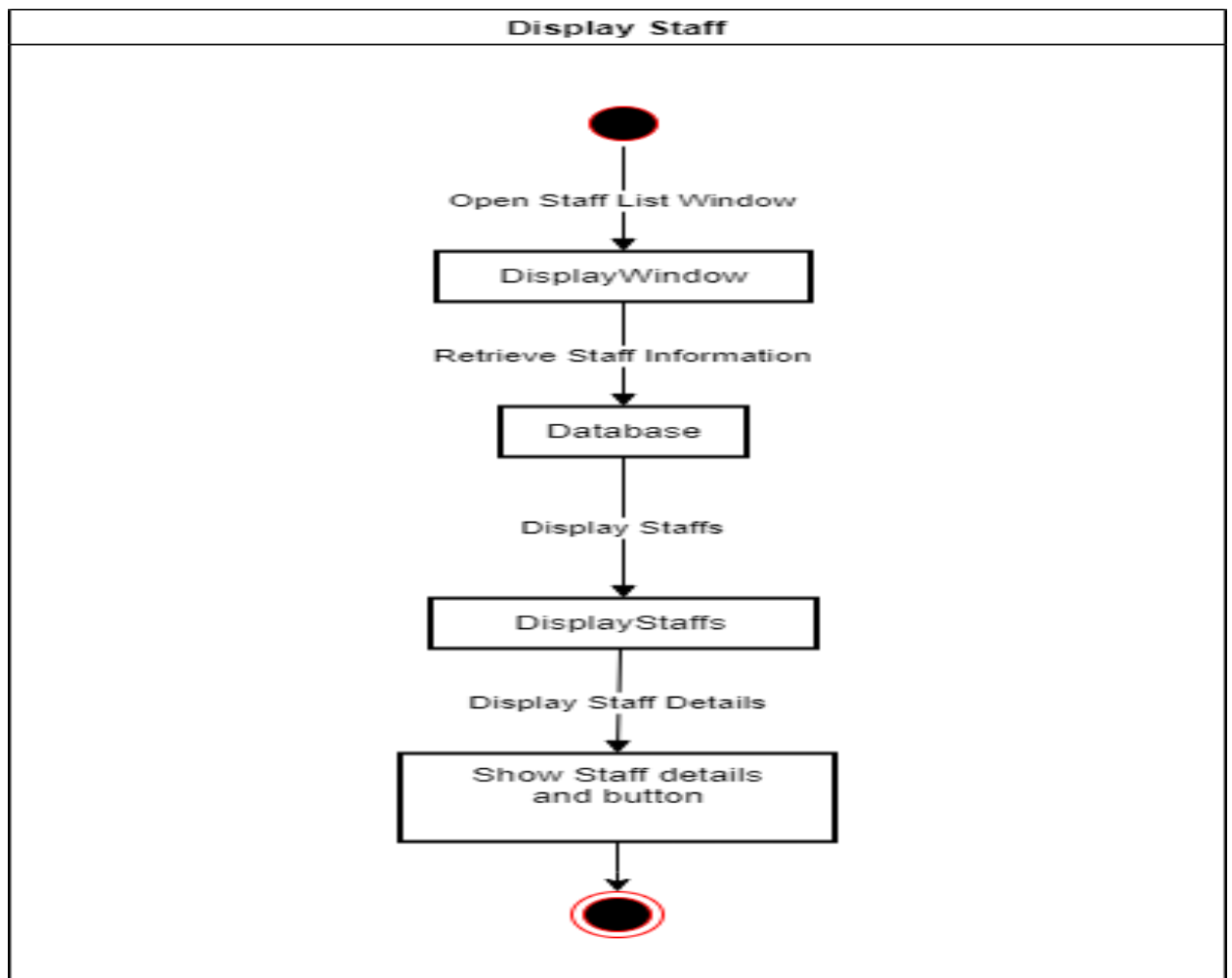
Use Case Name	DeleteCategory
Use Case ID	UC_DeleteCategory
Description	Marks a category and its associated products as inactive in the database.
Actor	Staff
Organizational Benefits	Ensures product and category records are up-to-date.
Triggers	Staff clicks delete category in Admin menu
Preconditions	Category must already exist in the database
Postconditions	Success: Remove category successfully Fail: Can't remove category
Main Course	- Step 1: Staff click delete category in Admin menu. - Step 2: Enter ID of the category that needs to be remove. - Step 3: Remove successfully.
Alternate Courses	Staff cancels the deletion: - Staff decides not to proceed with the deletion and cancels the operation.

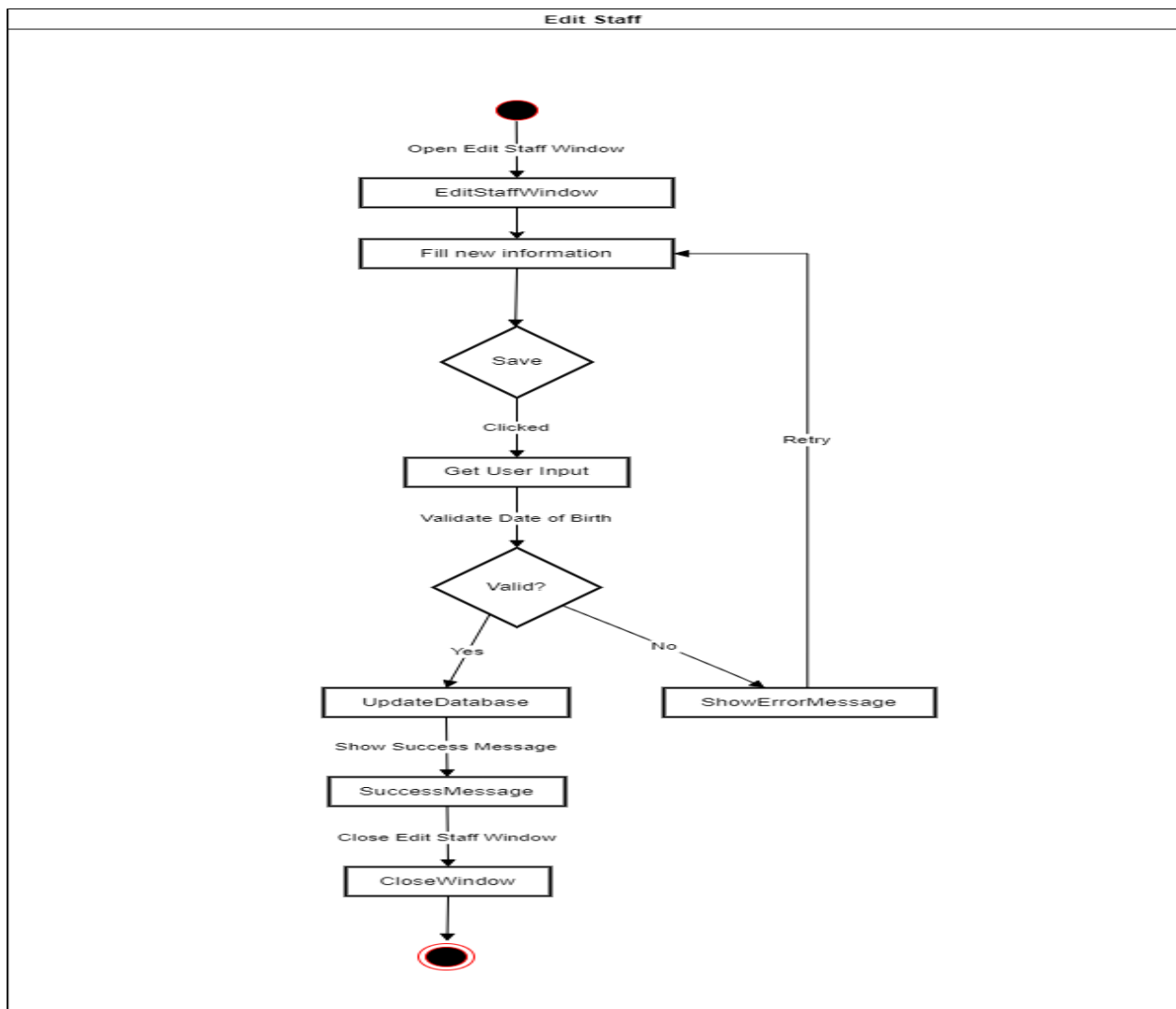
	- System does not perform any update operation on the database.
Exceptions	EX 1: Can't find category - Step 1: Display error message. - Step 2: Re-enter category ID. EX 2: Invalid data type filled in information - Step 1: Display error message. - Step 2: Re-fill the correct information. EX 3: Database connection error - If the system fails to establish a connection to the database, an error message is displayed indicating the failure.

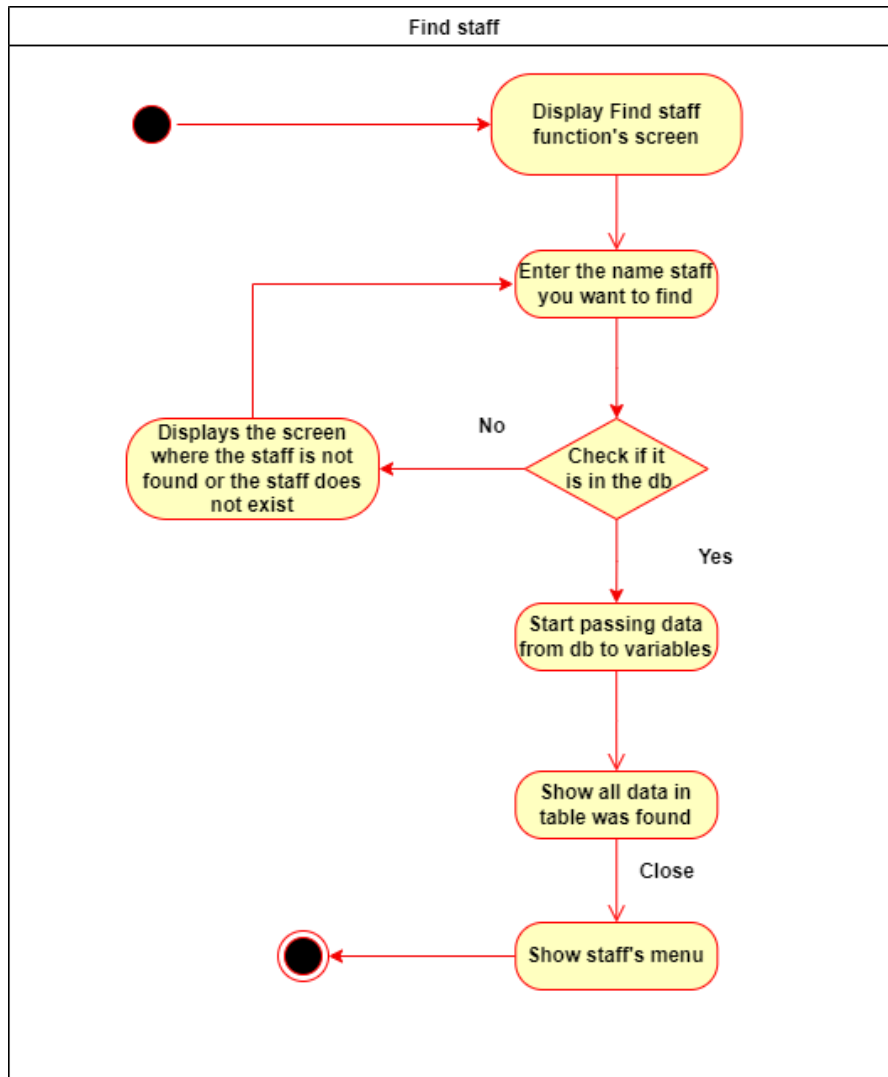
2. Activity Diagram:

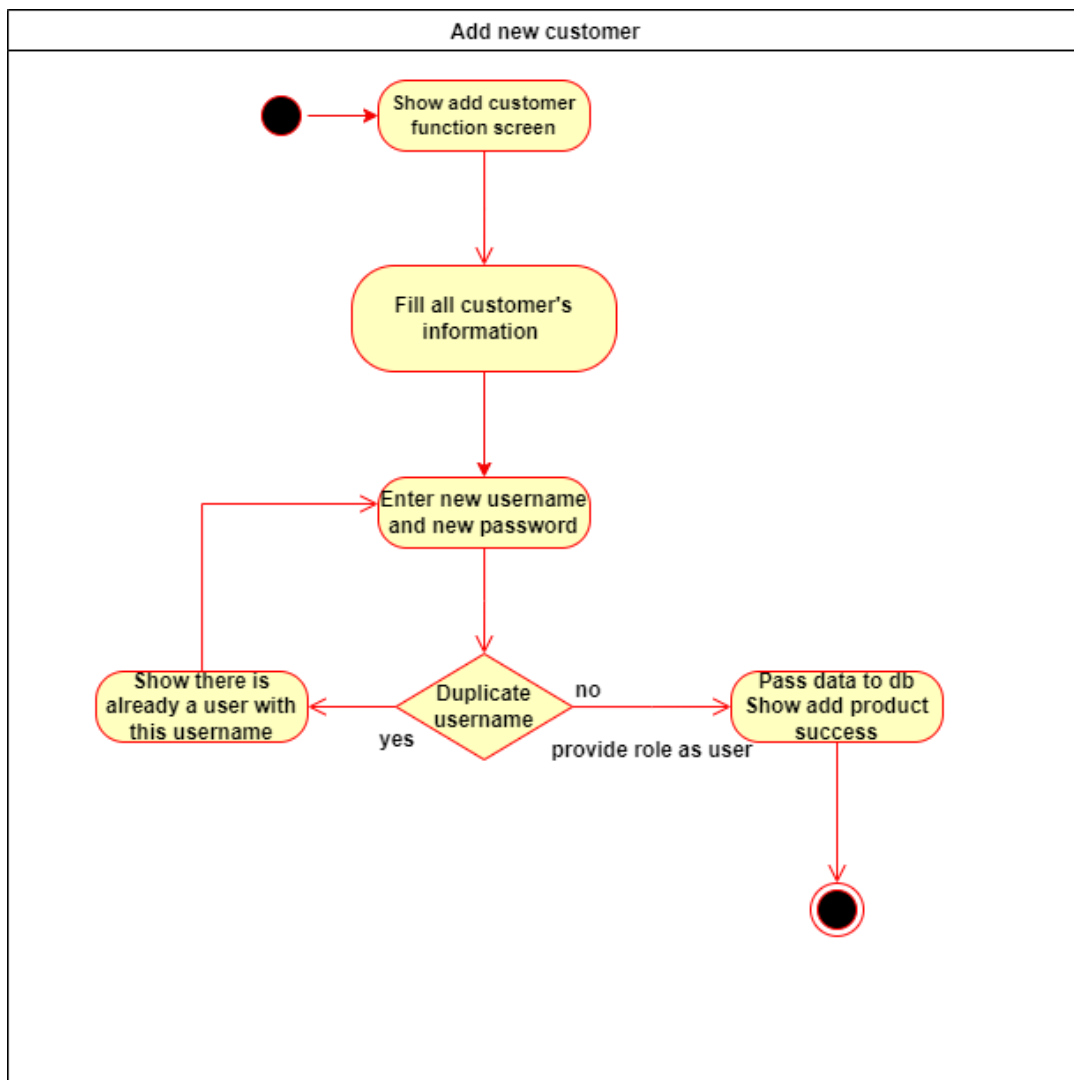


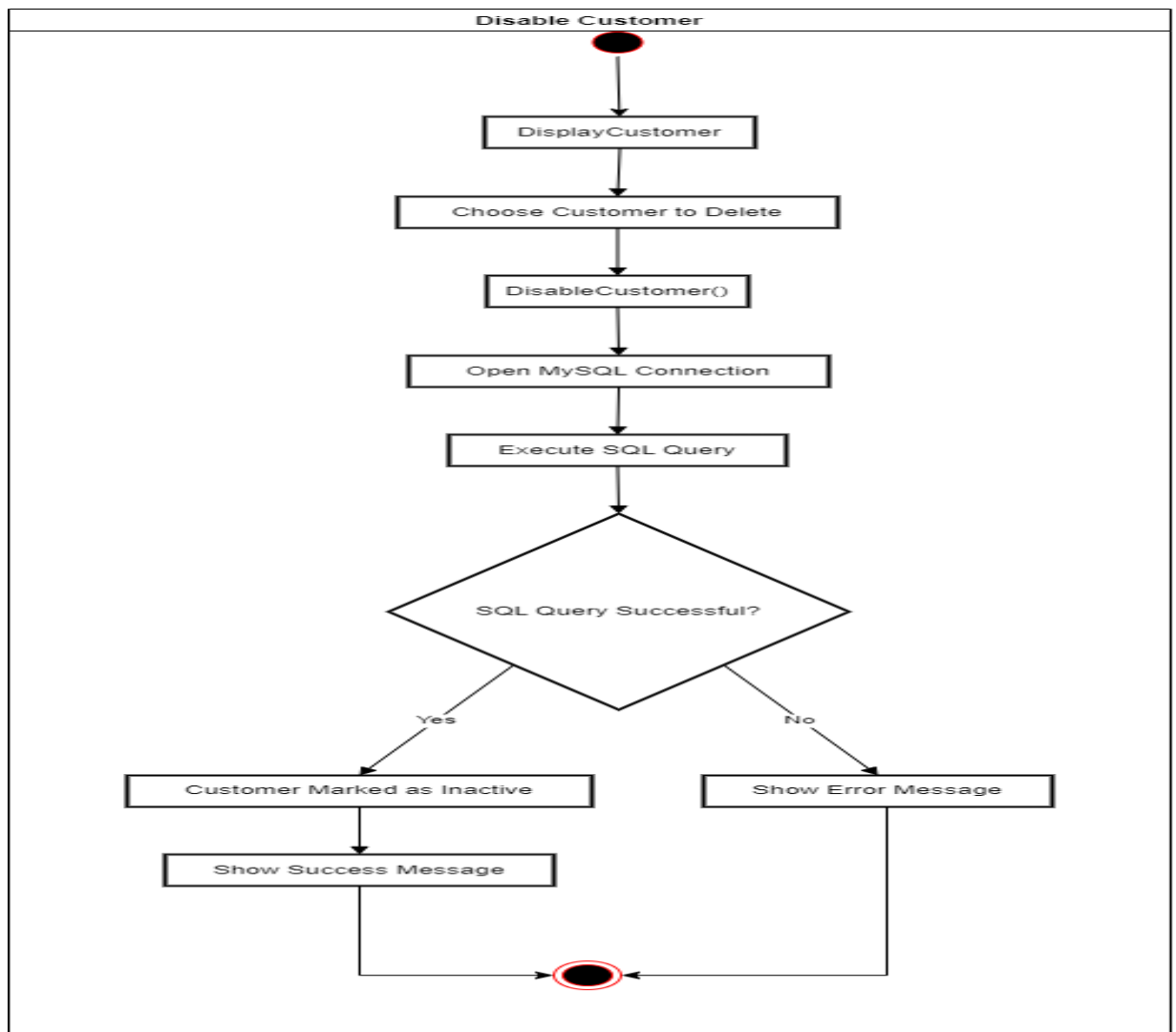


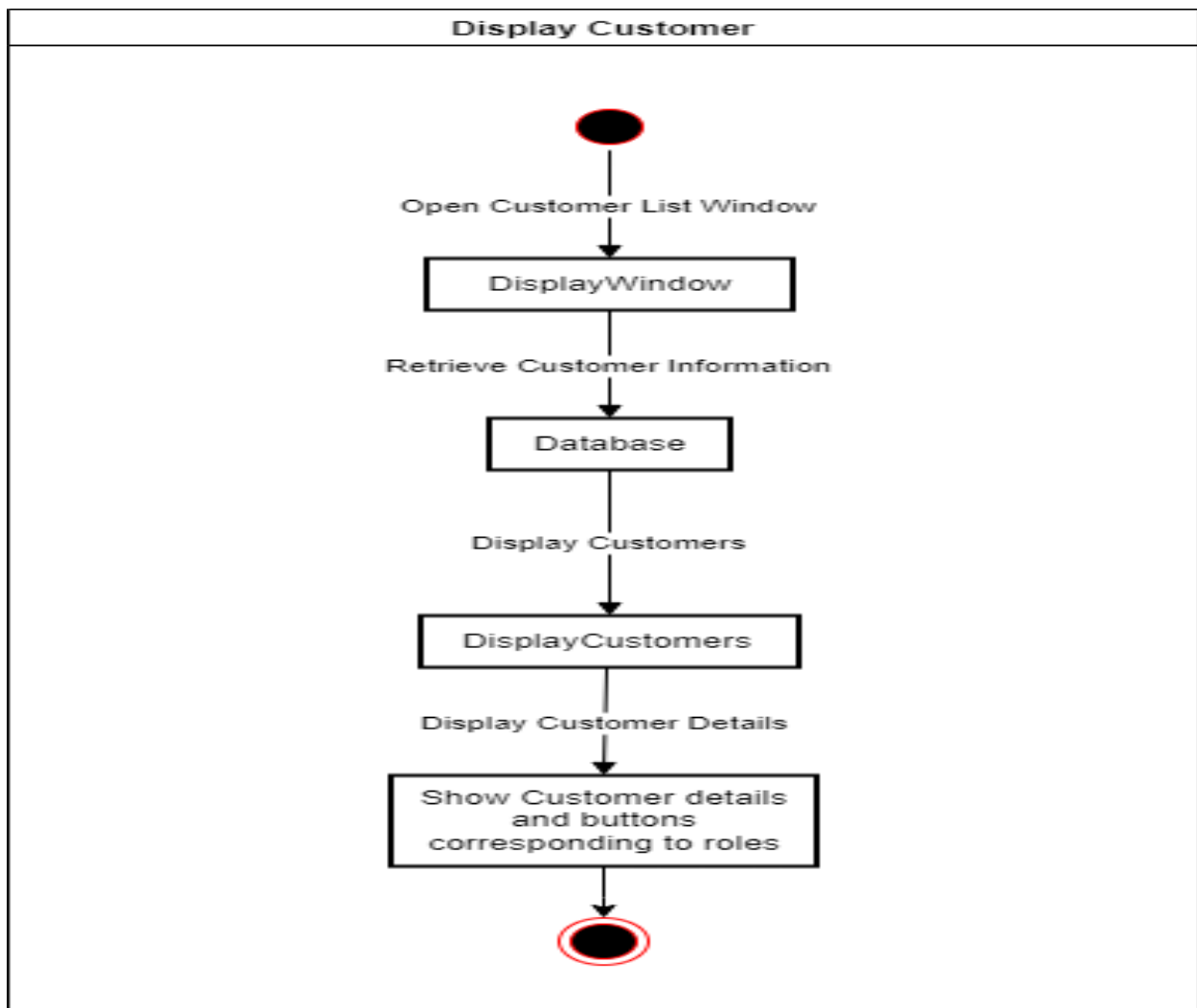


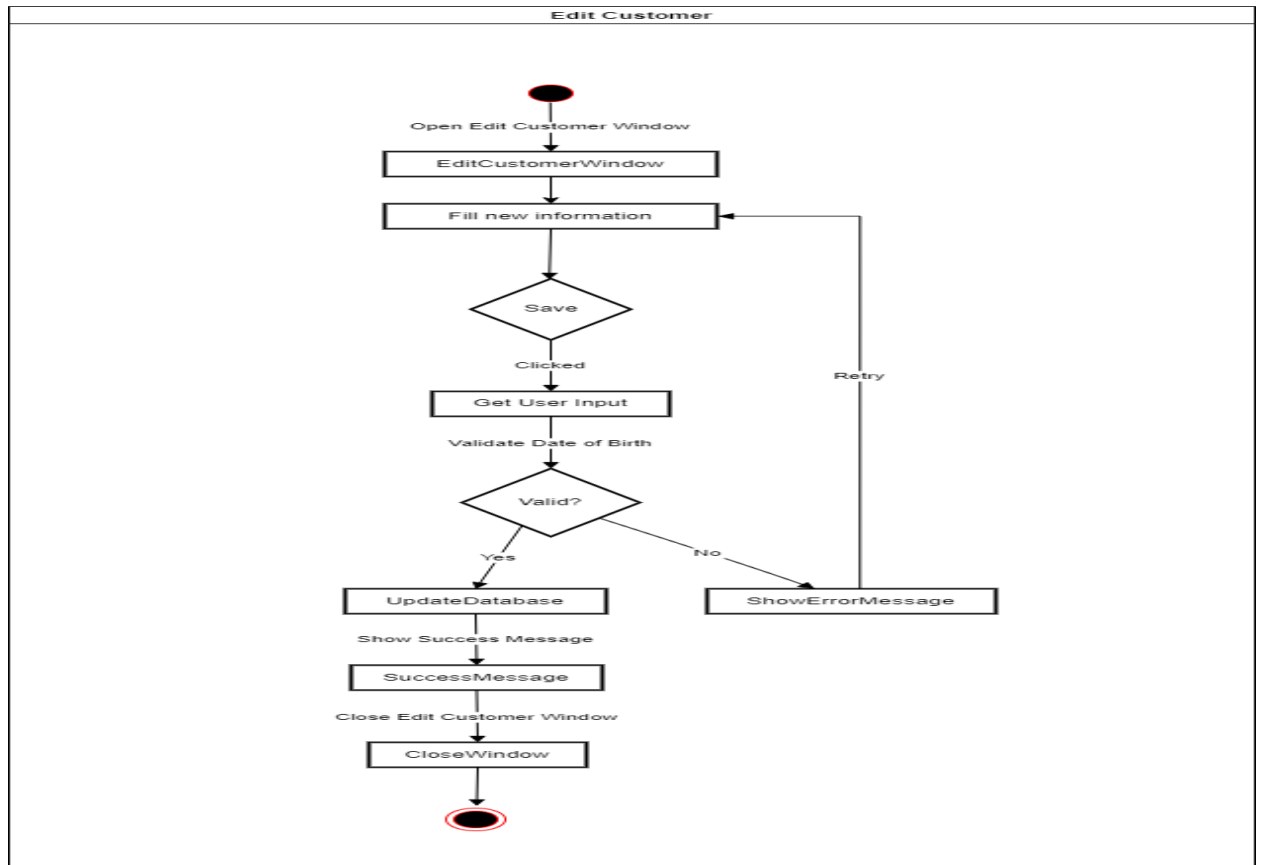


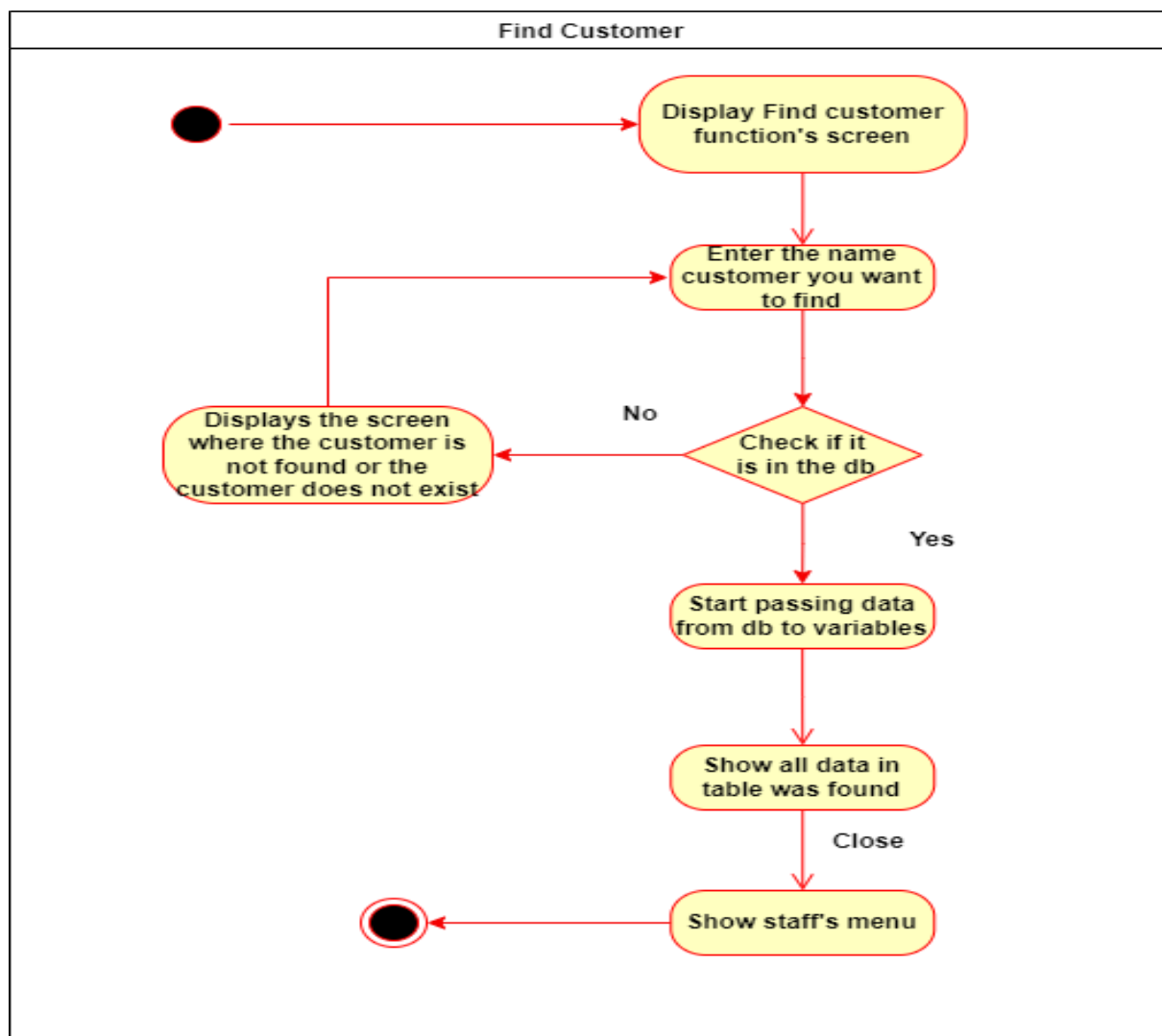


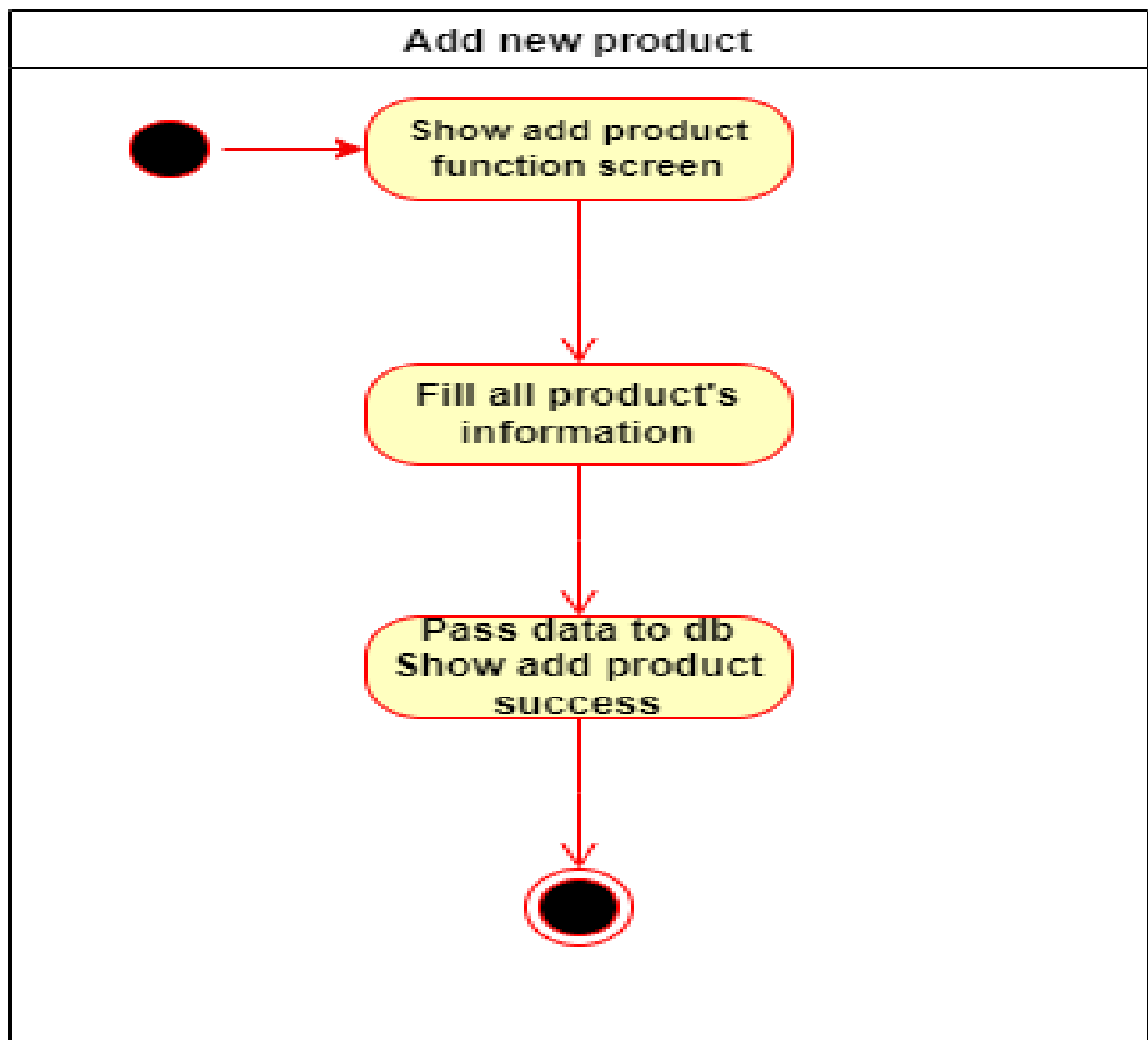


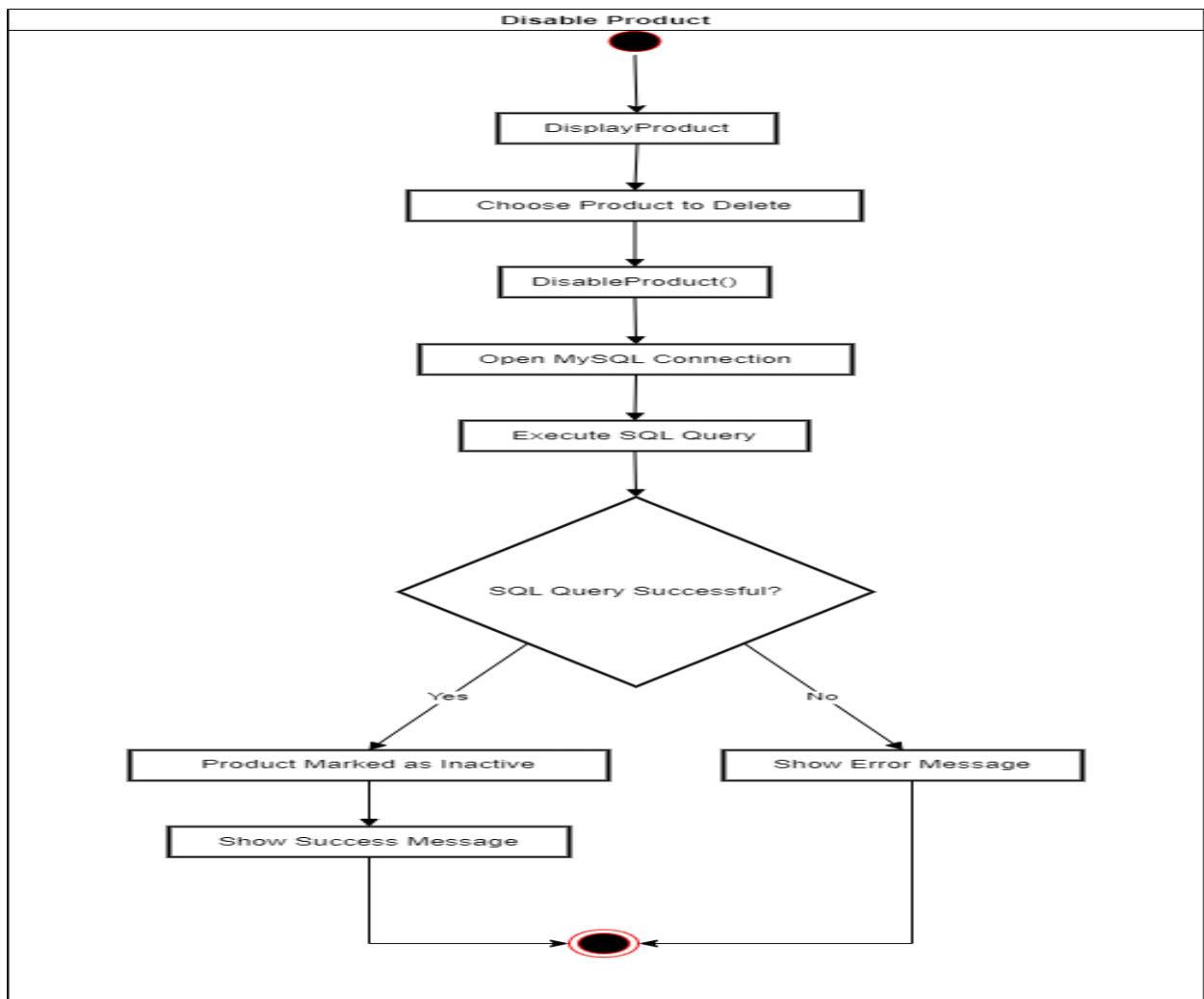


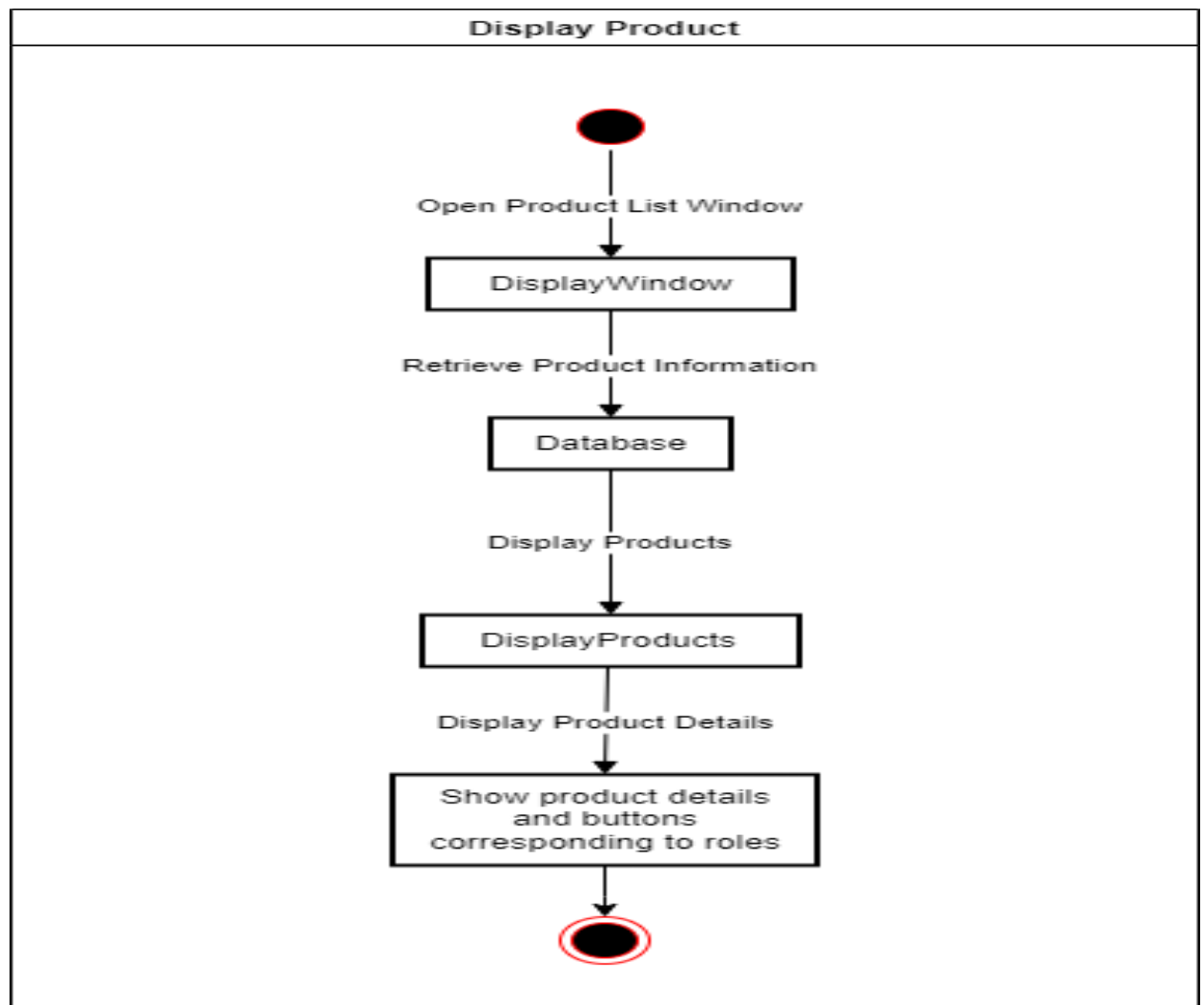


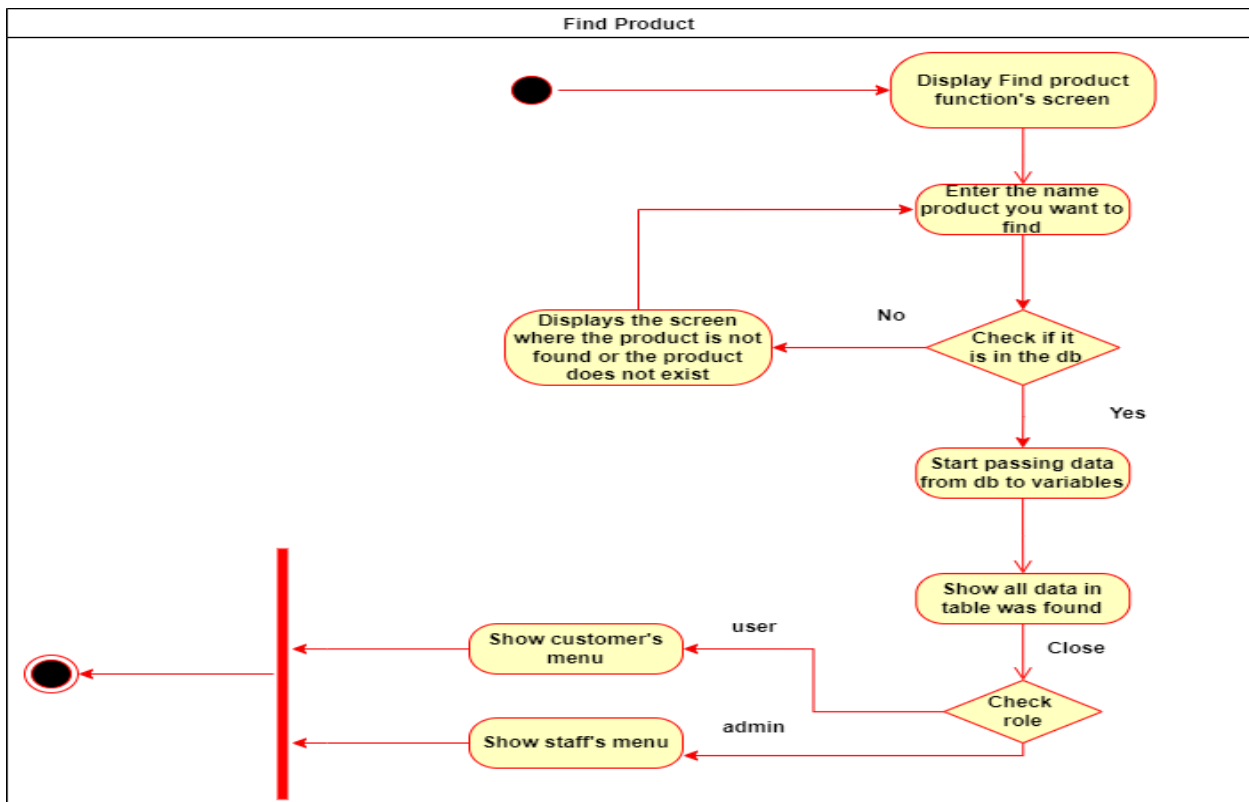
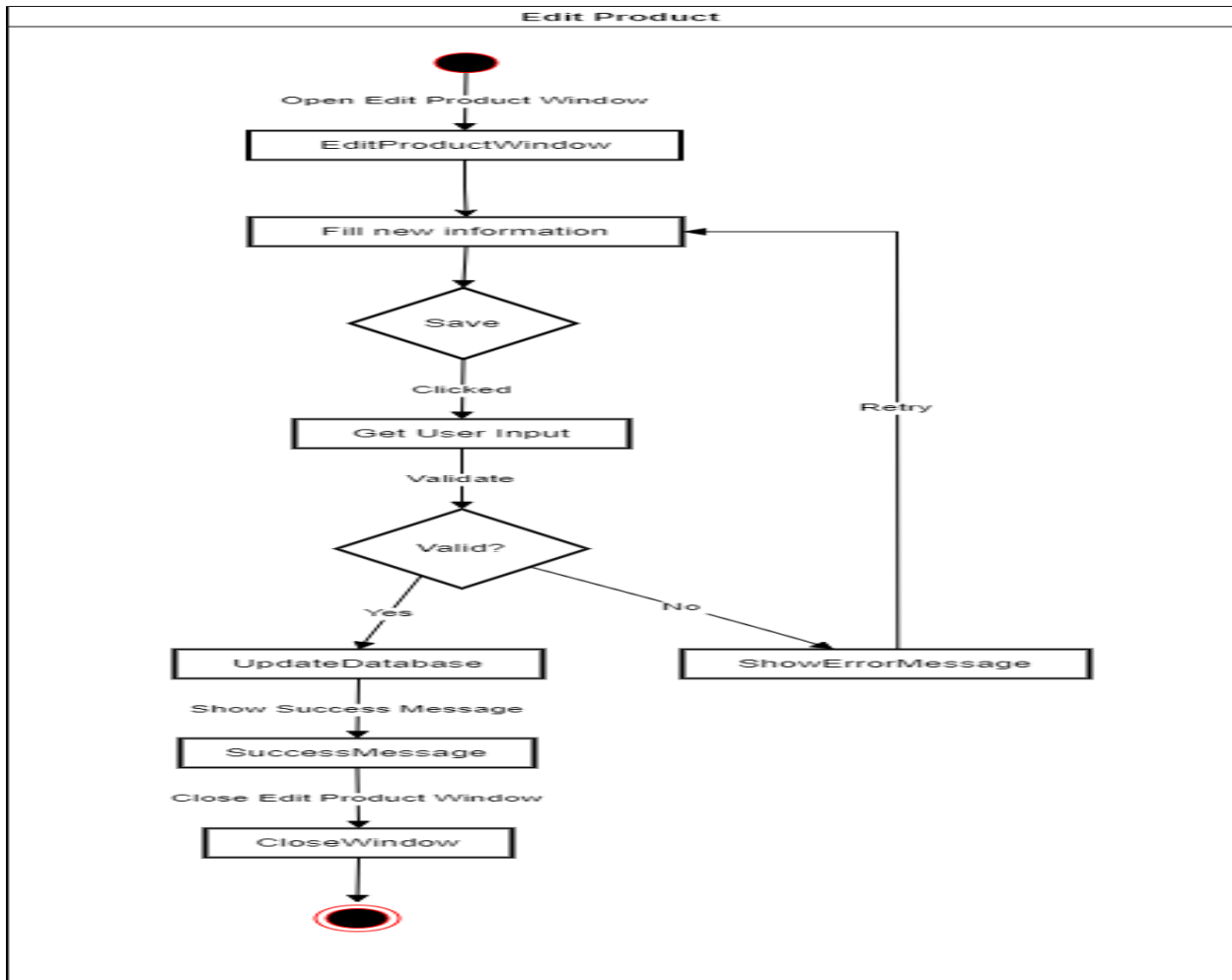


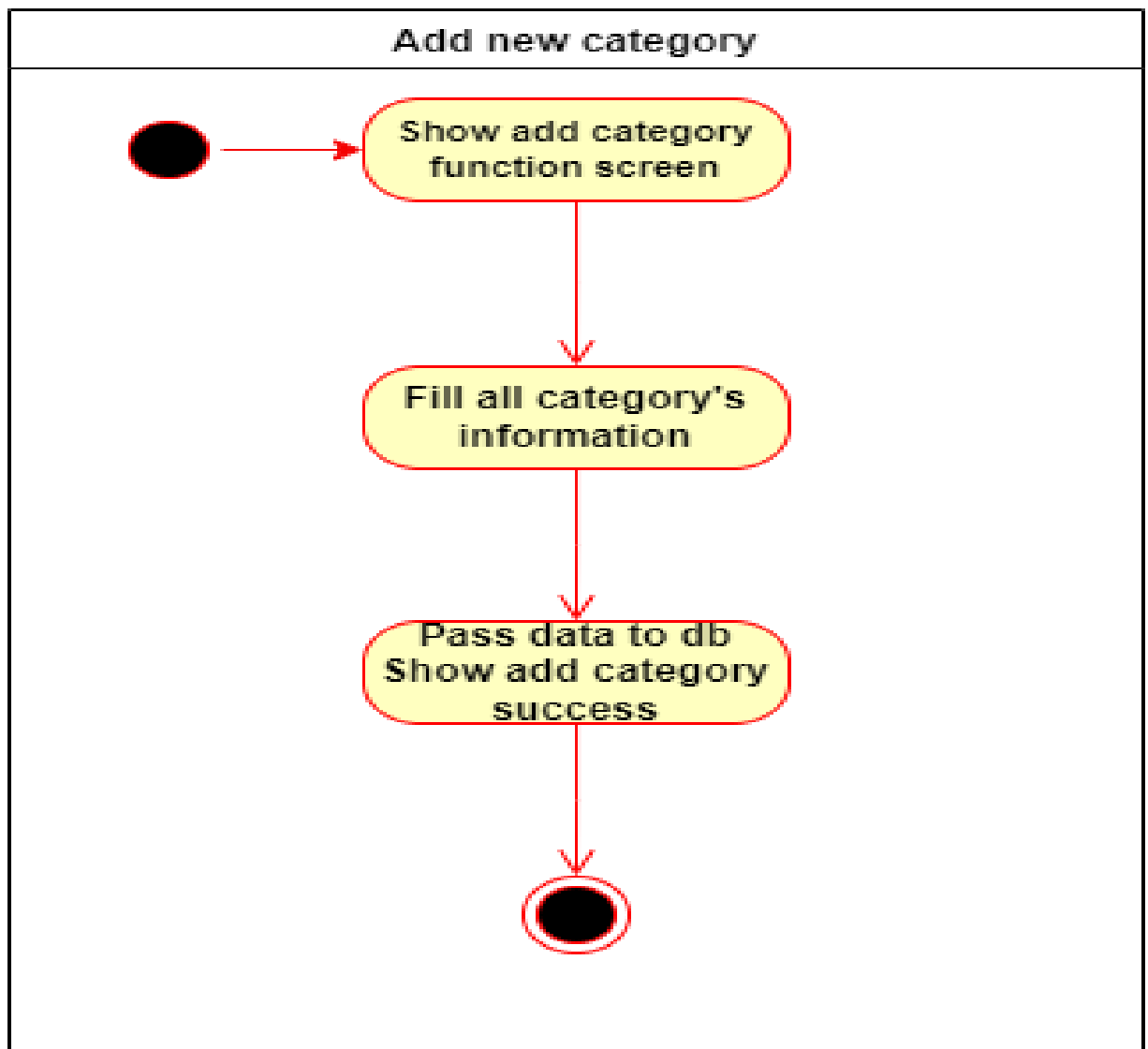


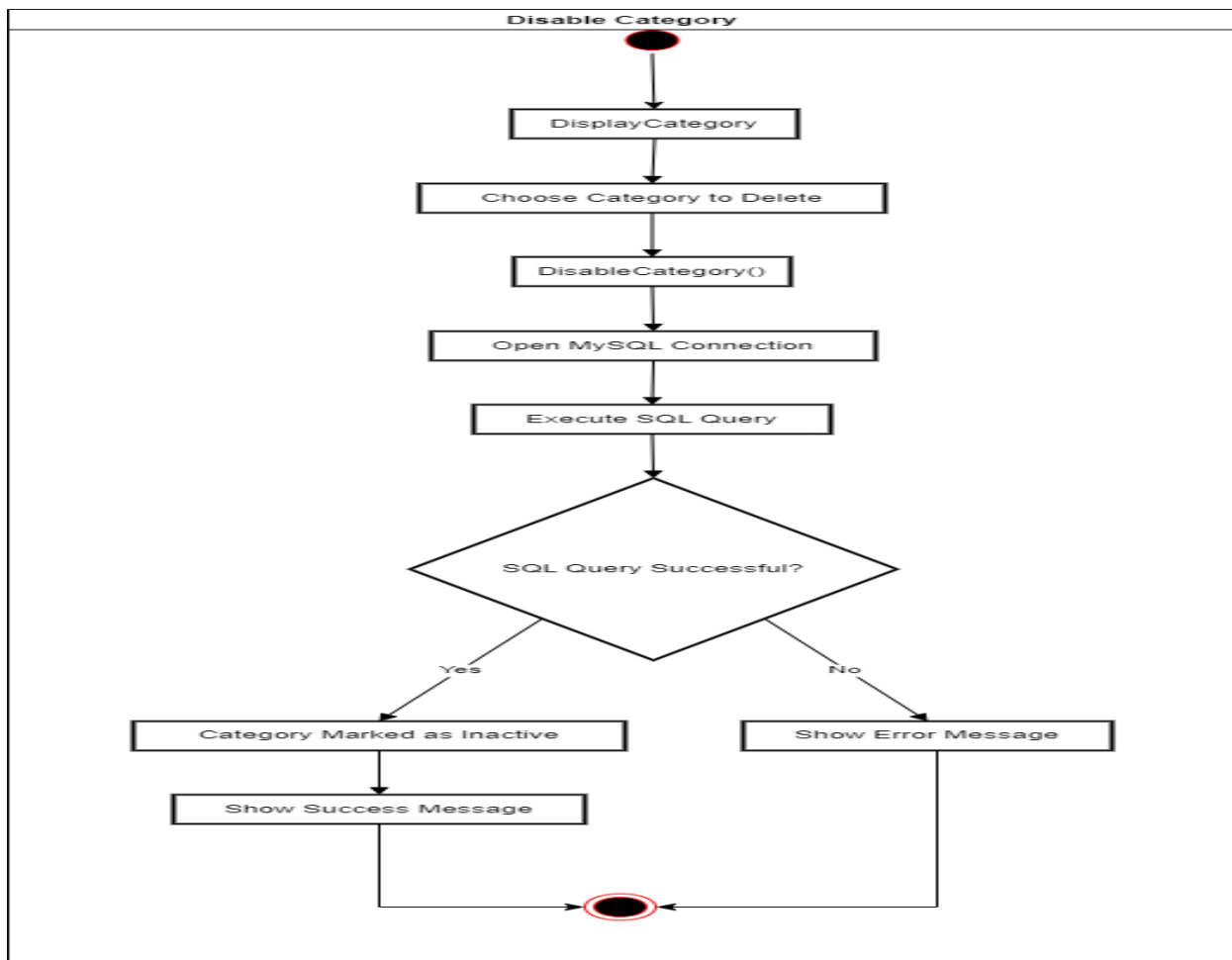


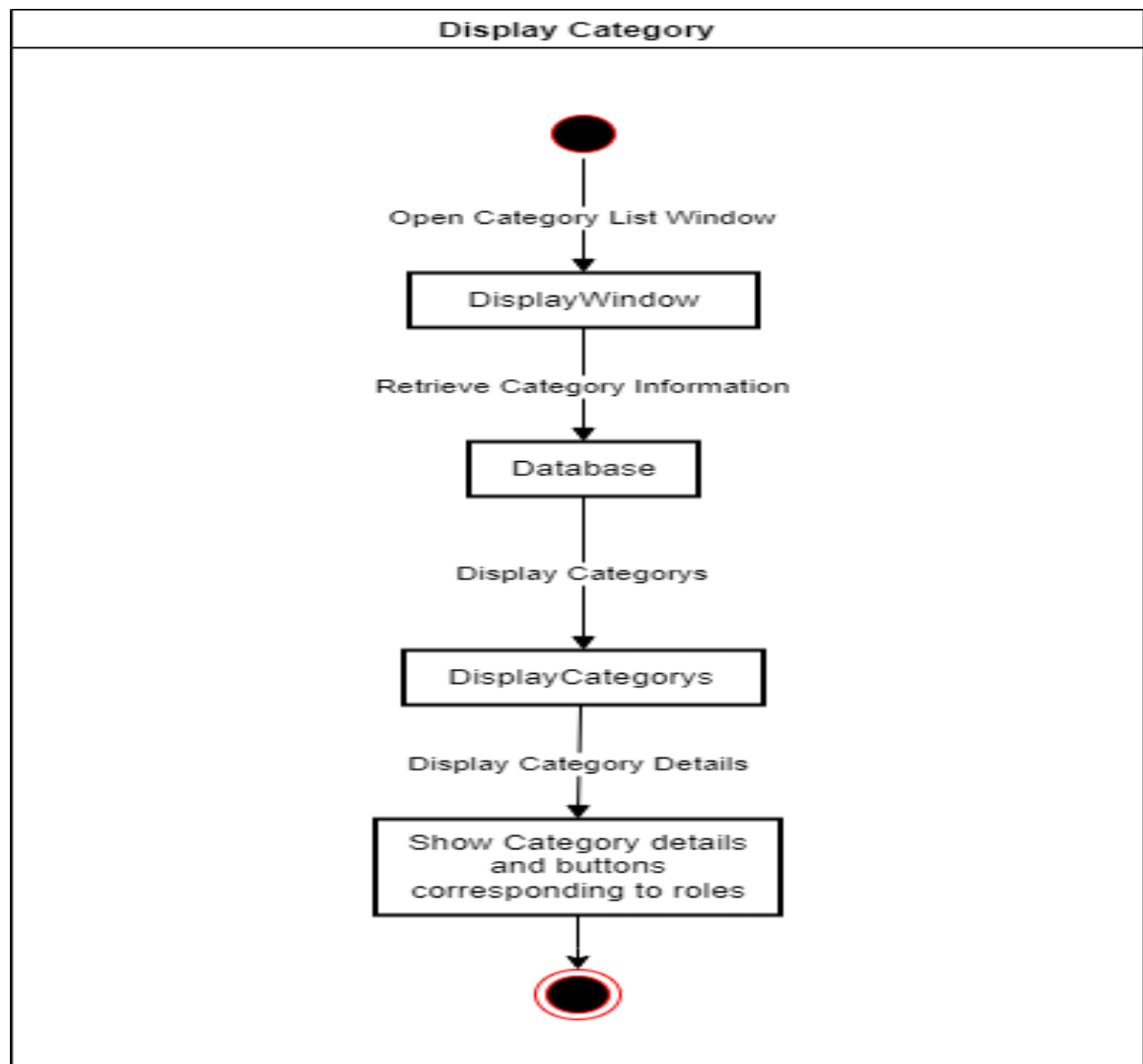


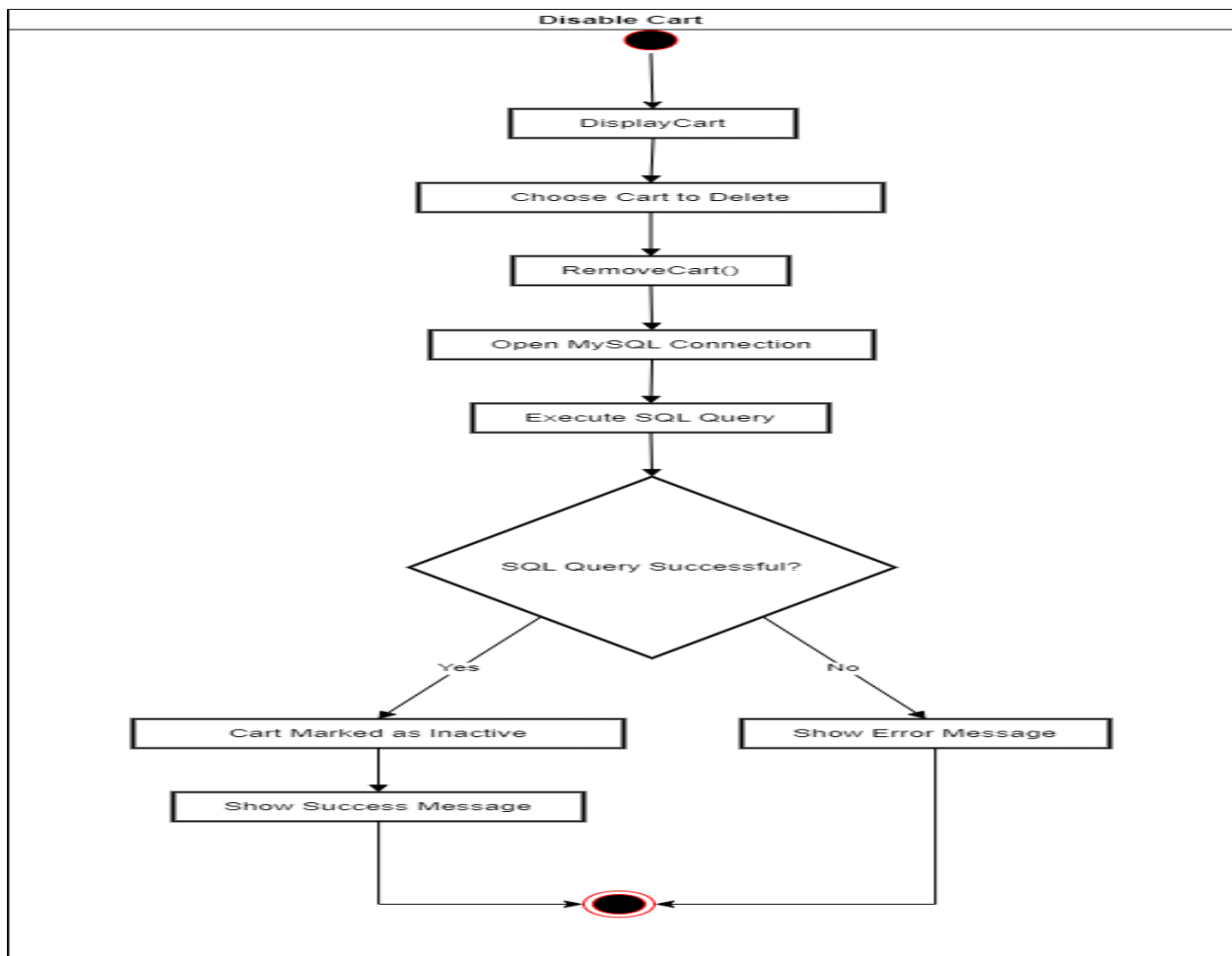


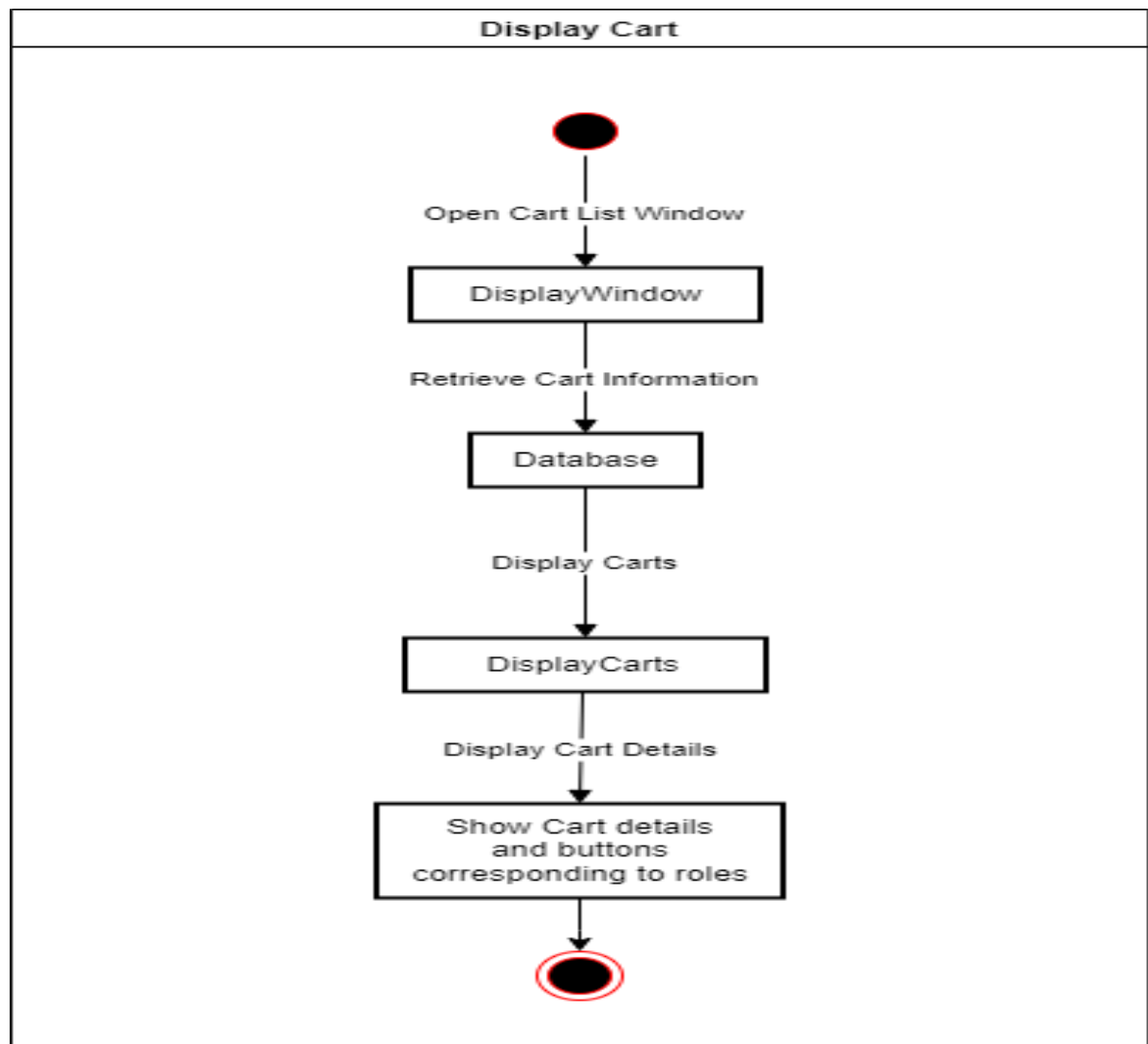


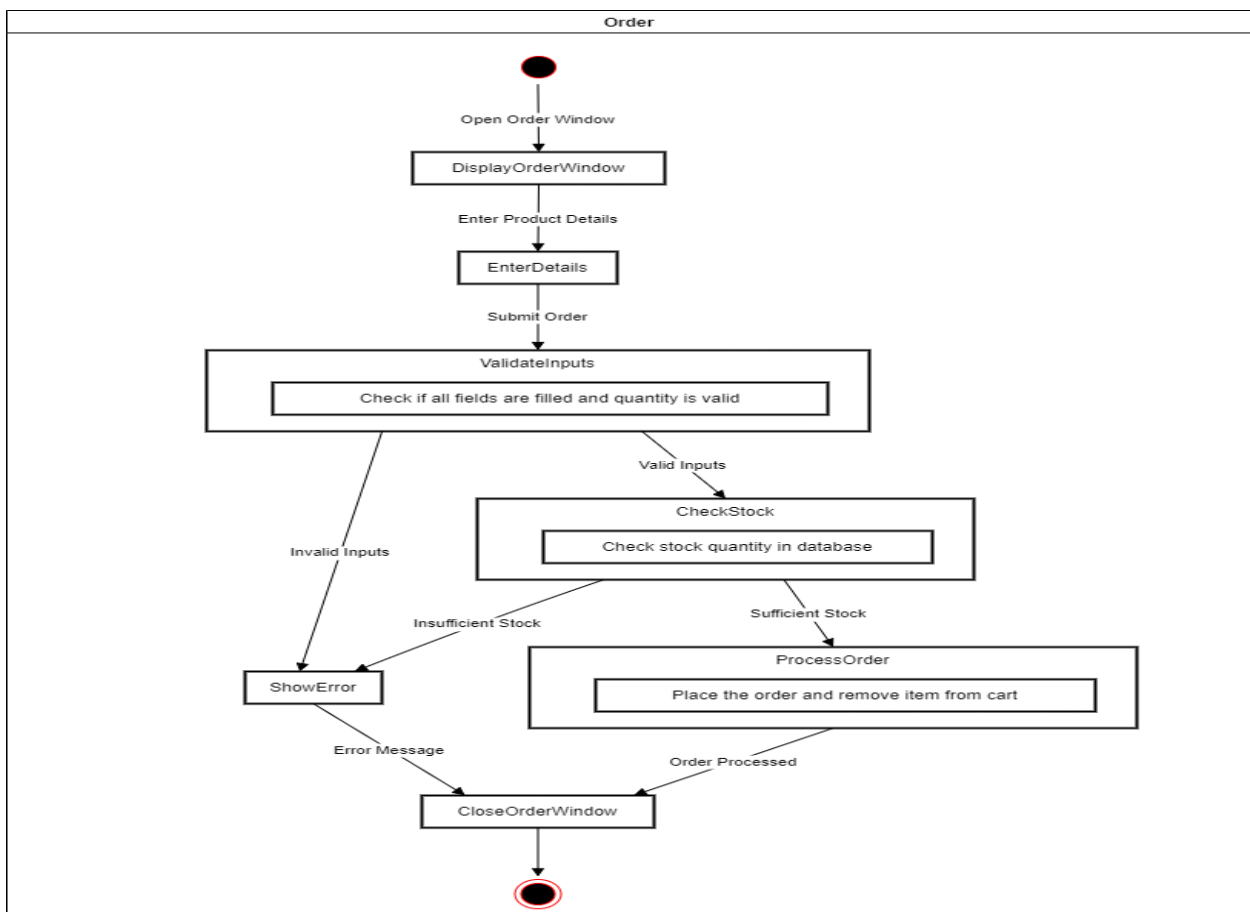
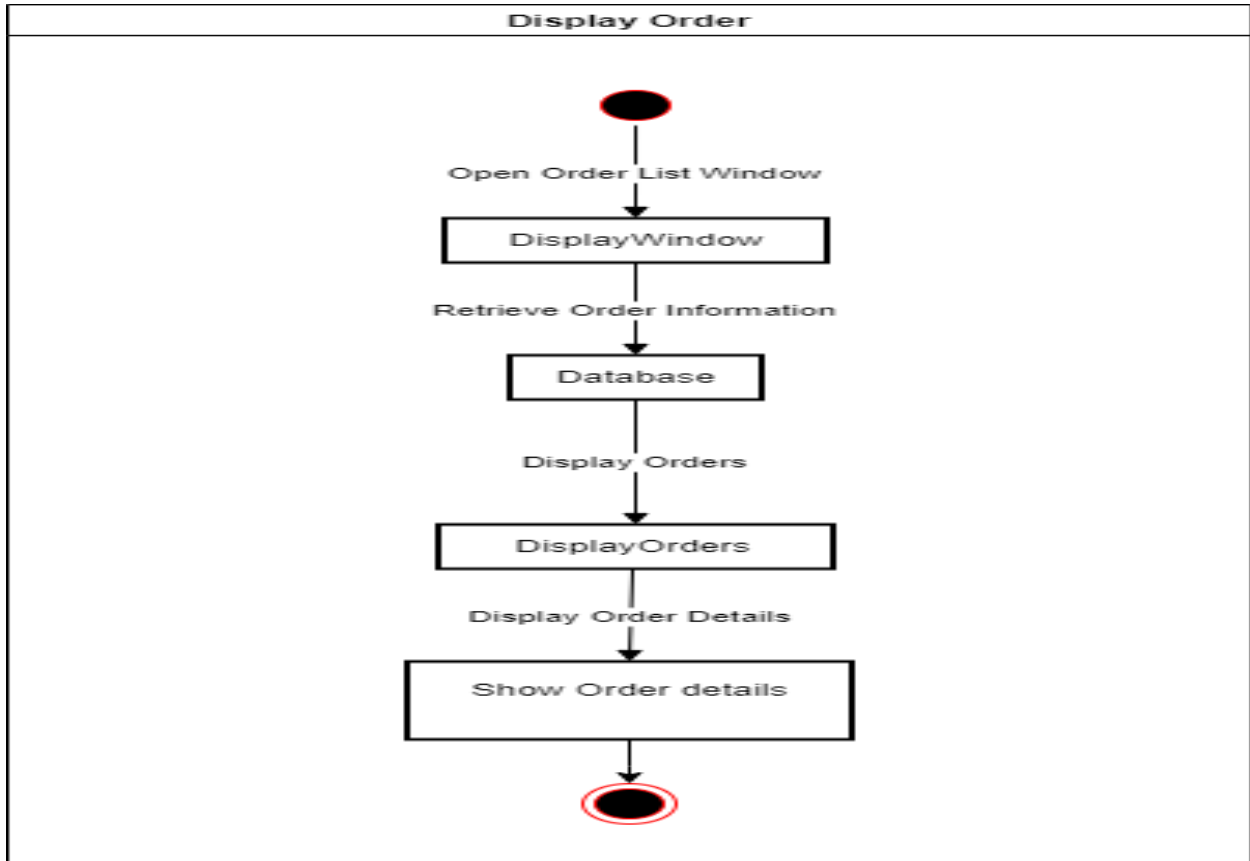








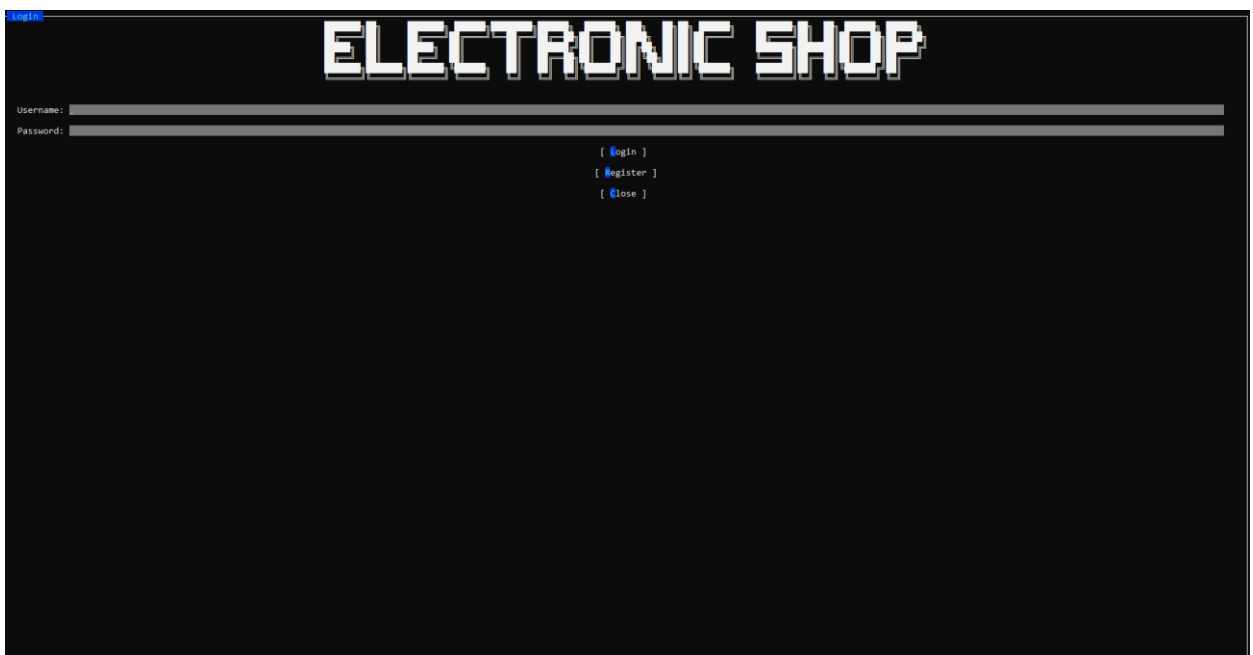




III. Design Details

1. UI Design

< Design the main user interface and for each features of the software >



Logout: 1234 Name: Long Phone: 1233 Address: QN Email: kl Gender: Male DOB: 11-06-2005 Orders: 2 Spent: \$2,849.96 Username: lololo Password: koko [Edit] [Delete]	Name: Jane Smith Phone: 123-456-0002 Address: 456 Oak Avenue, Springfield Email: janesmith@example.com Gender: F DOB: 22-07-1990 Orders: 2 Spent: \$7,659.87 Username: alicejohnson3 Password: hashed_password_3 [Edit] [Delete]	Name: Alice Johnson Phone: 123-456-0003 Address: 789 Pine Road, Springfield Email: alicejohnson@example.com Gender: F DOB: 11-03-1988 Orders: 0 Spent: \$0.00 Username: bobbrown4 Password: hashed_password_4 [Edit] [Delete]
Name: Carol Davis Phone: 123-456-0005 Address: 202 Birch Lane, Springfield Email: caroldavis@example.com Gender: F DOB: 05-01-1995 Orders: 0 Spent: \$0.00 Username: davidwilson6 Password: hashed_password_6 [Edit] [Delete]	Name: David Wilson Phone: 123-456-0006 Address: 303 Cedar Street, Springfield Email: davidwilson@example.com Gender: M DOB: 18-08-1982 Orders: 0 Spent: \$0.00 Username: emilyclark7 Password: hashed_password_7 [Edit] [Delete]	Name: Emily Clark Phone: 123-456-0007 Address: 404 Walnut Way, Springfield Email: emilyclark@example.com Gender: F DOB: 27-10-1993 Orders: 0 Spent: \$0.00 Username: frankharris8 Password: hashed_password_8 [Edit] [Delete]
Name: Frank Harris Phone: 123-456-0008 Address: 505 Cherry Hill, Springfield Email: frankharris@example.com Gender: M DOB: 09-02-1987 Orders: 0 Spent: \$0.00 Username: irenemartinez9 Password: hashed_password_9 [Edit] [Delete]	Name: Henry Walker Phone: 123-456-0010 Address: 707 Ash Street, Springfield Email: henrywalker@example.com Gender: Male DOB: 21-12-1986 Orders: 0 Spent: \$0.00 Username: irenemartinez11 Password: hashed_password_11 [Edit] [Delete]	Name: Irene Martinez Phone: 123-456-70 Address: 808 Maple Street, Springfield Email: irenemartinez1@example.com Gender: F DOB: 17-04-1989 Orders: 0 Spent: \$0.00 Username: jamesthompson12 Password: hashed_password_12 [Edit] [Delete]
Name: Katherine White Phone: 123-456-0013 Address: 1010 Oak Road, Springfield Email: katherinewhite13@example.com Gender: F DOB: 25-06-1984 Orders: 0 Spent: \$0.00	Name: Liam Anderson Phone: 123-456-0014 Address: 1111 Birch Lane, Springfield Email: liamanderson14@example.com Gender: M DOB: 15-07-1987 Orders: 0 Spent: \$0.00	Name: Megan Hall Phone: 123-456-0015 Address: 1212 Pine Hill, Springfield Email: meganhall15@example.com Gender: F DOB: 22-08-1993 Orders: 0 Spent: \$0.00 [Back]

EDIT CUSTOMER

Customer Name:

Customer Phone:

Customer Address:

Customer Email:

Customer Gender:

Customer Date of Birth (dd-mm-yyyy):

Username:

Password:

[Save]

[Lose]

Register for Customer

Username:

Password:

Name:

Phone number:

Address:

Email:

Gender:

Date of birth (DD-MM-YYYY):

[Register]

[Close]

Product List Samsung Galaxy S21 Name: Samsung Galaxy S21 Stock: 31 Price: 8999.00 \$ Category: Smartphones Brand: Samsung [Edit] [Order For Customer]	Google Pixel 7 Name: Google Pixel 7 Stock: 30 Price: 699.99 \$ Category: Smartphones Brand: Google [Edit] [Order For Customer]	OnePlus 10 Name: OnePlus 10 Stock: 25 Price: 749.99 \$ Category: Smartphones Brand: OnePlus [Edit] [Order For Customer]	Dell XPS 13 [Back] Name: Dell XPS 13 Stock: 30 Price: 1299.99 \$ Category: Laptops Brand: Dell [Edit] [Order For Customer]	MacBook Pro Name: MacBook Pro Stock: 15 Price: 2399.99 \$ Category: Laptops Brand: Apple [Edit] [Order For Customer]	HP Spectre x360 Name: HP Spectre x360 Stock: 10 Price: 1399.99 \$ Category: Laptops Brand: HP [Edit] [Order For Customer]
Lenovo ThinkPad X1 Name: Lenovo ThinkPad X1 Stock: 10 Price: 1599.99 \$ Category: Laptops Brand: Lenovo [Edit] [Order For Customer]	Apple iPad Air Name: Apple iPad Air Stock: 35 Price: 599.99 \$ Category: Tablets Brand: Apple [Edit] [Order For Customer]	Samsung Galaxy Tab S8 Name: Samsung Galaxy Tab S8 Stock: 10 Price: 649.99 \$ Category: Tablets Brand: Samsung [Edit] [Order For Customer]	Microsoft Surface Pro 8 Name: Microsoft Surface Pro 8 Stock: 17 Price: 999.99 \$ Category: Tablets Brand: Microsoft [Edit] [Order For Customer]	Lenovo Tab P11 Name: Lenovo Tab P11 Stock: 20 Price: 349.99 \$ Category: Tablets Brand: Lenovo [Edit] [Order For Customer]	Apple Watch Series 8 Name: Apple Watch Series 8 Stock: 30 Price: 399.99 \$ Category: Smartwatches Brand: Apple [Edit] [Order For Customer]
Samsung Galaxy Watch 5 Name: Samsung Galaxy Watch Stock: 25 Price: 349.99 \$ Category: Smartwatches Brand: Samsung [Edit] [Order For Customer]	Garmin Forerunner 945 Name: Garmin Forerunner 945 Stock: 10 Price: 599.99 \$ Category: Smartwatches Brand: Garmin [Edit] [Order For Customer]	Fitbit Versa 4 Name: Fitbit Versa 4 Stock: 10 Price: 229.99 \$ Category: Smartwatches Brand: Fitbit [Edit] [Order For Customer]	LG OLED C1 Name: LG OLED C1 Stock: 0 Price: 1499.99 \$ Category: Televisions Brand: LG [Edit] [Order For Customer]	Samsung QN90B Name: Samsung QN90B Stock: 10 Price: 1299.99 \$ Category: Televisions Brand: Samsung [Edit] [Order For Customer]	Sony Bravia X90J Name: Sony Bravia X90J Stock: 14 Price: 1099.99 \$ Category: Televisions Brand: Sony [Edit] [Order For Customer]
TCL 6-Series Name: TCL 6-Series Stock: 10 Price: 899.99 \$ Category: Televisions Brand: TCL [Edit] [Order For Customer]	Canon EOS R5 Name: Canon EOS R5 Stock: 0 Price: 3899.99 \$ Category: Cameras Brand: Canon [Edit] [Order For Customer]	Nikon Z6 II Name: Nikon Z6 II Stock: 10 Price: 1999.99 \$ Category: Cameras Brand: Nikon [Edit] [Order For Customer]	Sony Alpha A7 IV Name: Sony Alpha A7 IV Stock: 7 Price: 2499.99 \$ Category: Cameras Brand: Sony [Edit] [Order For Customer]	Fujifilm X-T4 Name: Fujifilm X-T4 Stock: 12 Price: 1699.99 \$ Category: Cameras Brand: Fujifilm [Edit] [Order For Customer]	Sony WH-1000XM4 Name: Sony WH-1000XM4 Stock: 45 Price: 349.99 \$ Category: Headphones Brand: Sony [Edit] [Order For Customer]
Bose QuietComfort 35 II Name: Bose QuietComfort 35 Stock: 50 Price: 299.99 \$	Apple AirPods Max Name: Apple AirPods Max Stock: 30 Price: 549.99 \$	Sennheiser Momentum 3 Name: Sennheiser Momentum 3 Stock: 25 Price: 399.99 \$	JBL Charge 5 Name: JBL Charge 5 Stock: 40 Price: 149.99 \$	Bose SoundLink Revolve Name: Bose SoundLink Revolve Stock: 35 Price: 229.99 \$	Sonos One Name: Sonos One Stock: 28 Price: 199.99 \$

Edit Product Information

Product Name:

Samsung Galaxy S21

Stock Quantity:

31

Category ID:

1

Price:

8999.00

Brand:

Samsung

[Save]

[Close]

Order Product

Product: Samsung Galaxy S21

Customer ID:

[Confirm]

[Cancel]

Add Product

Product Name:

Stock Quantity:

Category ID:

Price:

Brand:

[Save]

[Close]

Find Product

Product name:

[Find]

[Back]

File

Order Status Updater

Order ID:

[Update Status]

[Close]

Electronic Shop Functions

View Product

Find Product

View My Order

View My Cart

View Category

[Logout]

Welcome to

ELECTRONIC SHOP

HELLO

Name: Long

Number of times ordered: 2

Total amount spent: 2849.96

Product List Samsung Galaxy S21 Name: Samsung Galaxy S21 Stock: 31 Price: 8999.00 \$ Category: Smartphones Brand: Samsung Add to Cart Order	Google Pixel 7 Name: Google Pixel 7 Stock: 30 Price: 699.99 \$ Category: Smartphones Brand: Google Add to Cart Order	OnePlus 10 Name: OnePlus 10 Stock: 25 Price: 749.99 \$ Category: Smartphones Brand: OnePlus Add to Cart Order	Dell XPS 13 Back Name: Dell XPS 13 Stock: 20 Price: 1299.99 \$ Category: Laptops Brand: Dell Add to Cart Order	MacBook Pro Name: MacBook Pro Stock: 15 Price: 2399.99 \$ Category: Laptops Brand: Apple Add to Cart Order	HP Spectre x360 Name: HP Spectre x360 Stock: 10 Price: 1399.99 \$ Category: Laptops Brand: HP Add to Cart Order
Lenovo ThinkPad X1 Name: Lenovo ThinkPad X1 Stock: 18 Price: 1599.99 \$ Category: Laptops Brand: Lenovo Add to Cart Order	Apple iPad Air Name: Apple iPad Air Stock: 35 Price: 599.99 \$ Category: Tablets Brand: Apple Add to Cart Order	Samsung Galaxy Tab S8 Name: Samsung Galaxy Tab S8 Stock: 16 Price: 649.99 \$ Category: Tablets Brand: Samsung Add to Cart Order	Microsoft Surface Pro 8 Name: Microsoft Surface Pro 8 Stock: 17 Price: 999.99 \$ Category: Tablets Brand: Microsoft Add to Cart Order	Lenovo Tab P11 Name: Lenovo Tab P11 Stock: 28 Price: 349.99 \$ Category: Tablets Brand: Lenovo Add to Cart Order	Apple Watch Series 8 Name: Apple Watch Series 8 Stock: 30 Price: 399.99 \$ Category: Smartwatches Brand: Apple Add to Cart Order
Samsung Galaxy Watch 5 Name: Samsung Galaxy Watch Stock: 25 Price: 349.99 \$ Category: Smartwatches Brand: Samsung Add to Cart Order	Garmin Forerunner 945 Name: Garmin Forerunner 945 Stock: 16 Price: 599.99 \$ Category: Smartwatches Brand: Garmin Add to Cart Order	Fitbit Versa 4 Name: Fitbit Versa 4 Stock: 16 Price: 229.99 \$ Category: Smartwatches Brand: Fitbit Add to Cart Order	LG OLED C1 Name: LG OLED C1 Stock: 8 Price: 1499.99 \$ Category: Televisions Brand: LG Add to Cart Order	Samsung QN90B Name: Samsung QN90B Stock: 18 Price: 1299.99 \$ Category: Televisions Brand: Samsung Add to Cart Order	Sony Bravia X90J Name: Sony Bravia X90J Stock: 14 Price: 1099.99 \$ Category: Televisions Brand: Sony Add to Cart Order
TCL 6-Series Name: TCL 6-Series Stock: 16 Price: 899.99 \$ Category: Televisions Brand: TCL Add to Cart Order	Canon EOS R5 Name: Canon EOS R5 Stock: 8 Price: 3899.99 \$ Category: Cameras Brand: Canon Add to Cart Order	Nikon Z6 II Name: Nikon Z6 II Stock: 18 Price: 1999.99 \$ Category: Cameras Brand: Nikon Add to Cart Order	Sony Alpha A7 IV Name: Sony Alpha A7 IV Stock: 7 Price: 2499.99 \$ Category: Cameras Brand: Sony Add to Cart Order	Fujifilm X-T4 Name: Fujifilm X-T4 Stock: 12 Price: 1699.99 \$ Category: Cameras Brand: Fujifilm Add to Cart Order	Sony WH-1000XM4 Name: Sony WH-1000XM4 Stock: 45 Price: 349.99 \$ Category: Headphones Brand: Sony Add to Cart Order
Rose QuietComfort 35 II Name: Rose QuietComfort 35 Stock: 58 Add to Cart Order	Apple AirPods Max Name: Apple AirPods Max Stock: 36 Add to Cart Order	Sennheiser Momentum 3 Name: Sennheiser Momentum 3 Stock: 25 Add to Cart Order	JBL Charge 5 Name: JBL Charge 5 Stock: 40 Add to Cart Order	Bose SoundLink Revolve Name: Bose SoundLink Revolve Stock: 35 Add to Cart Order	Sonos One Name: Sonos One Stock: 28 Add to Cart Order

Order Product

Product: Samsung Galaxy S21

Quantity:

Delivery Address:

Payment Method:

[Submit Order](#) [Cancel](#)

File
Find Product

Product name:

[Find]

[Back]

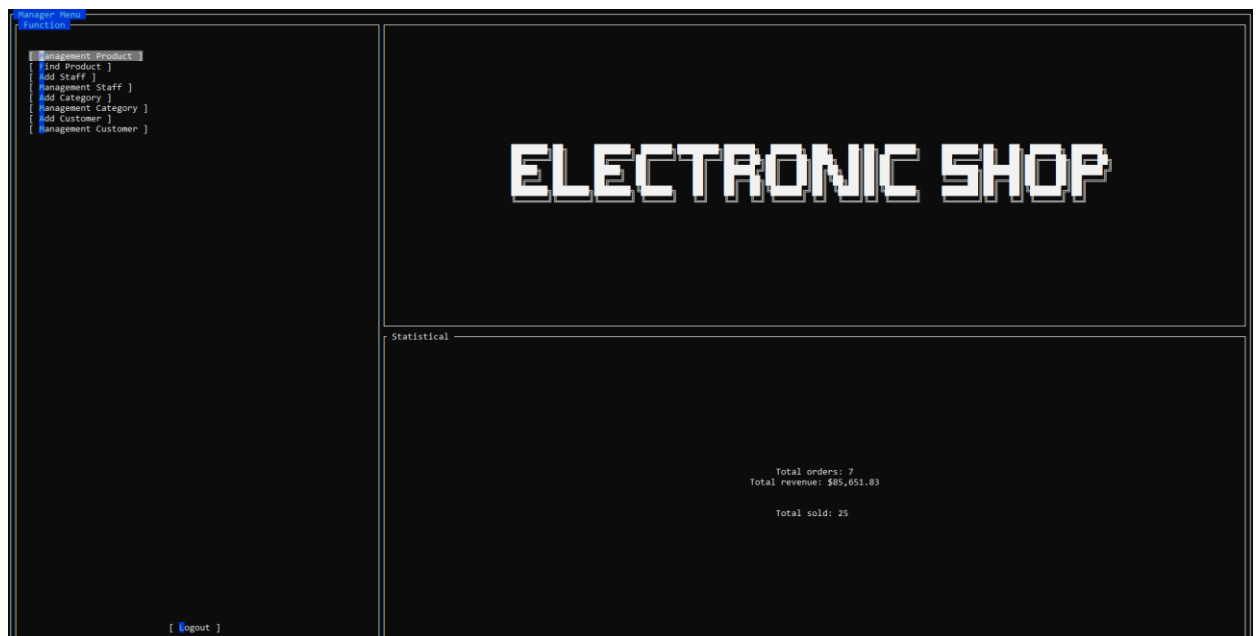
File
My Orders
Order 1

Product Name: Samsung Galaxy Tab S8
Quantity: 3
Payment: Tm
Status: Pending
Total Price: \$1,949.97

Order 2

Product Name: Samsung Galaxy S21
Quantity: 1
Payment: TM
Status: Pending
Total Price: \$899.99

[Close]



Register for Staff

Username:

Password:

Name:

Phone number:

Email:

Gender:

[Register]

[Close]

Staff List

Name	Phone	Email	Gender	Username	Password	Actions
HI	321	add	fgf	admin	admin123	[Edit] [Delete]
hello	123	fg	hgfd	admin1	admin456	[Edit] [Delete]

[Back]

File

Add Category

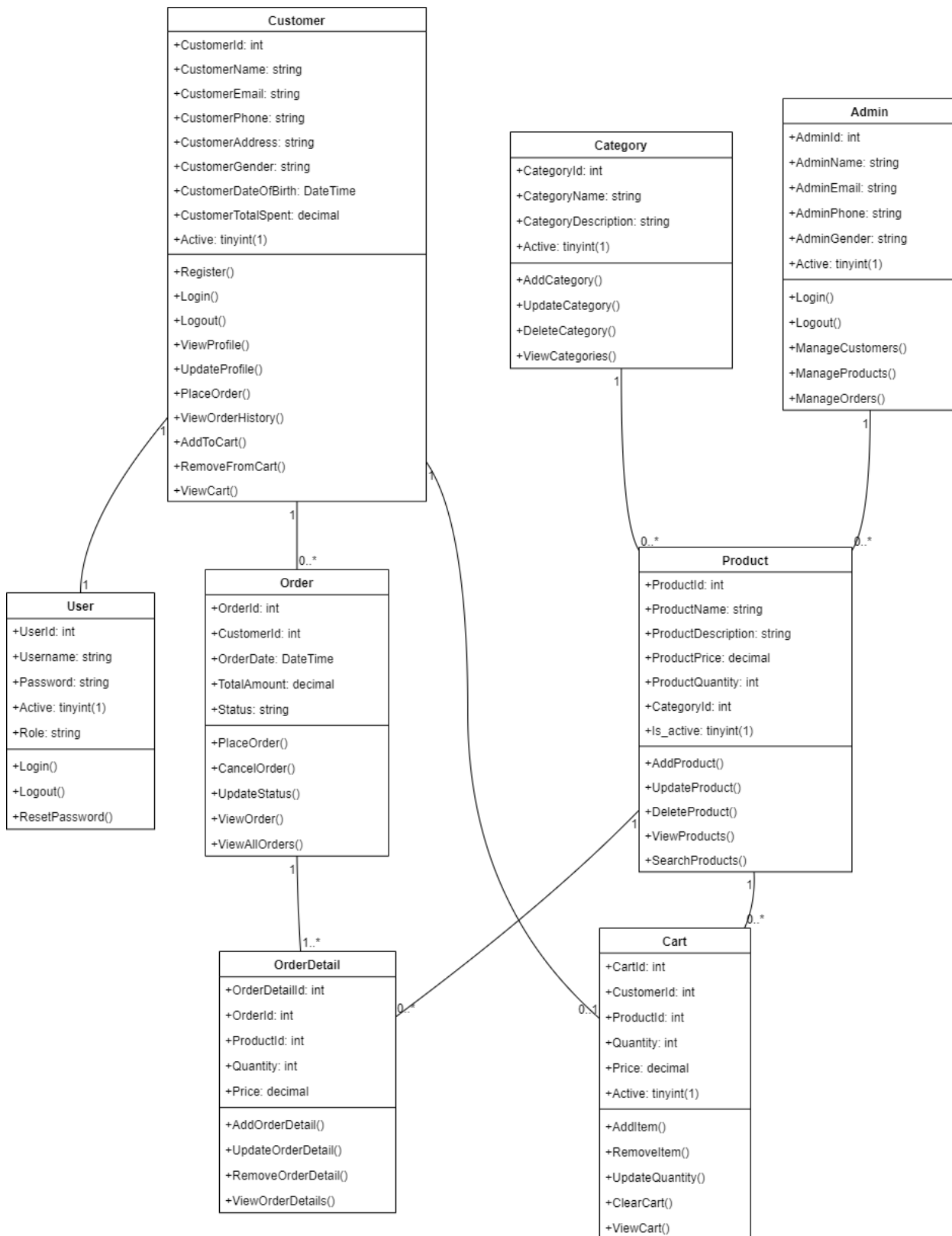
Category Name:

Description:

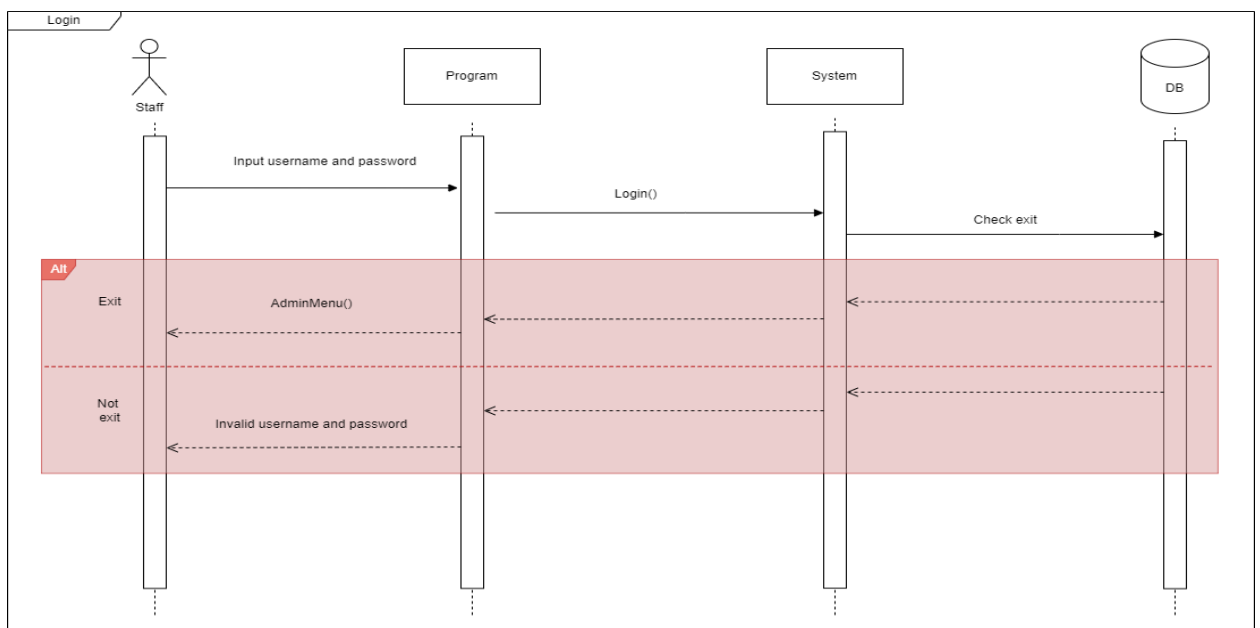
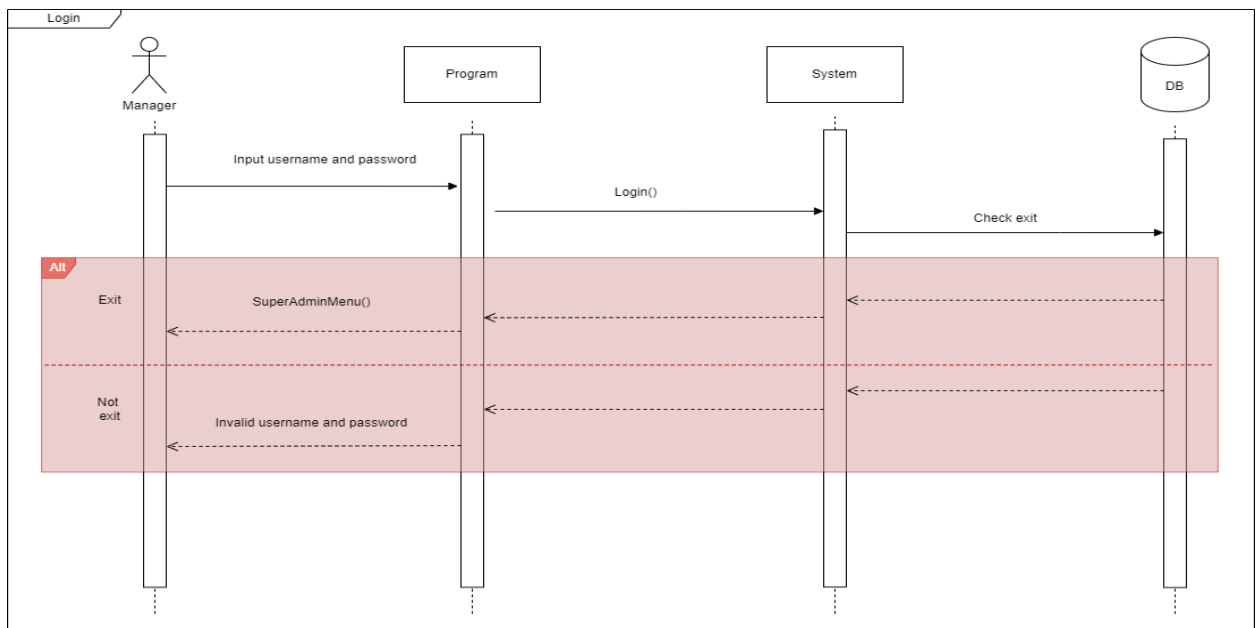
[Save]

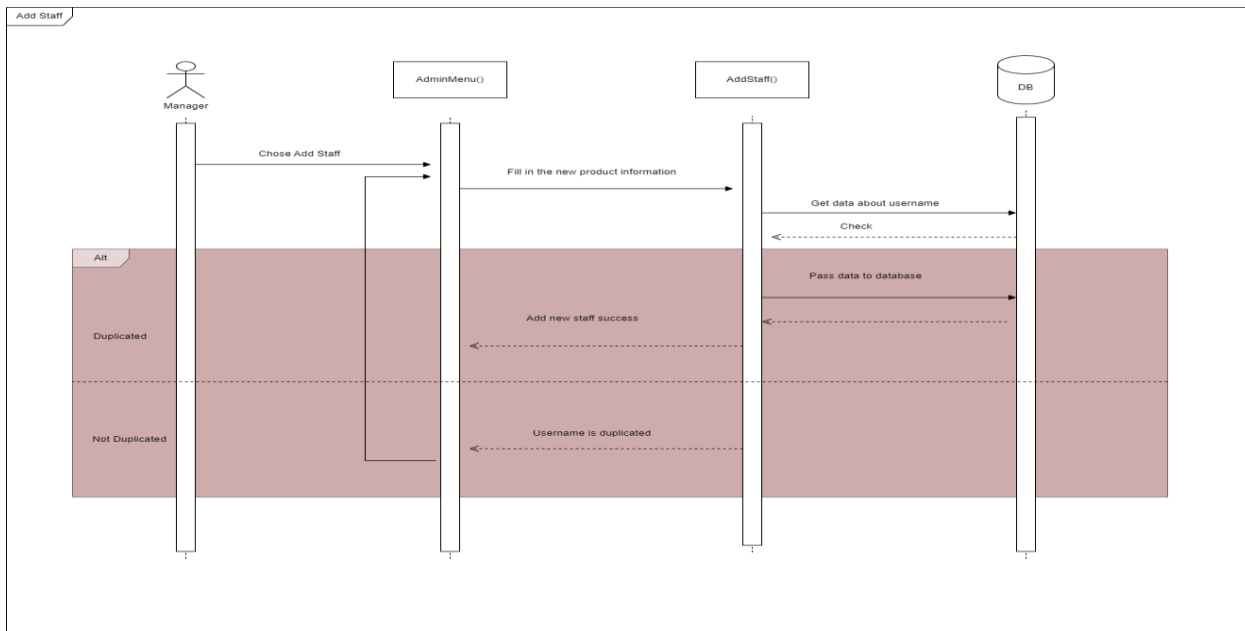
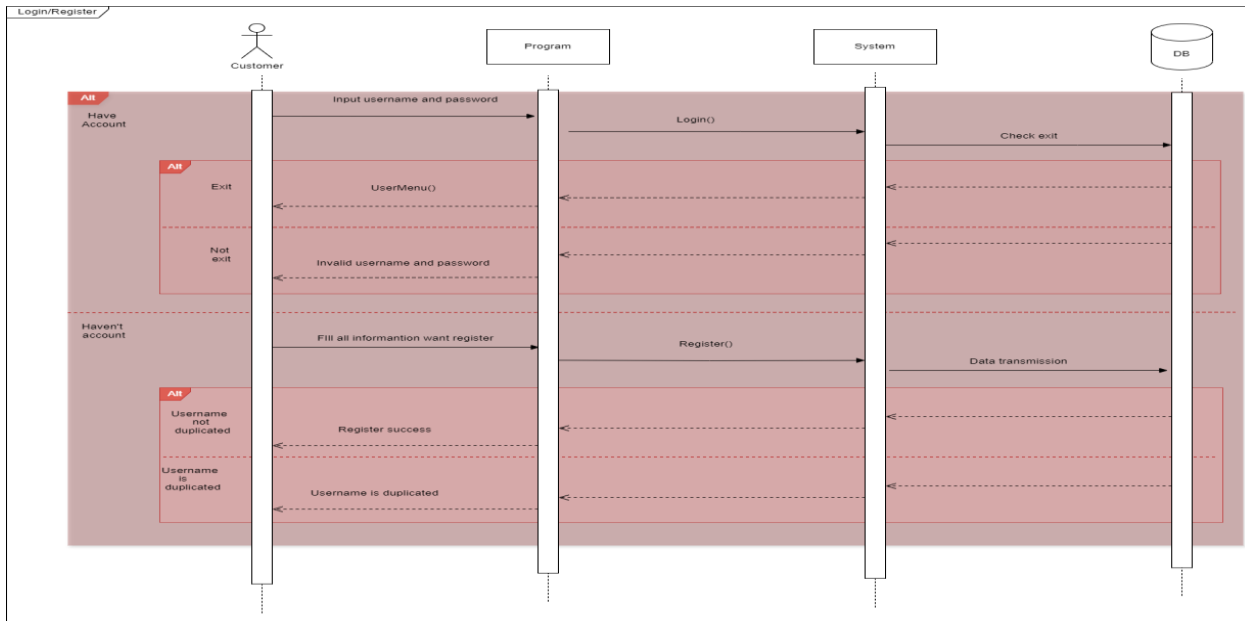
[Close]

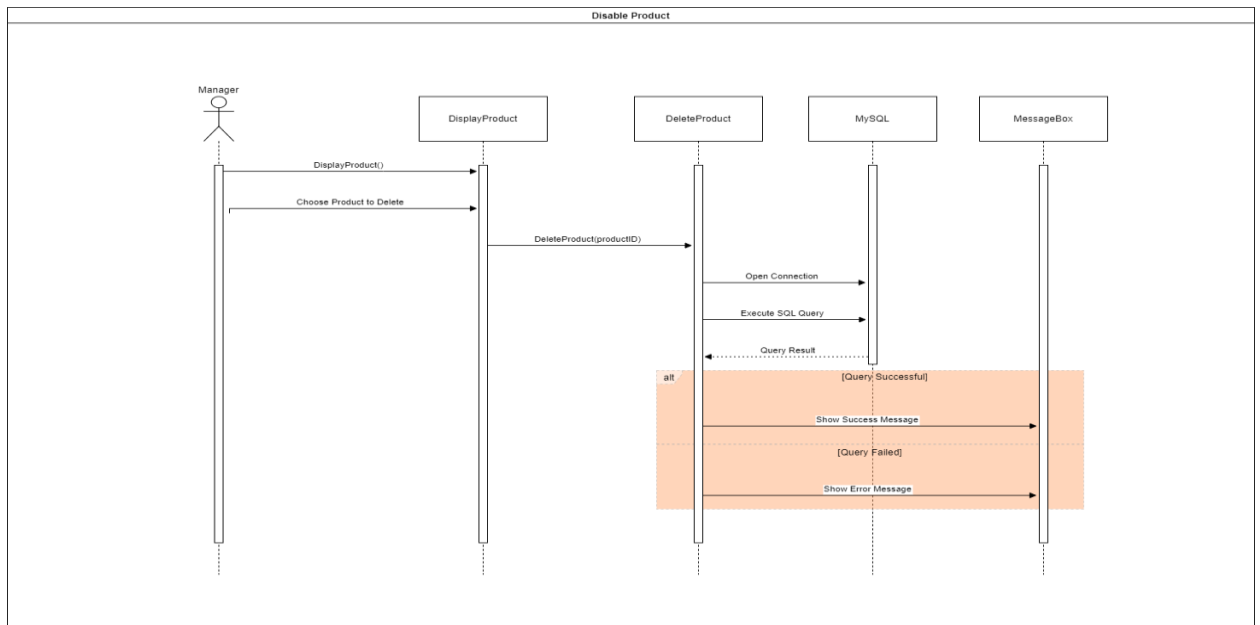
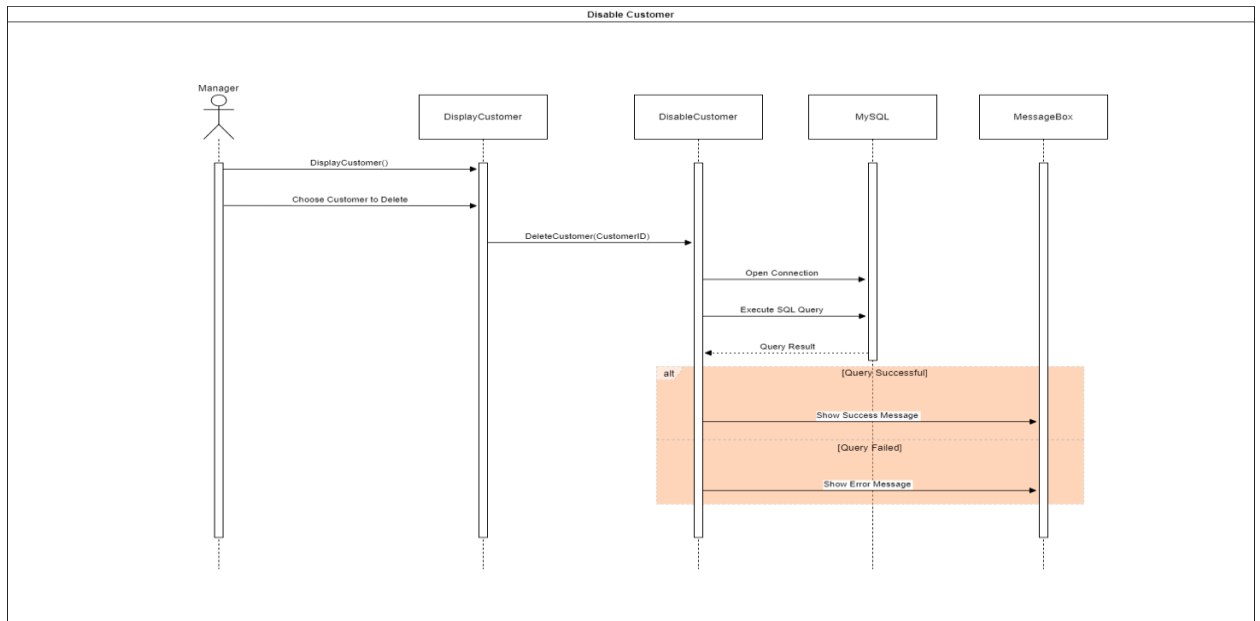
2. Code Design (Class Diagram)

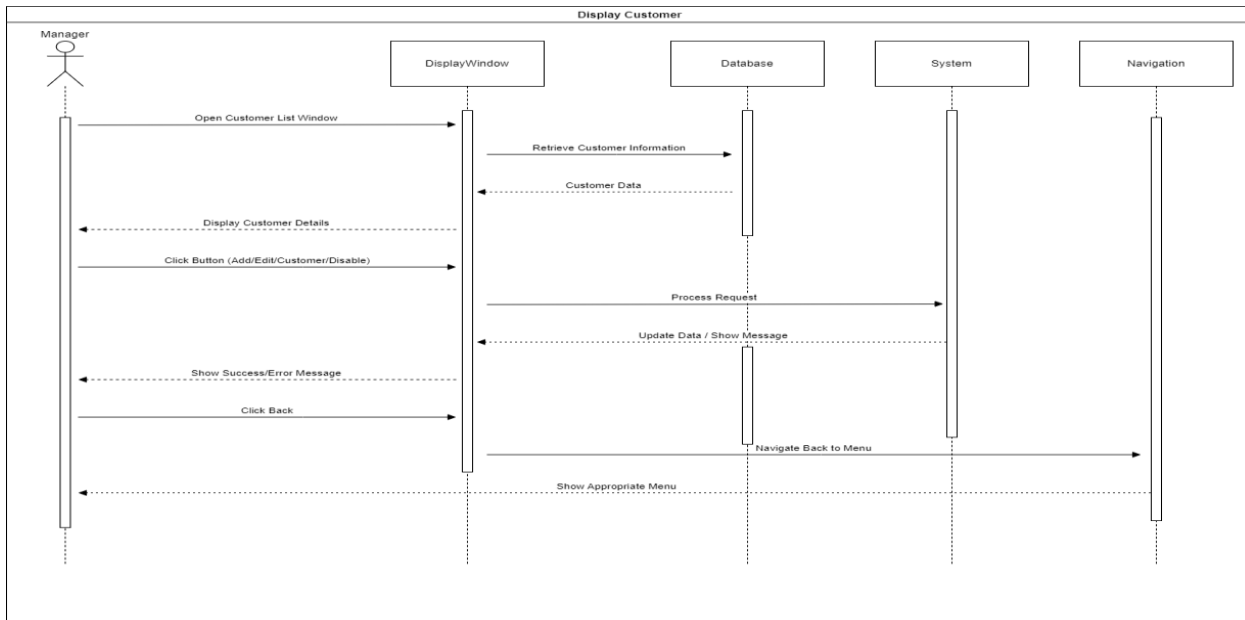
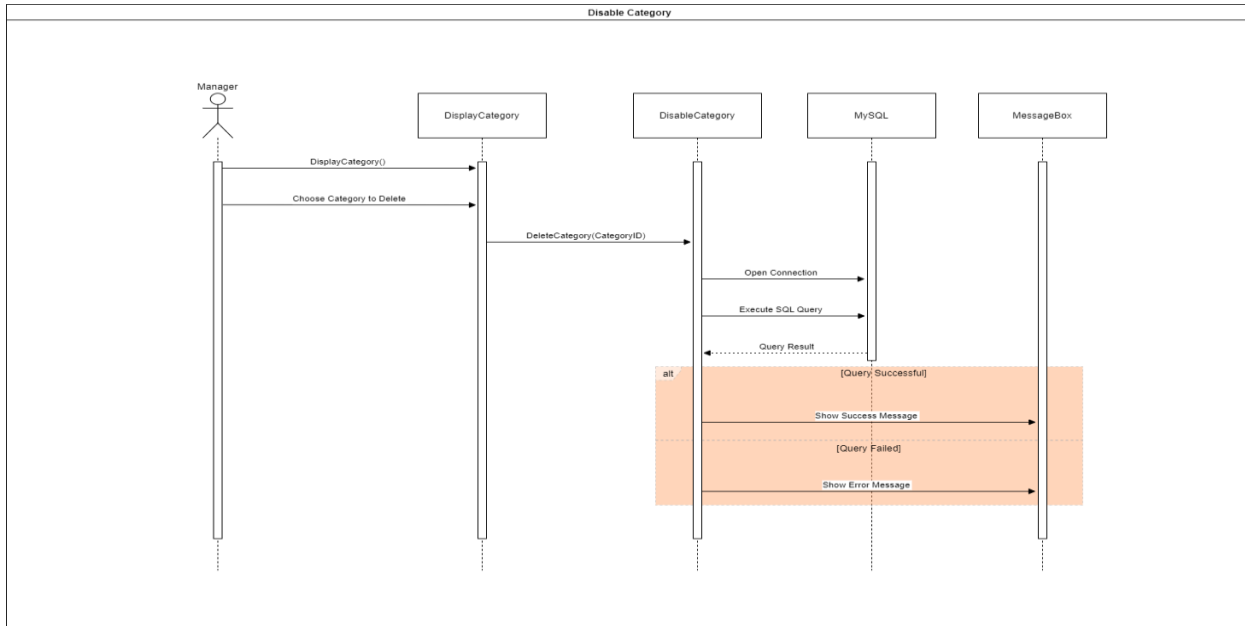


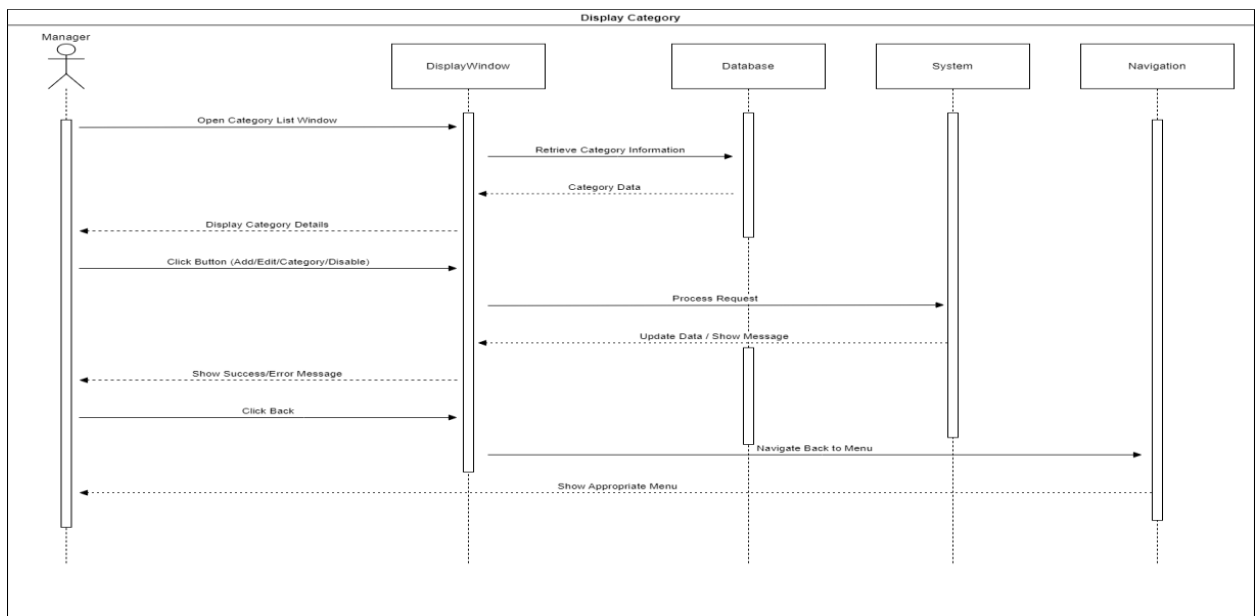
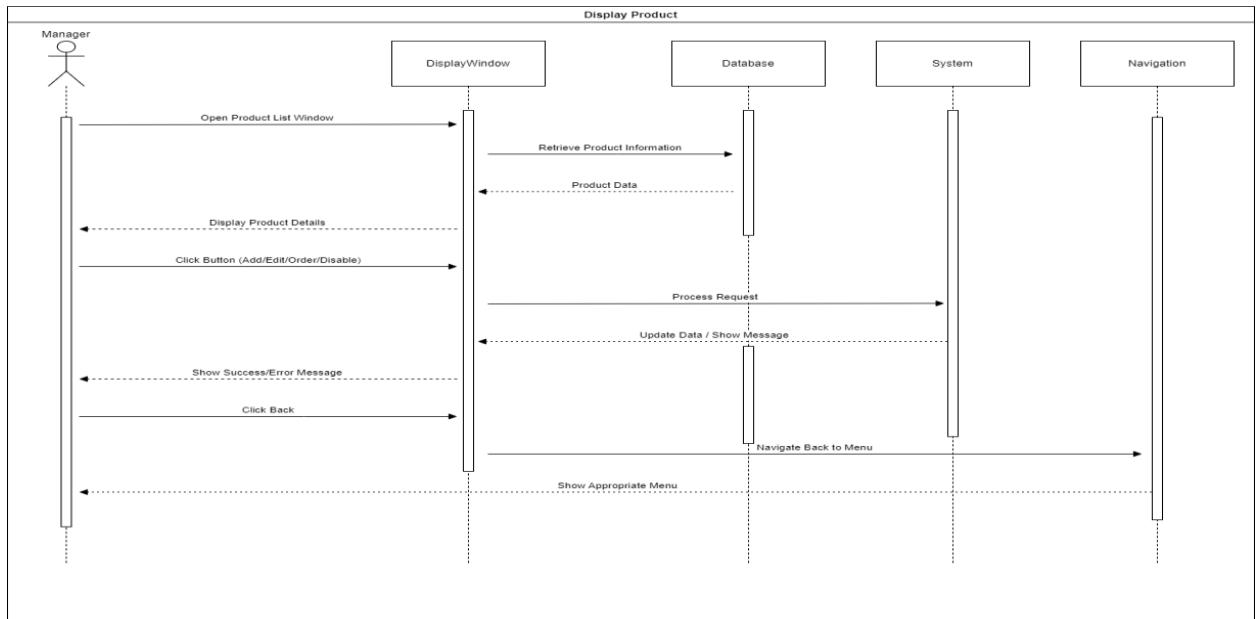
3. Sequence Diagram

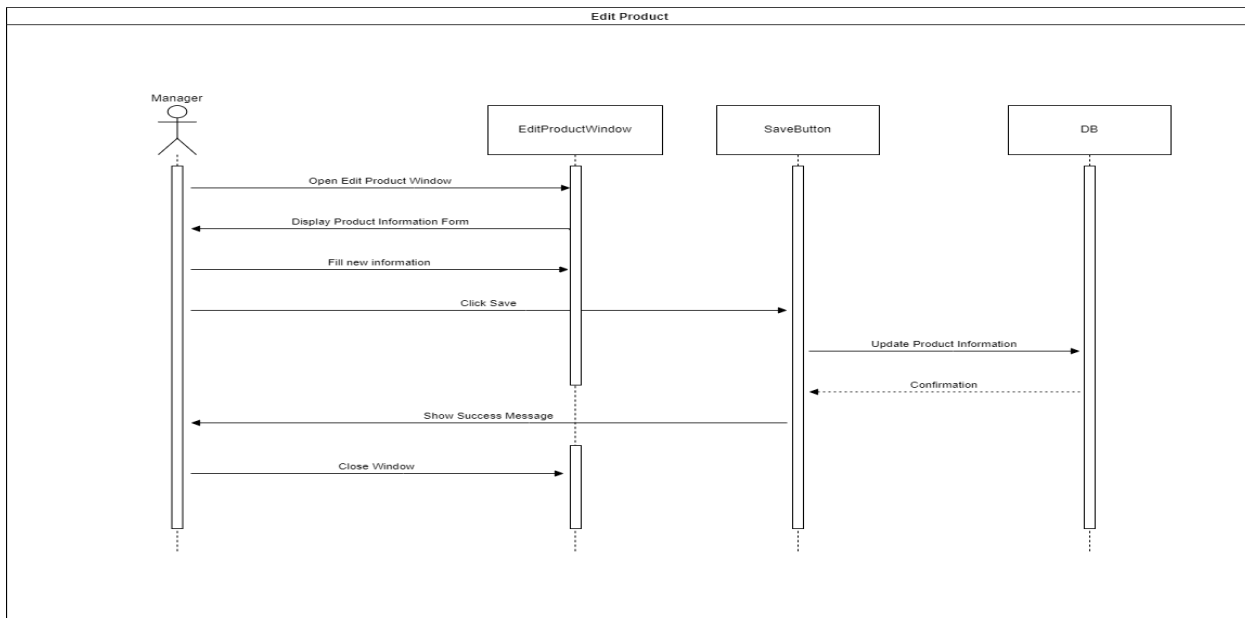
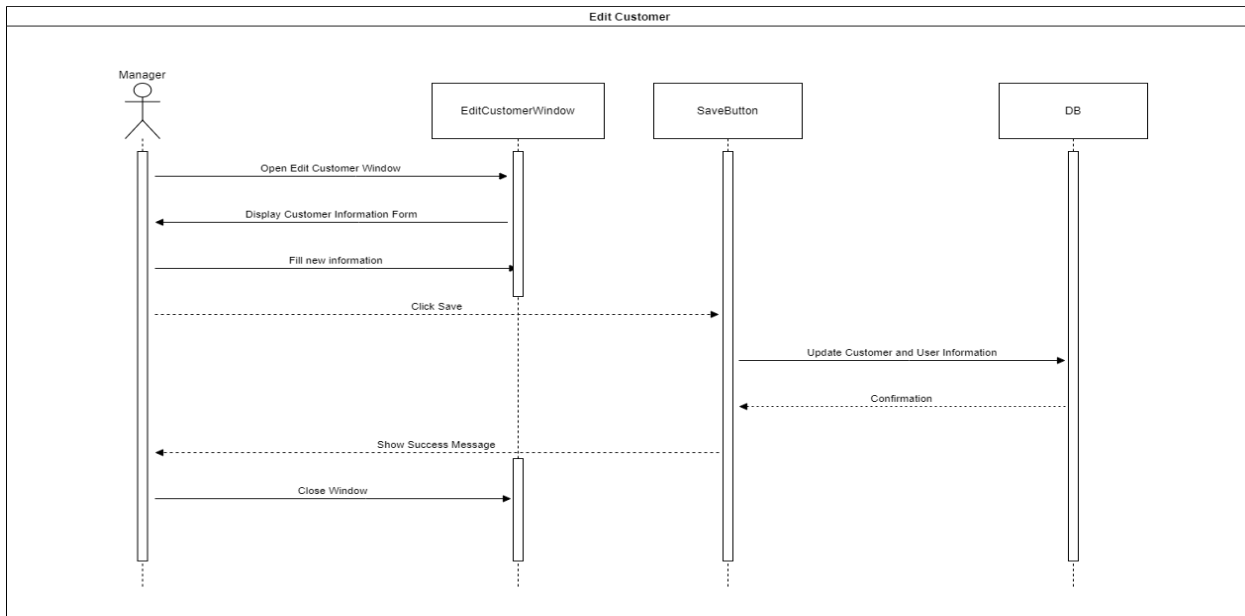


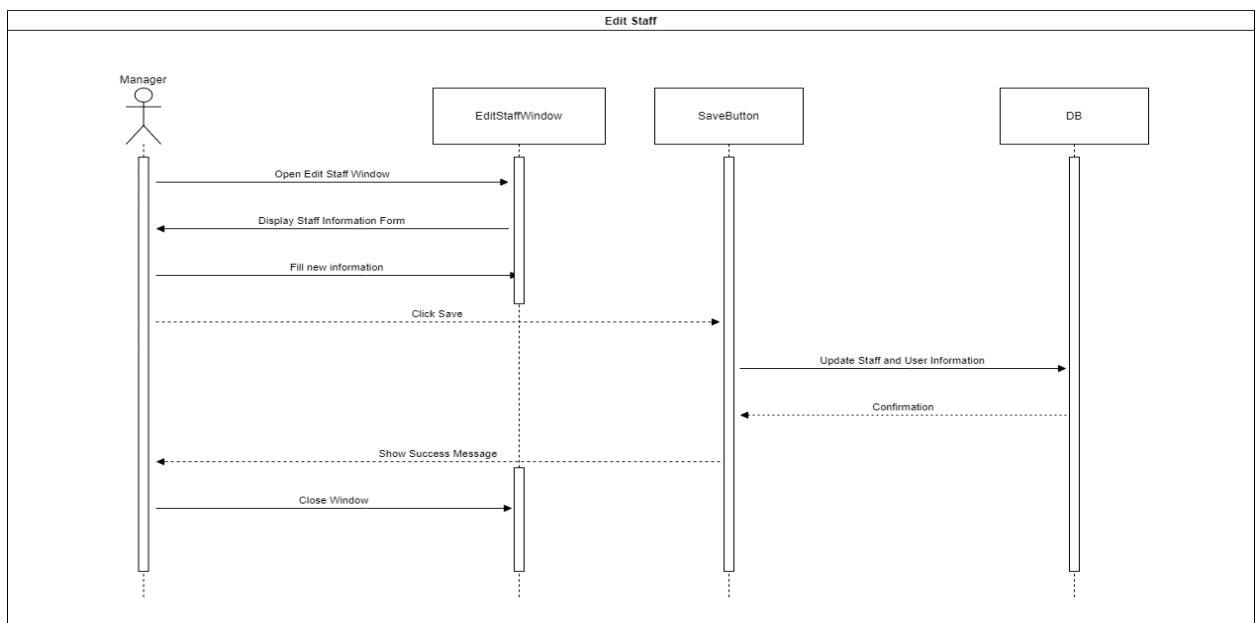
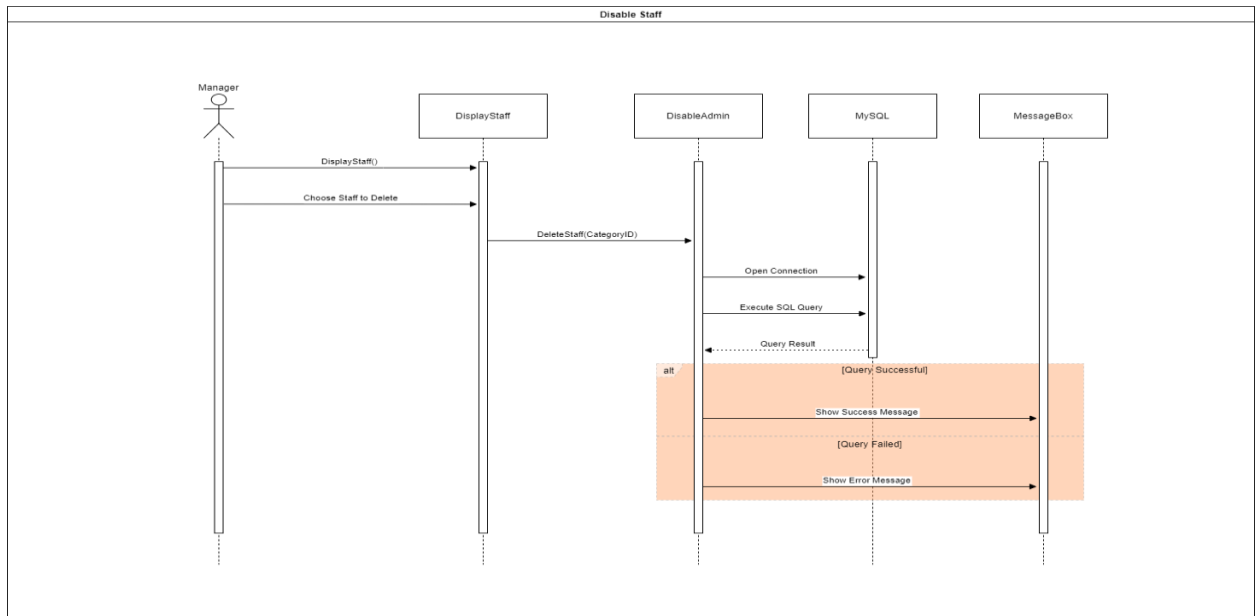


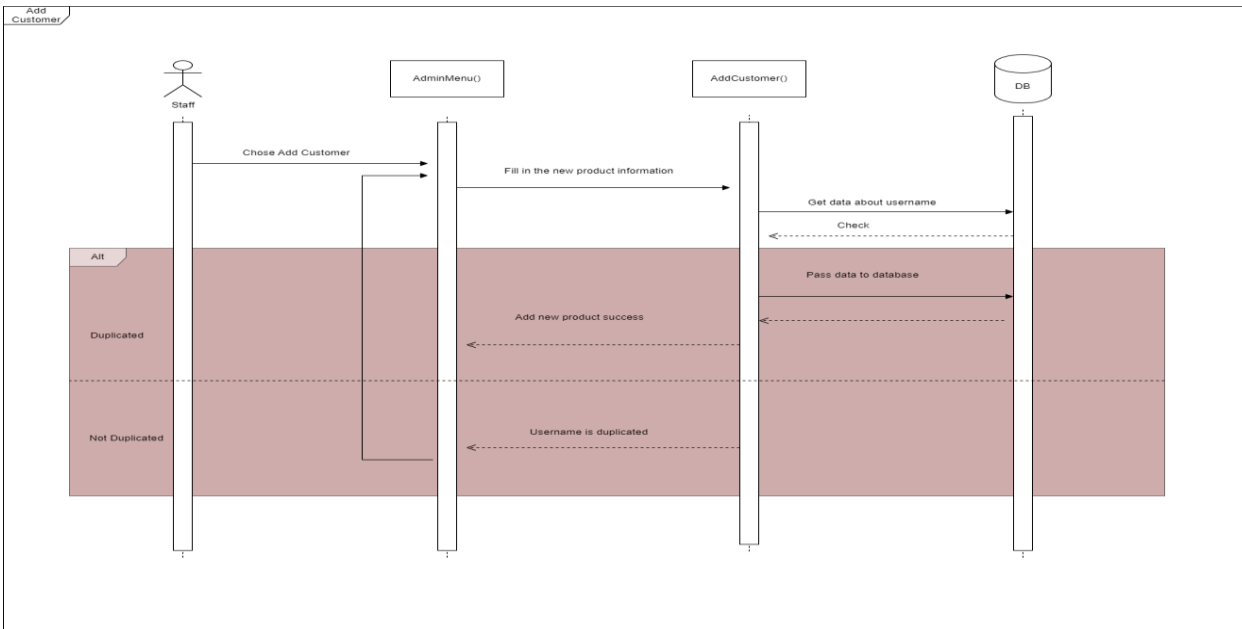
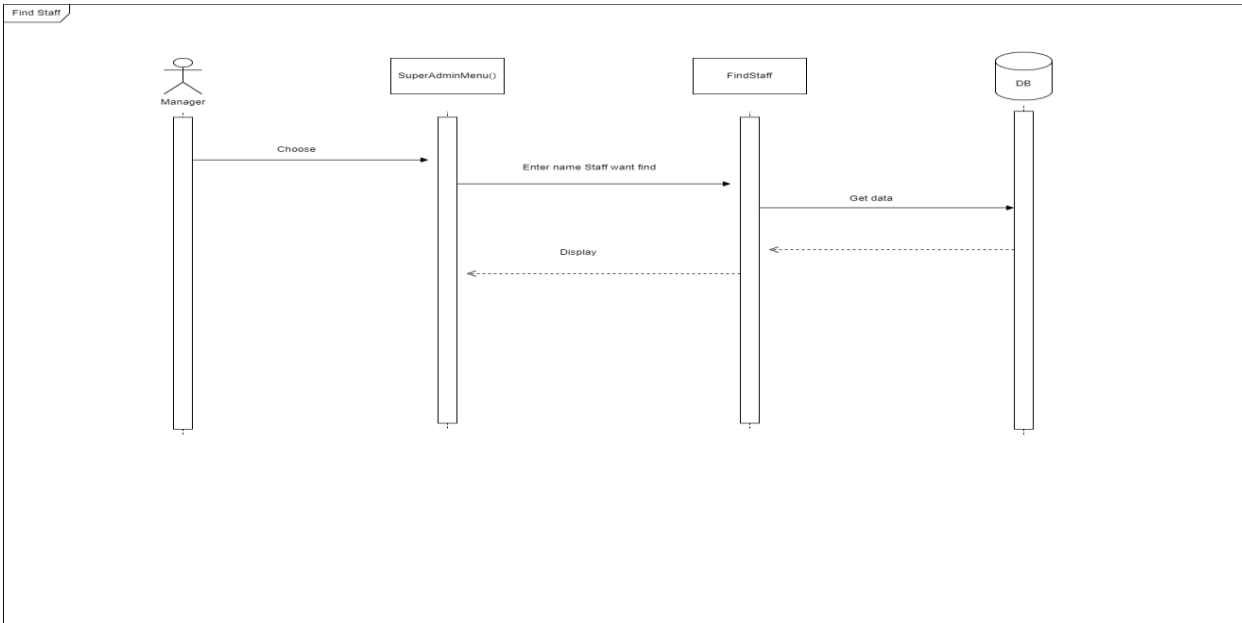


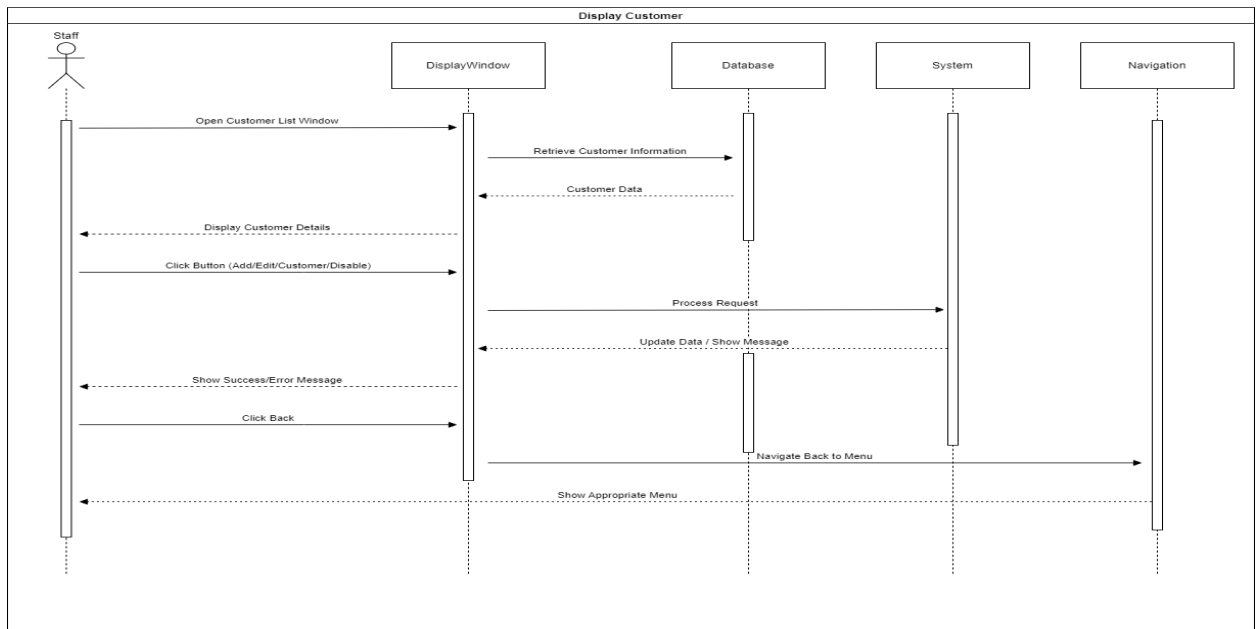
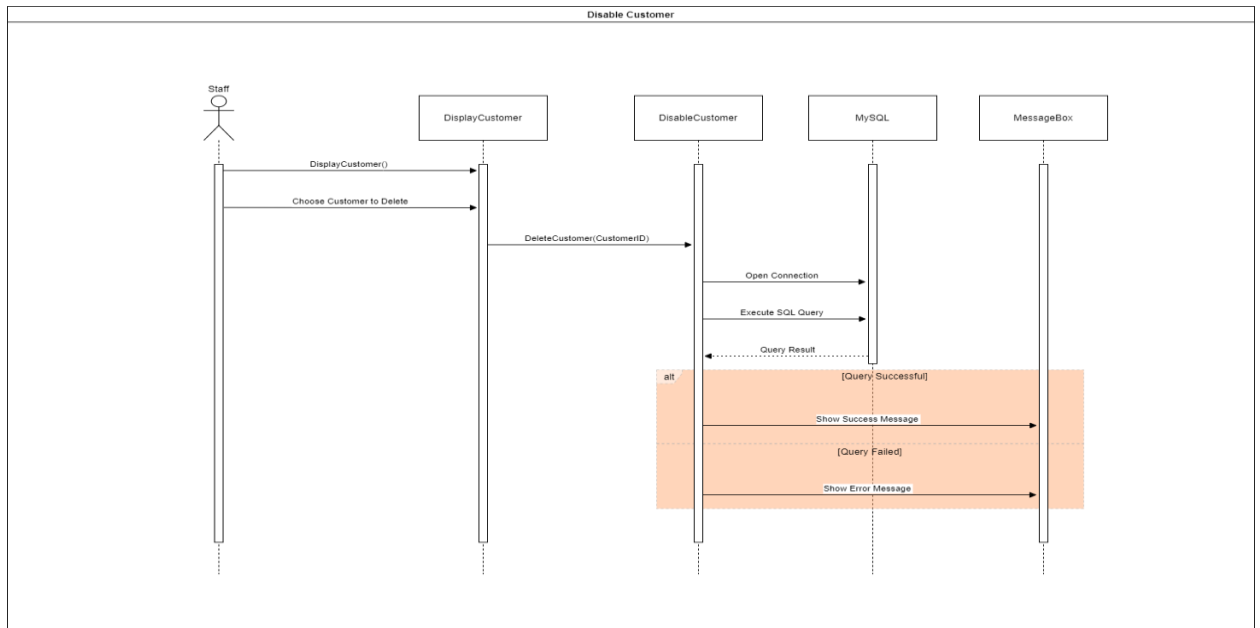


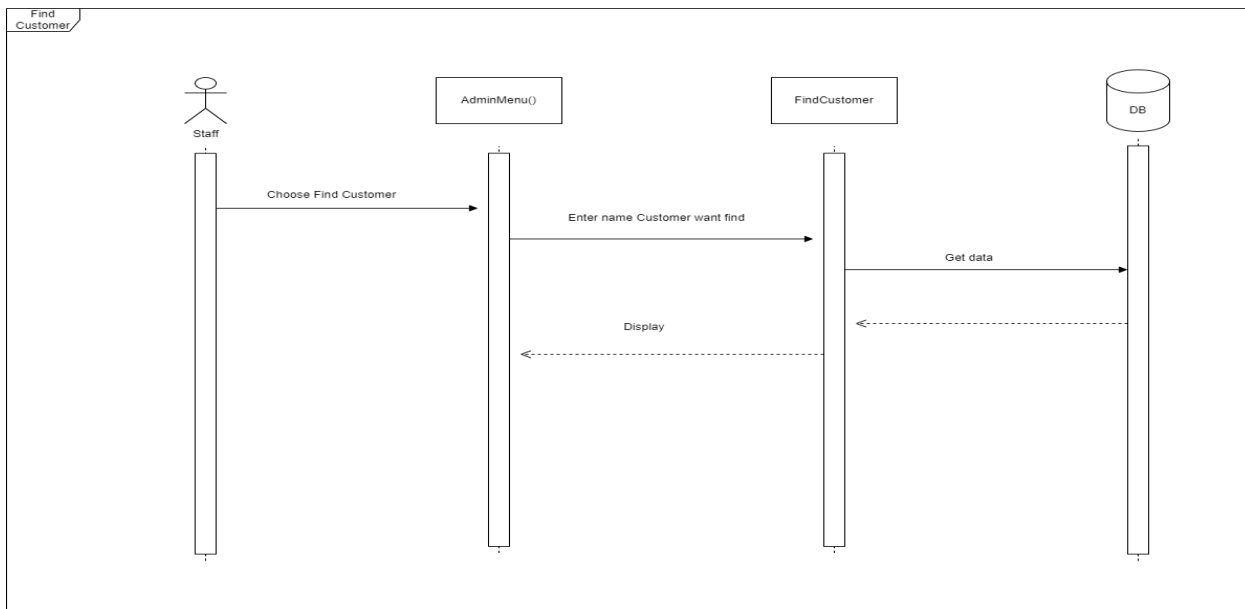
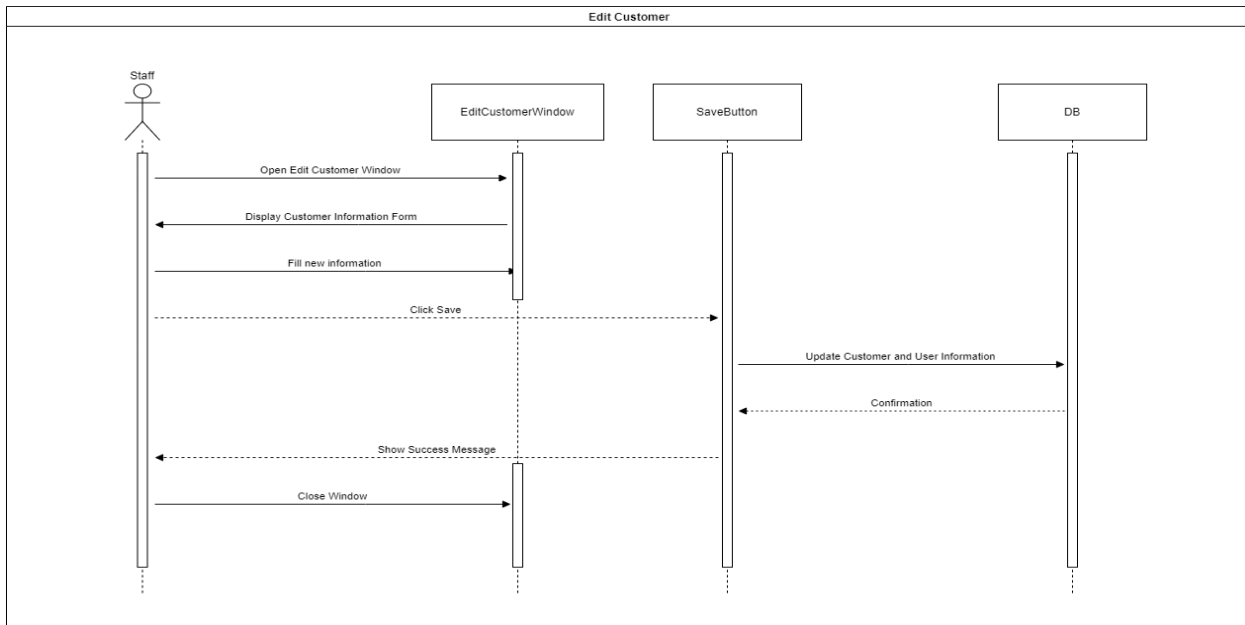


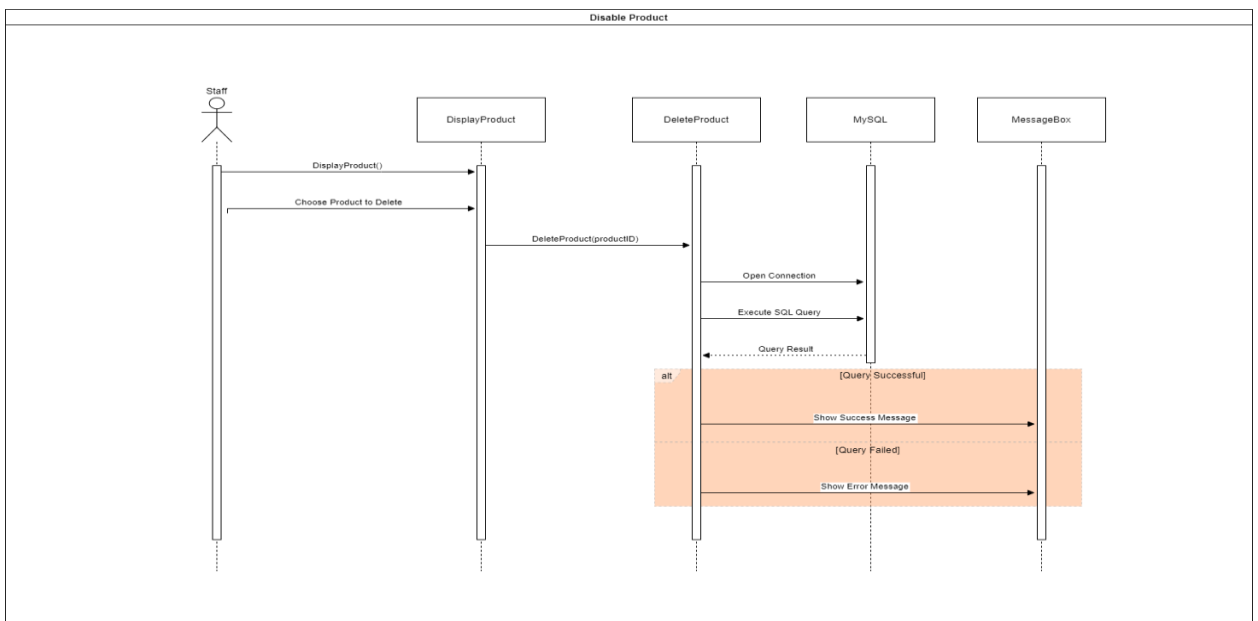
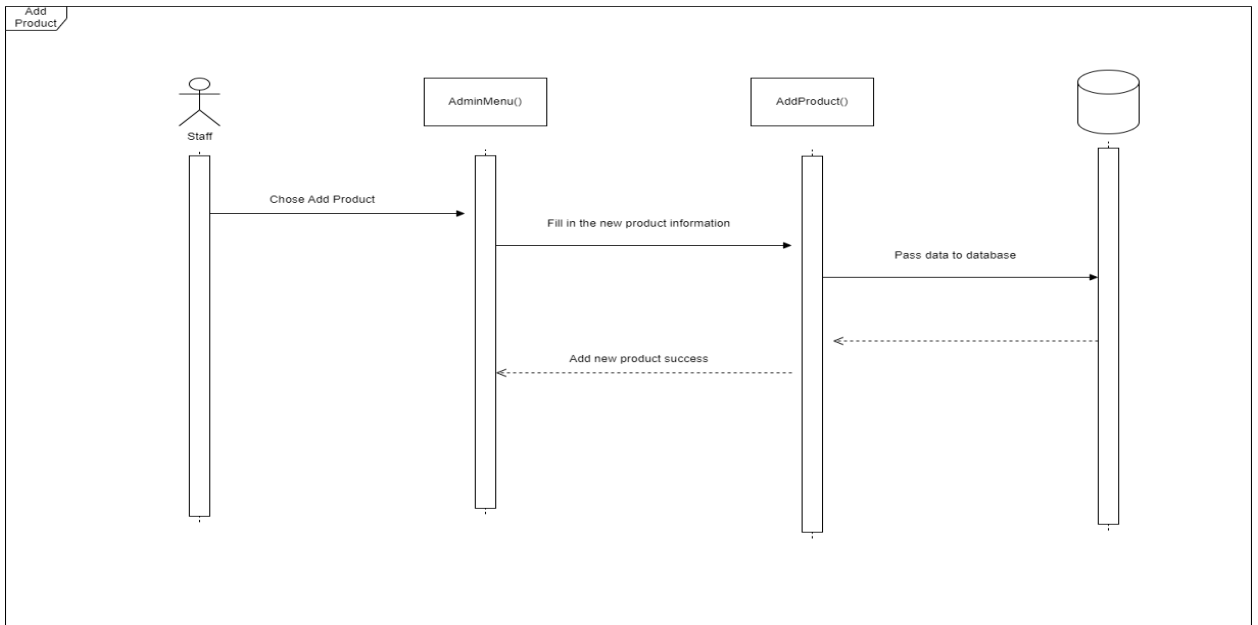


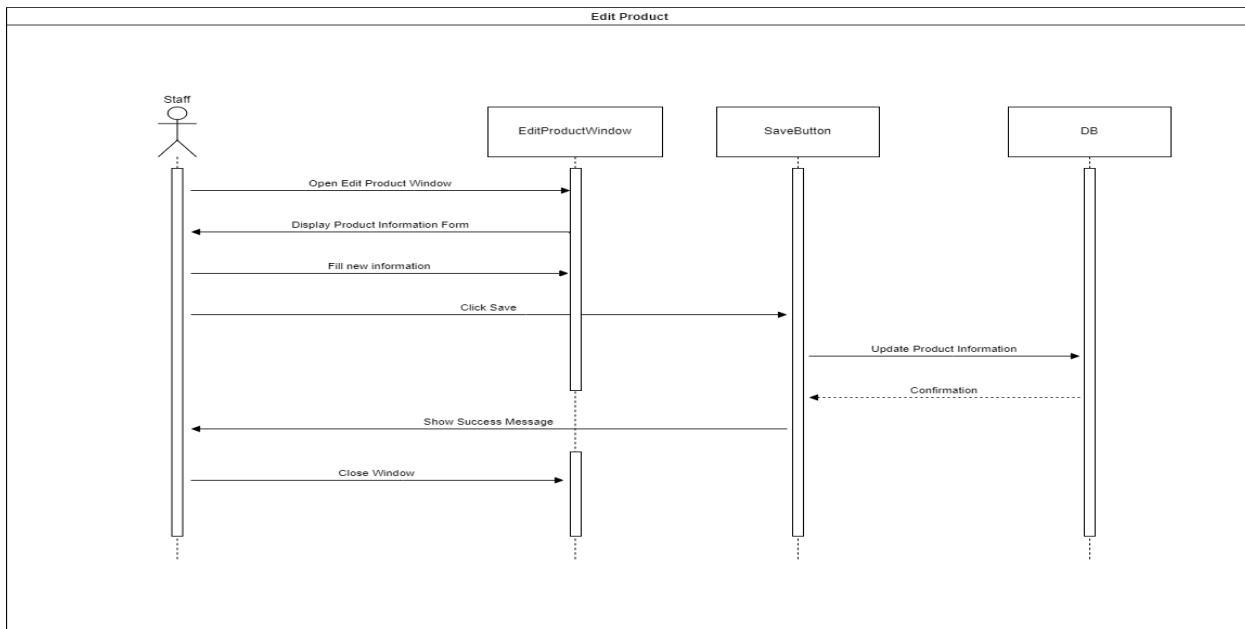
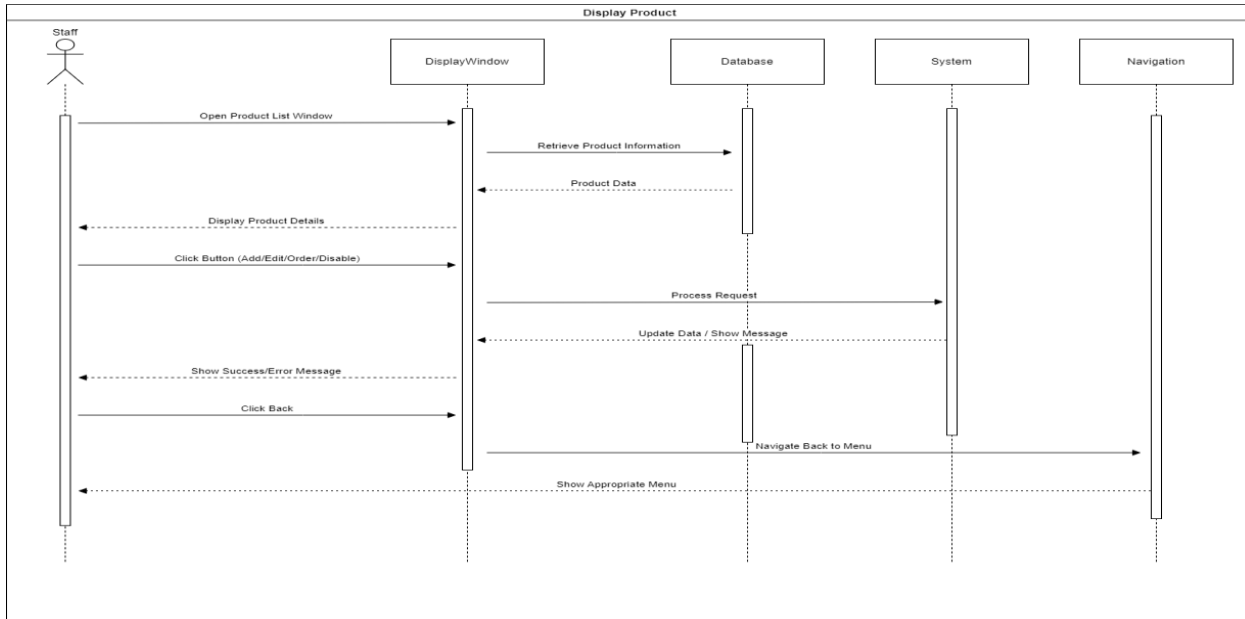


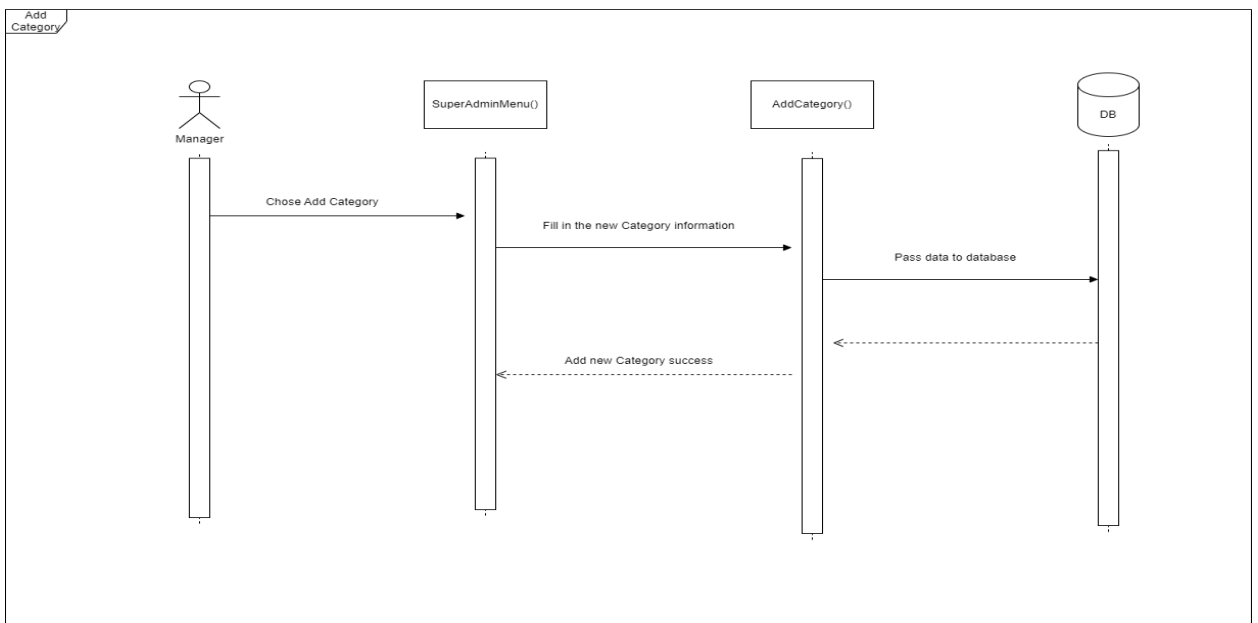
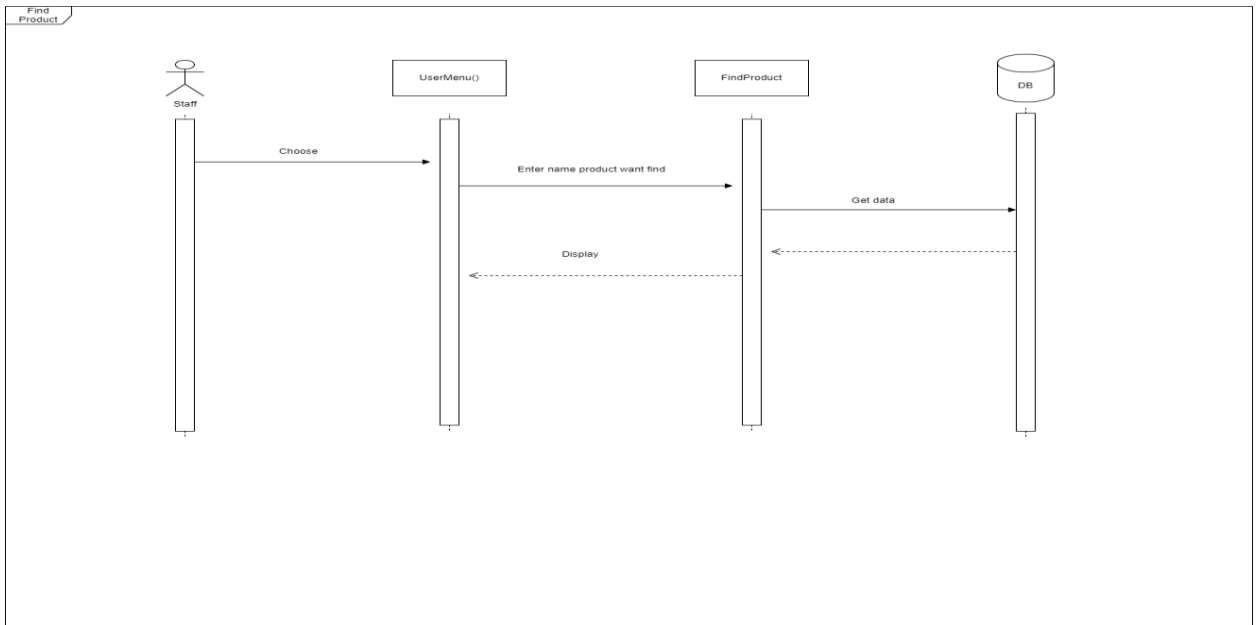


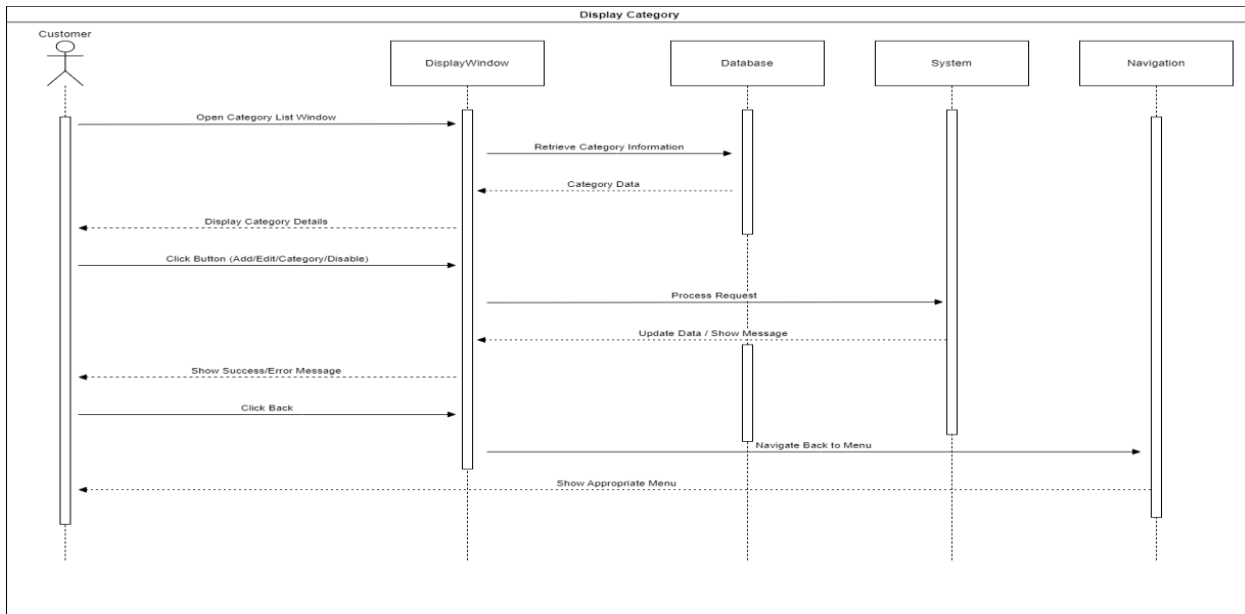
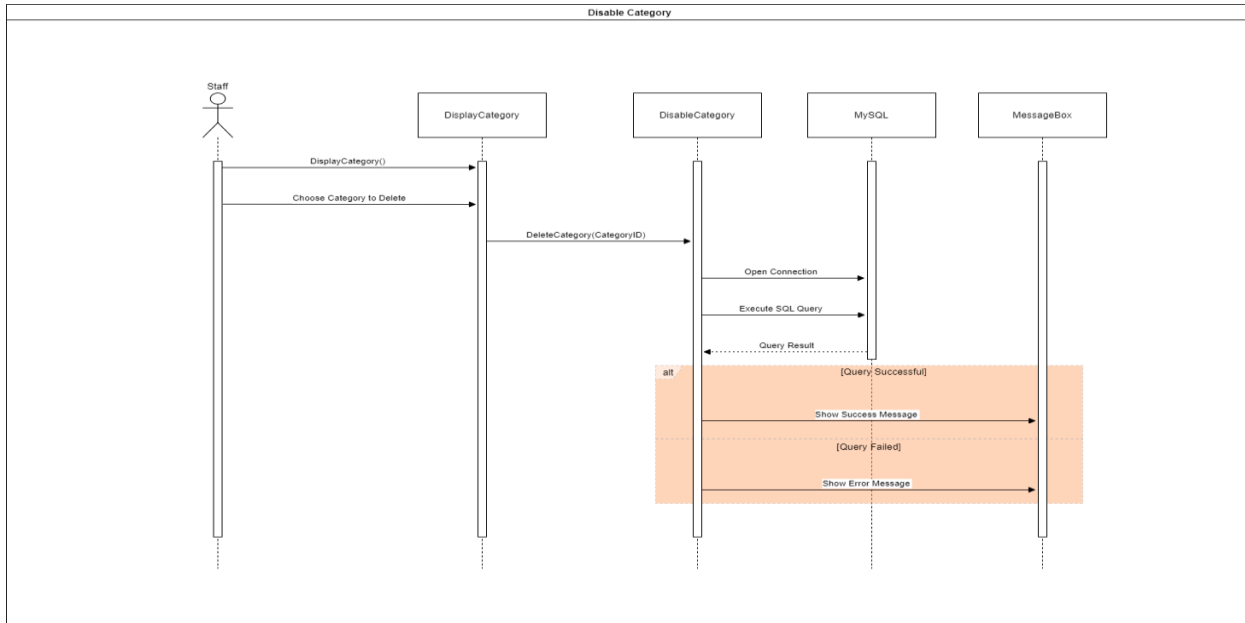


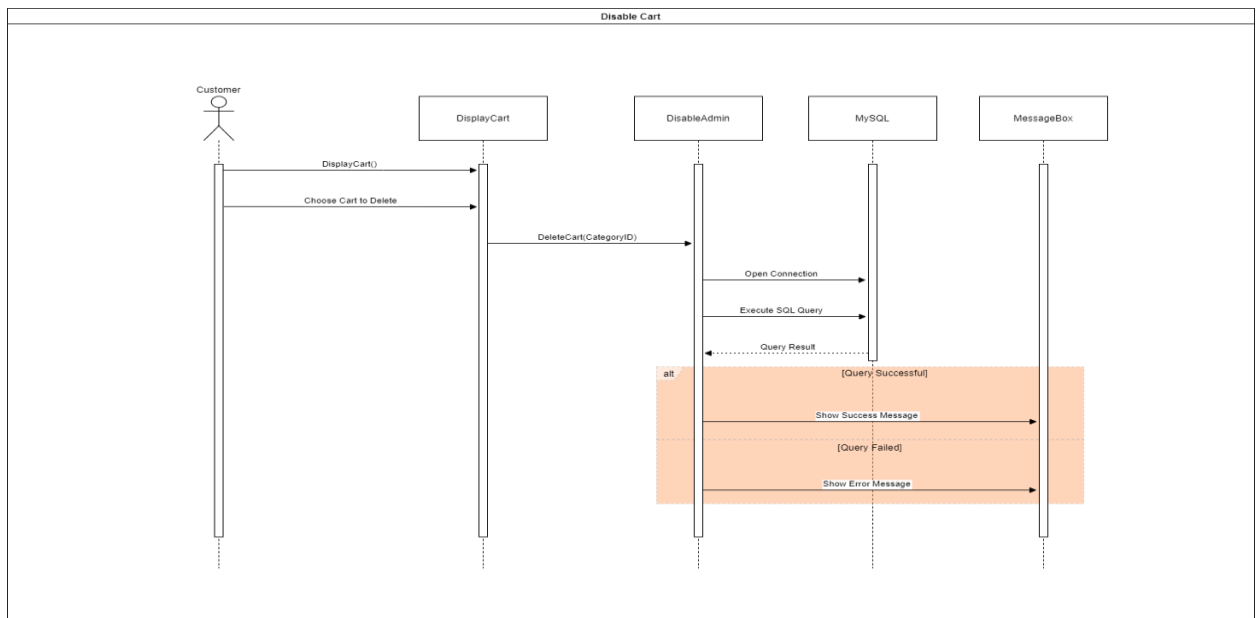
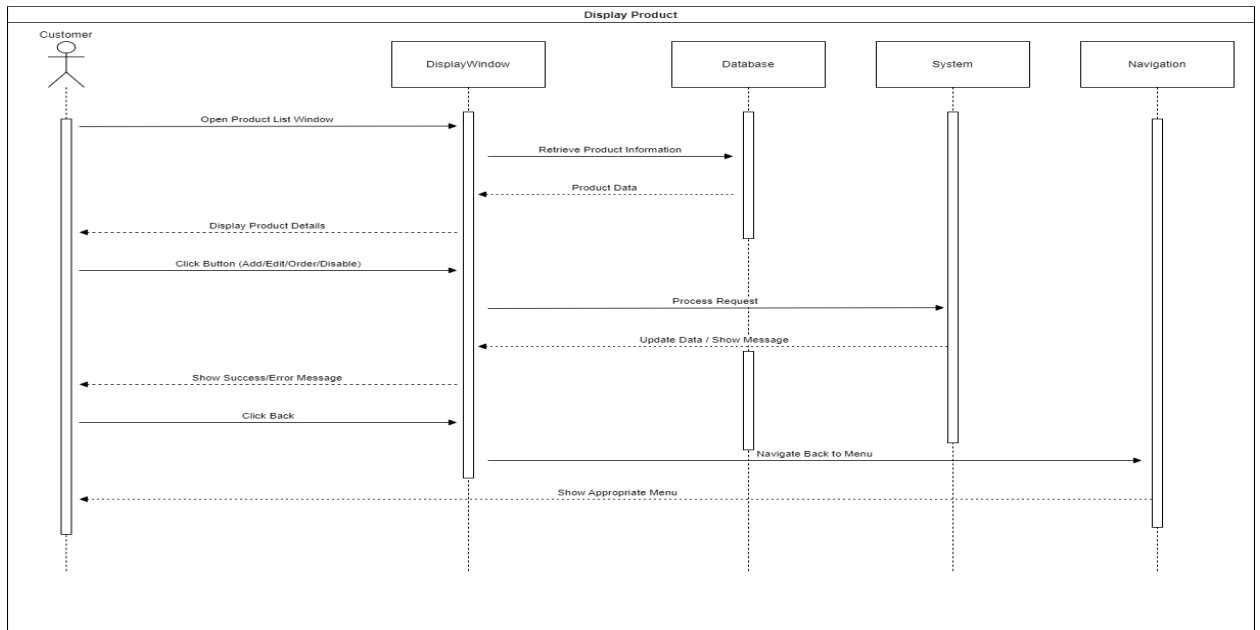


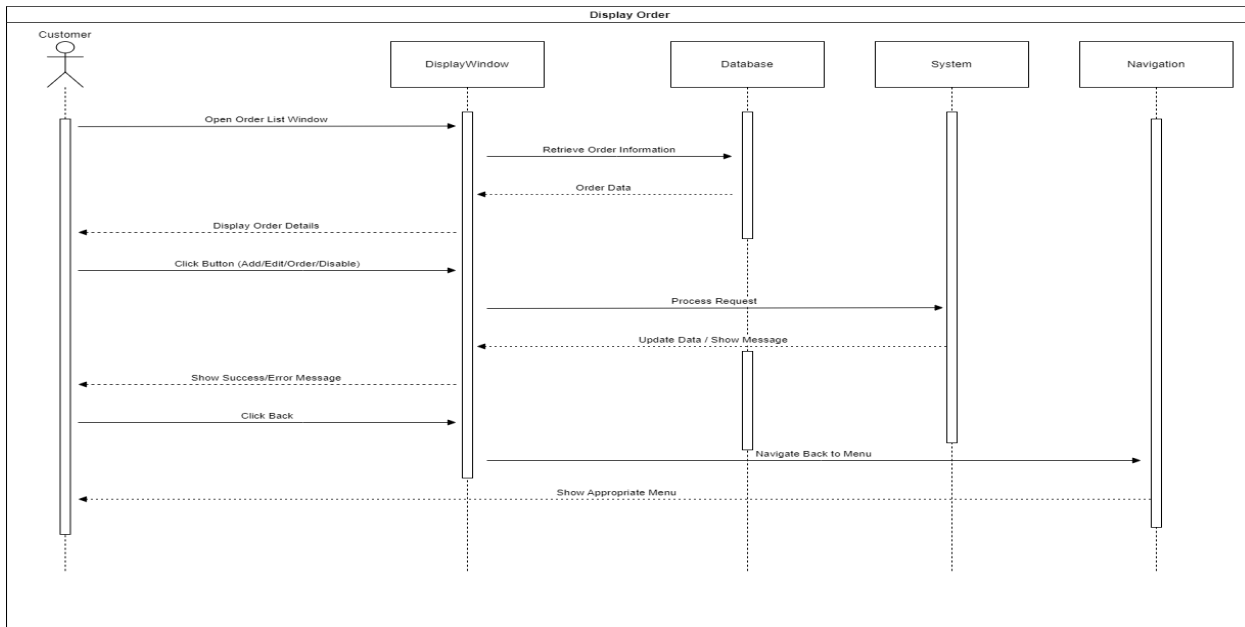
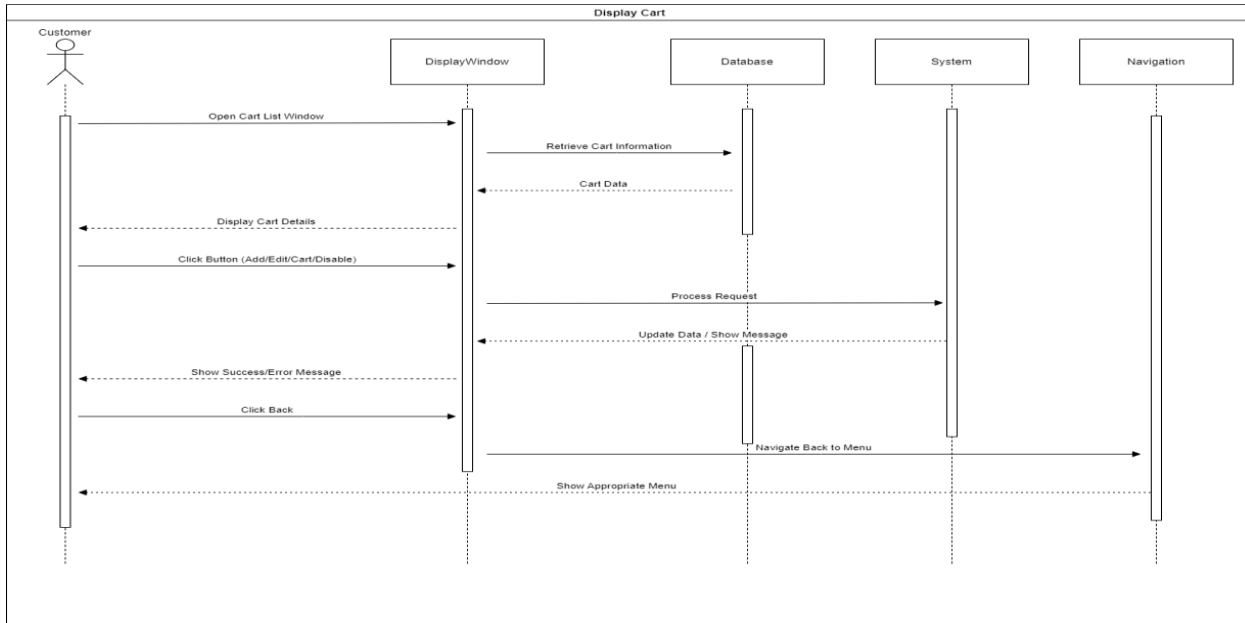


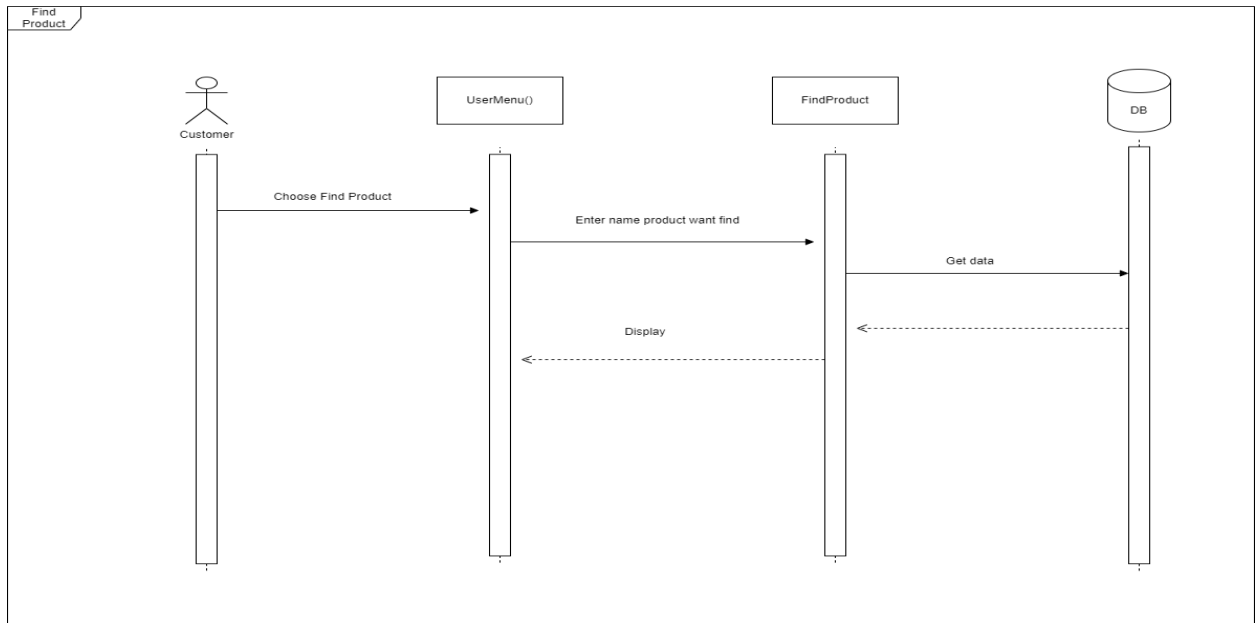






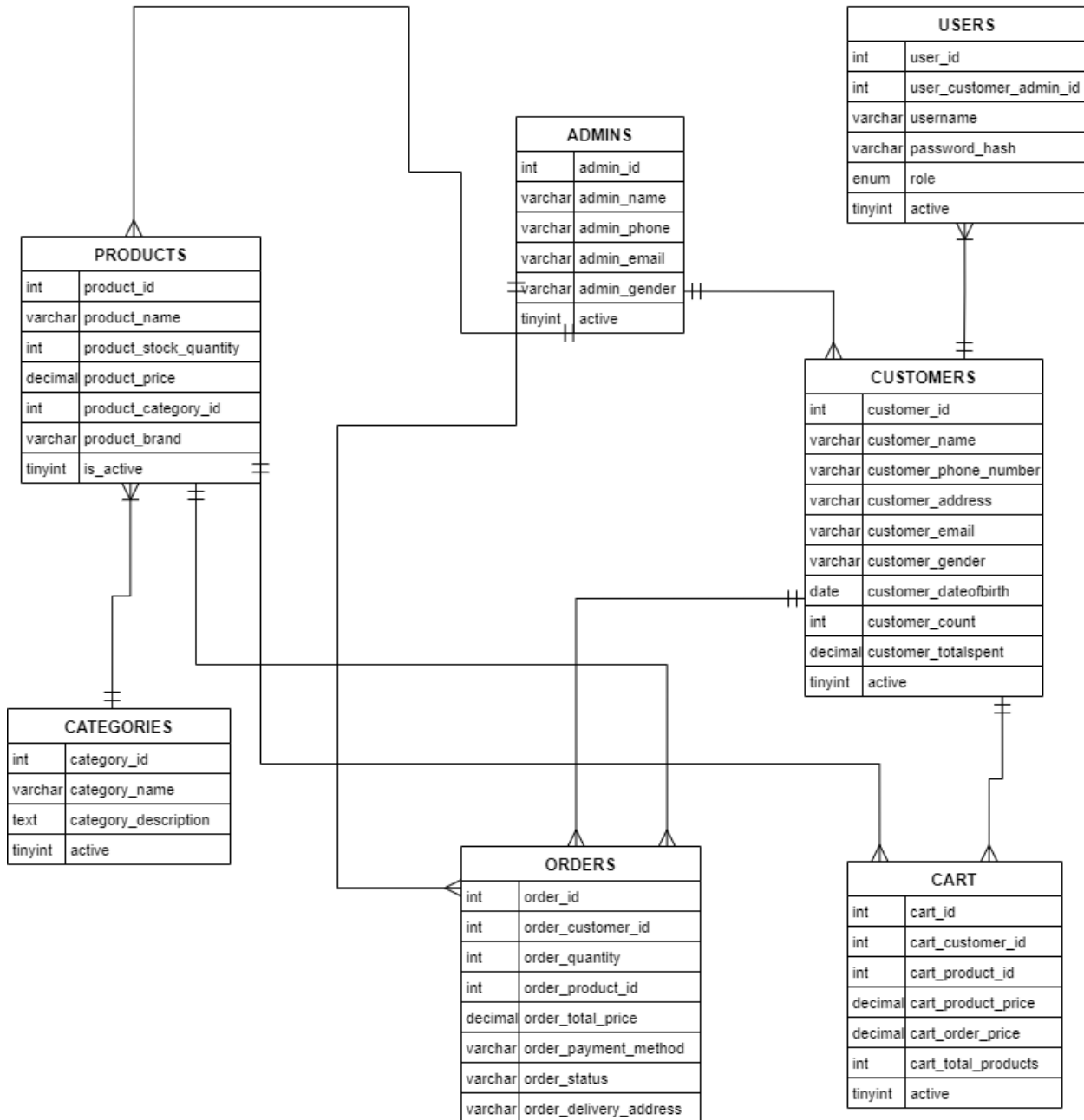




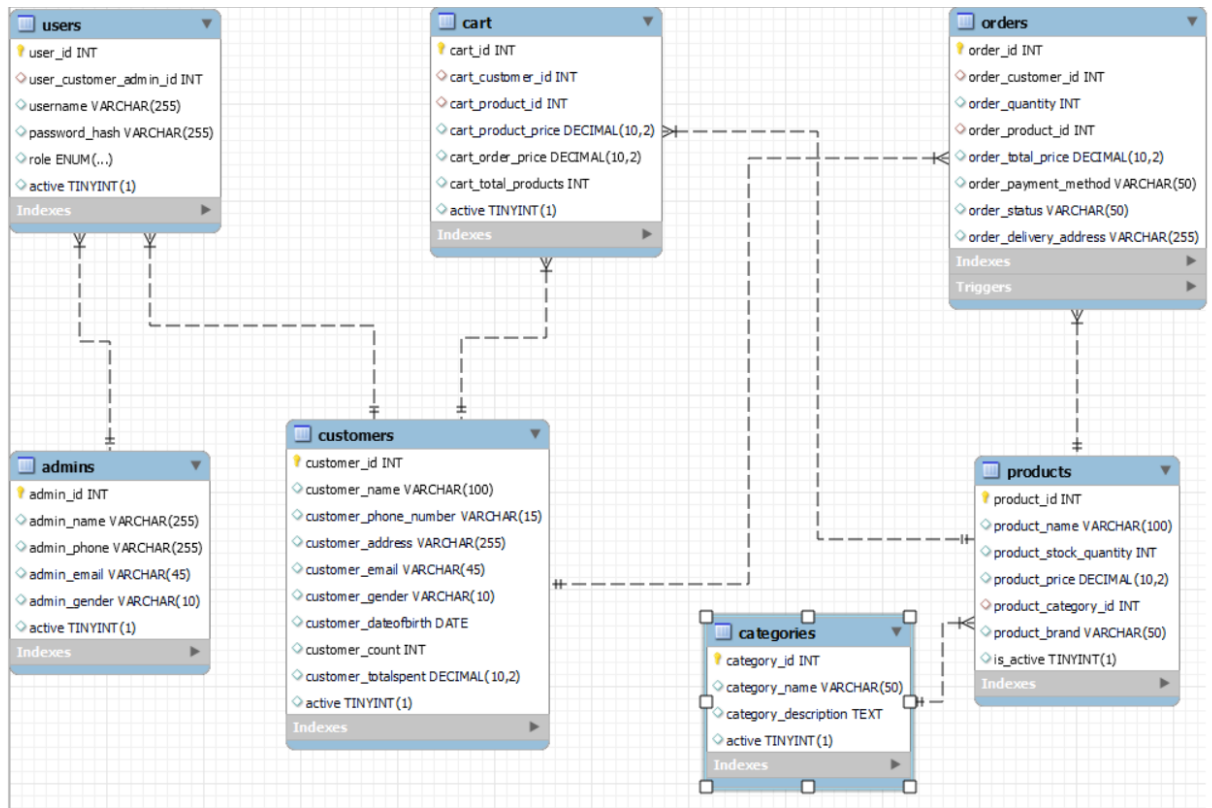


4. Database Design

a. Entity Relationship Diagram



b. Database Design Details



users			
Column Name	Data Type	Constraints	Description
user_id	INT	PRIMARY KEY, NOT NULL, AUTO INCREMENTAL	
user_customer_id	INT	FOREIGN KEY	
username	VARCHAR(255)	UNIQUE INDEX	
password_hash	VARCHAR(255)		
role	ENUM(user, admin, superadmin)		
active	TINYINT(1)		1

products			
Column Name	Data Type	Constraints	Description
product_id	INT	PRIMARY KEY, NOT NULL, AUTO INCREMENTAL	
product_name	VARCHAR(100)		
product_stock_quantity	INT		
product_price	DEMICAL(10,2)		
product_category_id	INT	FOREIGN KEY	Connect to category_id in table categories
product_brand	VARCHAR(255)		
is_active	TINYINT(1)		1

orders			
Column Name	Data Type	Constraints	Description
order_id	INT	PRIMARY KEY, NOT NULL, AUTO INCREMENTAL	
order_customer_id	INT	FOREIGN KEY	Connect to customer_id in table customers
order_quantity	INT		
order_product_id	INT	FOREIGN KEY	Connect to product_id in table products
order_total_price	DEMICAL(10,2)		
order_payment_method	VARCHAR(50)		
order_status	VARCHAR(50)		
order_delivery_address	VARCHAR(255)		

customers			
Column Name	Data Type	Constraints	Description
customer_id	INT	PRIMARY KEY, NOT NULL, AUTO INCREMENTAL	
customer_name	VARCHAR(100)		
customer_phone_number	VARCHAR(15)	UNIQUE INDEX	
customer_address	VARCHAR(255)		
customer_email	VARCHAR(45)	UNIQUE INDEX	
customer_gender	VARCHAR(10)		
customer_dateofbirth	DATE		
customer_count	INT		0
customer_totalspent	DECIMAL(10,2)		0.00
active	TINYINT(1)		1

categories			
Column Name	Data Type	Constraints	Description
category_id	INT	PRIMARY KEY, NOT NULL, AUTO INCREMENTAL	
category_name	VARCHAR(50)		
category_description	TEXT		
active	TINYINT(1)		1

cart			
Column Name	Data Type	Constraints	Description
cart_id	INT	PRIMARY KEY, NOT NULL, AUTO INCREMENTAL	
cart_customer_id	INT	FOREIGN KEY	Connect to customer_id in table customers

cart_product_id	INT	FOREIGN KEY	Connect to product_id in table products
cart_order_id	INT	FOREIGN KEY	Connect to order_id in table orders
cart_product_price	DEMICAL(10,2)		
cart_order_price	DEMICAL(10,2)		
cart_total_products	INT		
active	TINYINT(1)		1

IV. Test

1. Login

Test Case Number	001
Test Case Name	Login()
Test Case Description	Verify successful login with valid credentials
Preconditions	User has valid credentials in DB
Test Case Input	“lolo”, “koko”. “admin1”, “admin456”. “superadmin”, “superadmin123”.
Test Case Expected Output	Displays welcome message and navigates to the appropriate menu
Test Case Steps	1. Launch the application 2. Enter valid username and password 3. Click "Login" button 4. Verify the welcome message and redirection to the correct menu
Default Value Preverving	Yes

2. Register

Test Case Number	002
Test Case Name	Register()
Test Case Description	Verify successful registration with valid inputs

Preconditions	All fields are empty
Test Case Input	newuser , P@ssw0rd, John Doe , 1234567890, 123 Elm Street, john.doe@example.com, Male, 01-01-2000
Test Case Expected Output	Displays "Registration successful!" message and returns to login screen
Test Case Steps	1. Launch the application 2. Fill in all fields with valid data 3. Click "Register" button 4. Verify success message and redirection to login screen
Default Value Preverving	Yes

3. Display product

Test Case Number	003
Test Case Name	DisplayProduct()
Test Case Description	Verify product list display and button visibility for user role
Preconditions	User is logged in as a user
Test Case Input	Role: user
Test Case Expected Output	Displays products with "Add to Cart" and "Order" buttons; "Edit", "Disable", "Order For Customer" hidden
Test Case Steps	1. Launch the application 2. Log in as a user 3. Navigate to product display 4. Verify product list and button visibility
Default Value Preverving	Yes

4. Add product

Test Case Number	004
Test Case Name	AddProduct()

Test Case Description	Verify adding a product with valid inputs
Preconditions	User is logged in as admin
Test Case Input	Sample Product, 100, 1, 29.99, Sample Brand
Test Case Expected Output	Product is successfully added; success message shown; redirected to admin menu
Test Case Steps	1. Launch the application 2. Log in as admin 3. Navigate to add product form 4. Fill in all fields with valid data 5. Click "Save" button 6. Verify success message and redirection to admin menu
Default Value Preverving	Yes

5. Edit product

Test Case Number	005
Test Case Name	EditProduct()
Test Case Description	Verify updating product information with valid inputs
Preconditions	User is logged in as admin
Test Case Input	1, Updated Product, 150, 19.99, 2, Updated Brand
Test Case Expected Output	Product information is successfully updated; success message shown; redirected to product list
Test Case Steps	1. Launch the application 2. Log in as admin 3. Navigate to edit product form 4. Enter valid data for all fields 5. Click "Save" button 6. Verify success message and redirection to product list
Default Value Preverving	Yes

6. Disable Product

Test Case Number	006
Test Case Name	EditProduct()
Test Case Description	Verify that a product is marked as inactive successfully
Preconditions	User is logged in as admin
Test Case Input	1, Updated Product, 150, 19.99, 2, Updated Brand
Test Case Expected Output	1
Test Case Steps	1. Launch the application 2. Log in as admin 3. Call DeleteProduct(1) 4. Verify success message
Default Value Preverving	Yes

7. Find product

Test Case Number	007
Test Case Name	FindProduct()
Test Case Description	Verify that a product is marked as inactive successfully
Preconditions	User is logged in as admin
Test Case Input	1, Updated Product, 150, 19.99, 2, Updated Brand
Test Case Expected Output	1
Test Case Steps	1. Launch the application 2. Log in as admin 3. Call DeleteProduct(1) 4. Verify success message
Default Value Preverving	Yes

8. Add customer

Test Case Number	008
Test Case Name	AddCustomer()
Test Case Description	Verify that a product is marked as inactive successfully
Preconditions	Application is running and accessible. Database is connected and tables users and customers exist.
Test Case Input	Testuser, password123, John Doe, 123-456-7890, 123 Elm Street, john.doe@example.com, Male, 01-01-1990
Test Case Expected Output	- UI Feedback: Success message "Registration successful!" and navigation to admin menu. - Database: New entries in customers and users tables with correct data.
Test Case Steps	1) Open registration form. 2) Fill in fields with input values. 3) Click "Register" button. 4) Verify success message and UI navigation. 5) Check database for new entries.
Default Value Preverving	

9. Edit Customer

Test Case Number	009
Test Case Name	EditCustomer()
Test Case Description	Verifies the EditCustomer method for correct customer and user data editing, validation, database update, and UI interactions.
Preconditions	Application is running and accessible. Database is connected and contains customer and user data with the provided customerID.
Test Case Input	Admin, 1, Jane Doe, 098-765-4321, 456 Oak Street, jane.doe@example.com , Female, 02-02-1992, janedoe, newpassword

Test Case Expected Output	- UI Feedback: Success message "Customer information has been updated successfully!" and navigation to customer list. - Database: Updated entries in customers and users tables with the provided data.
Test Case Steps	1) Open the edit customer form with existing data. 2) Update fields with new input values. 3) Click "Save" button. 4) Verify success message and UI navigation. 5) Check database for updated entries.
Default Value Preverving	

10. Disable customer

Test Case Number	010
Test Case Name	DisableCustomer()
Test Case Description	Tests the DeleteCustomer method for marking a customer as inactive and ensuring proper database update and UI notification.
Preconditions	Application is running and connected to the database. Customer with the specified customerID exists and is active.
Test Case Input	1
Test Case Expected Output	- UI Feedback: Success message "Customer successfully marked as inactive!" - Database: active field in customers and users tables set to FALSE for the specified customerID.
Test Case Steps	1) Invoke the DeleteCustomer method with a valid customerID. 2) Verify the success message displayed. 3) Check the database to ensure the active field is set to FALSE for the customerID in both customers and users tables.
Default Value Preverving	

11. Display Customer

Test Case Number	011
Test Case Name	DisplayCustomer()
Test Case Description	Tests the DisplayCustomers method for correctly displaying a list of active customers in a window with their details and action buttons.
Preconditions	Application is running and connected to the database. The database contains active customers and users with the role 'user'.
Test Case Input	Role: "admin" or "superadmin"
Test Case Expected Output	- UI: A window listing customers with: - o Customer details: Name, Phone, Address, Email, Gender, Date of Birth, Orders, Total Spent, Username, Password - o Buttons for "Edit" and "Delete" for each customer - o A "Back" button to navigate to the previous menu - Database: Customers with active = TRUE and users with role = 'user' should be displayed.
Test Case Steps	1) Invoke the DisplayCustomers method with a valid role. 2) Verify the display window appears with the customer list. 3) Check each customer's details and action buttons are correctly displayed. 4) Click on "Edit" and ensure it opens the edit customer window with pre-filled details. 5) Click on "Delete", confirm deletion, and ensure the customer is marked as inactive. 6) Click "Back" and ensure it navigates to the appropriate menu based on the role.
Default Value Preserving	

12. Find Customer

Test Case Number	012
Test Case Name	FindCustomer()
Test Case Description	Tests the FindCustomer method for its ability to search and display customer details based on the input name.

Preconditions	Application is running and connected to the database. The database contains customers and users with the role 'user'.
Test Case Input	Role: "admin" or "superadmin" Customer Name: A valid customer name or partial name for searching.
Test Case Expected Output	- UI: A window with: - o A text field for entering the customer name. - o A "Find" button to initiate the search. - o A "Close" button to close the search window. - If Customer is Found: - o A new window with a list of customers matching the search term, displaying: ▪ Customer details: Name, Phone, Address, Email, Gender, Date of Birth, Orders Count, Total Spent, Username, Password ▪ A "Close" button to close this window. - If No Customer is Found: - o An error message indicating that no customer was found.
Test Case Steps	Open Find Customer Window Search for a Customer Verify Search Results Close the Find Customer Window
Default Value Preverving	

13. Add Staff

Test Case Number	013
Test Case Name	AddStaff()
Test Case Description	Tests the AddStaff method to ensure it correctly registers a new staff member and handles input validation, database interactions, and UI updates.
Preconditions	Application is running and connected to the database.

	Database schema includes users and admins tables with appropriate fields.
Test Case Input	1, ,1, 1 ,1 , 1, 1
Test Case Expected Output	1) Success message. 2) Validation Errors 3) Database Errors
Test Case Steps	1) Open Add Staff Window 2) Register New Staff 3) Close the Window
Default Value Preverving	

14. Find Staff

Test Case Number	014
Test Case Name	FindStaff()
Test Case Description	Verify that the FindStaff method correctly searches for staff members and displays the results in a new window. Ensure that invalid inputs and errors are handled properly.
Preconditions	The application is running and connected to the database
Test Case Input	1
Test Case Expected Output	Valid Input Invalid Input No Matching Staff
Test Case Steps	1) Fill name want find 2) Click find button
Default Value Preverving	

15. Display Staff

Test Case Number	015
Test Case Name	DisplayStaff()
Test Case Description	Verify that the DisplayStaff method successfully displays a list of staff members in a new window, with appropriate staff details, and handles user interactions for editing and deleting staff records.
Preconditions	- The application is running and connected to the database. - Staff data exists in the admins and users tables.
Test Case Input	
Test Case Expected Output	Staf Display Edit button Delete button
Test Case Steps	
Default Value Preverving	

16. Edit Staff

Test Case Number	016
Test Case Name	EditStaff()
Test Case Description	Verify that the EditStaff method correctly displays an editable window with staff details, allows modifications, and updates the database with new information upon saving
Preconditions	The application is running and connected to the database.
Test Case Input	1, 1, 1, 1, 1, 1, 1
Test Case Expected Output	Edit button, Save button, Close button, Error handling
Test Case Steps	1) Modify Staff Details 2) Verify Update Functionality 3) Test Save Without Changes
Default Value Preverving	

17. Order product

Test Case Number	017
Test Case Name	OrderProduct()
Test Case Description	Verify that the PlaceOrderDirectly method correctly places an order by inserting it into the database, updating the total price, and handling exceptions
Preconditions	The application is running and connected to the database.
Test Case Input	1, 1, 1, 1
Test Case Expected Output	❖ Success Message ❖ Error Handling ➤ Database Error ➤ General Error
Test Case Steps	1) Successful Order Placement 2) Database Error Handling 3) General Error Handling
Default Value Preverving	

18. Order for customer

Test Case Number	018
Test Case Name	OrderProduct()
Test Case Description	Verify that the OrderProductForCustomer method correctly handles the ordering process by validating customer information, managing product stock, and placing the order
Preconditions	• The cart and pd objects are initialized and accessible. • The role parameter is correctly set to either "admin", "superadmin", or "user".
Test Case Input	1, 1, 1, 1, 1
Test Case Expected Output	❖ Success Message ❖ Error Handling
Test Case Steps	1) Order Placement with Valid Data 2) Field Validation 3) Cancellation
Default Value Preverving	

19. Update Status

Test Case Number	019
Test Case Name	UpdateStatus
Test Case Description	Verify that the UpdateStatus method correctly initializes and displays the user interface for updating the status of an order.
Preconditions	The application is running and connected to the database.
Test Case Input	1
Test Case Expected Output	❖ Success Message ❖ Error Handling ➤ Invalid Order ID ➤ Cancellation
Test Case Steps	1) Interface Initialization 2) Update Status Functionality 3) Closing the Window 4) Cancelling the Close Operation
Default Value Preverving	

20. Add category

Test Case Number	020
Test Case Name	AddCategory()
Test Case Description	Verify that the AddCategory method correctly initializes and displays the user interface for adding a new category
Preconditions	The application is running and connected to the database.
Test Case Input	1, 1
Test Case Expected Output	
Test Case Steps	1) Interface Initialization 2) Saving Category 3) Handling Database Errors
Default Value Preverving	

21. Disable category

Test Case Number	021
Test Case Name	DisableCategory()
Test Case Description	Verify that the DeleteCategory method correctly marks a category and its associated products as inactive in the database
Preconditions	The application is running and connected to the database.
Test Case Input	1
Test Case Expected Output	
Test Case Steps	
Default Value Preverving	• Ensure that no unintended records are modified if the CategoryID is not found. • Ensure that the application continues to function normally after the method execution.

22. Display category

Test Case Number	022
Test Case Name	DisableCategory()
Test Case Description	Verify that the DisplayCategories method displays categories correctly based on the user role and provides appropriate functionality for each role
Preconditions	The application is running and connected to the database.
Test Case Input	Role: The role of the current user (e.g., "user" or "superadmin").
Test Case Expected Output	
Test Case Steps	
Default Value Preverving	• Ensure that no unintended changes are made to the database. • Verify that the application continues to function normally after navigating back to the menu.

V. Task Assign (to each team member)

HCL	HCL Electronic Shop					
No	Task name	Description	Start Date	End Date	Member	Self assessment
1	Code logic	Come up with most of the ideas for how the system of the application will work.	May 29 th	June 27 th	Long	
2	Draw diagram	Drawing flowcharts and the diagrams for the code logic.	May 29 th		Long, Huy	
3	Code	Write the program.	May 29 th	June 27 th	Long, Cuong, Huy	
4	Write report	Make the detailed report about the project progression and explain all the functions.	June 28 th		Long, Cuong, Huy	
5	Slide design	Make power point slides of the report for presentation.	June 28 th		Cuong, Huy	

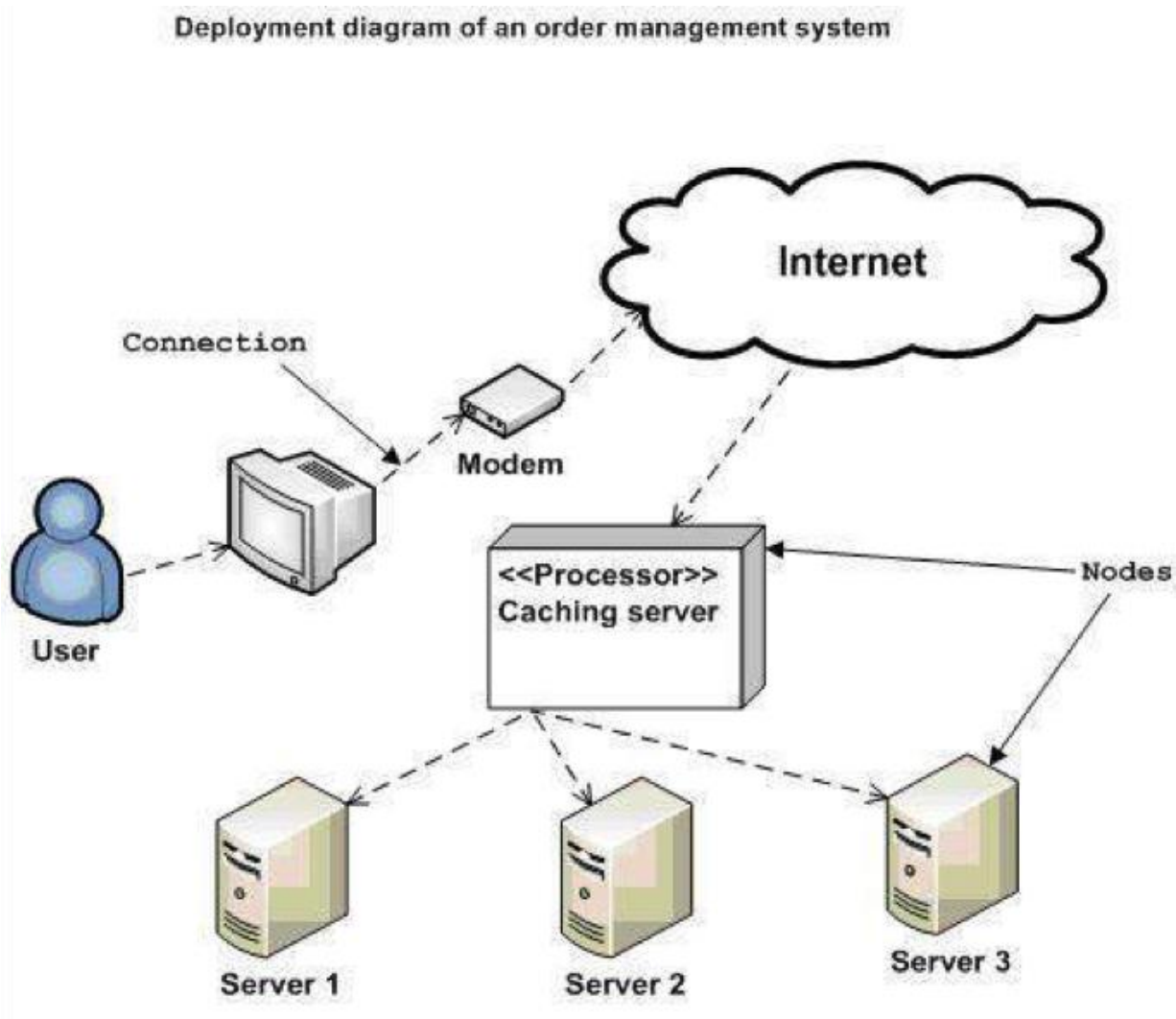
VI. Installation Instructions

INSTALLATION INSTRUCTIONS FOR HCL APPLICATION

System Requirements:

- Operating System: Windows

1. Deployment Diagram



2. Installation steps

- Database Install
 - Download MySQL Community Server from the website:
<https://dev.mysql.com/downloads/mysql/>

MySQL Community Downloads

MySQL Community Server

General Availability (GA) Releases
Archives

MySQL Community Server 9.0.0 Innovation

Select Version:
9.0.0 Innovation

Select Operating System:
Microsoft Windows

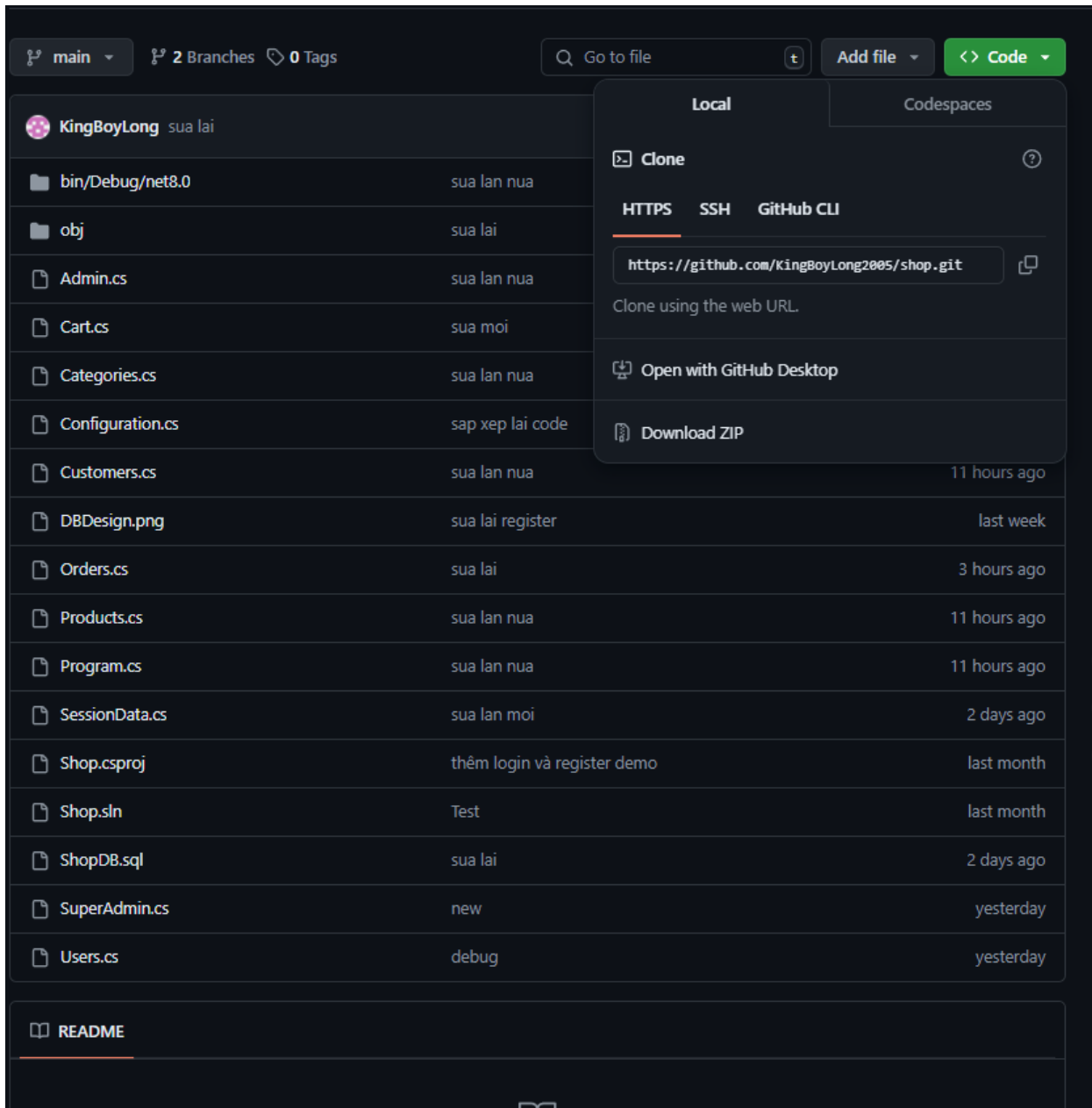
Windows (x86, 64-bit), MSI Installer (mysql-9.0.0-winx64.msi)	9.0.0	131.7M	Download MD5: eeeb88f0844bd346bbc5894dc6e5e99a Signature
Windows (x86, 64-bit), ZIP Archive (mysql-9.0.0-winx64.zip)	9.0.0	284.5M	Download MD5: e51f9650fc22d52b42f2e19245056e8b Signature
Windows (x86, 64-bit), ZIP Archive Debug Binaries & Test Suite (mysql-9.0.0-winx64-debug-test.zip)	9.0.0	772.6M	Download MD5: 26b872444c223e17eb520c51931c6bf7 Signature

We suggest that you use the MD5 checksums and GnuPG signatures to verify the integrity of the packages you download.

ORACLE © 2024 Oracle

[Privacy](#) / [Do Not Sell My Info](#) | [Terms of Use](#) | [Trademark Policy](#) | [Cookie Preferences](#)

- Choose the version that's most suitable for your device and download.
 - Extract the file and run the installation.
 - Adjust the installation options like installation directory, port, root, passwords, etc... to your liking then click "Next".
 - Finish configuring your MySQL and start the service.
 - Open Command Prompt and run the command: `mysql --version` to check the currently installed MySQL version.
- Application install
- Download the application from this link: [GitHub - KingBoyLong2005/shop](https://github.com/KingBoyLong2005/shop) and click the green < > Code button. Next choose Download ZIP to start downloading the application.



- Once downloaded, extract the ZIP file to any folder you want then follow the path shop-main/bin/Debug/net8.0. Here find the Shop.exe file, right click and choose move to Desktop.

Shop.dll	7/22/2024 8:42 PM	Application exten...	130 KB
Shop.exe	7/22/2024 8:42 PM	Application	140 KB
Shop.pdb	7/22/2024 8:42 PM	PDB File	46 KB
Shop.runtimeconfig.json	7/22/2024 8:42 PM	JSON File	1 KB
SixLabors.ImageSharp.dll	7/22/2024 8:42 PM	Application exten...	2,043 KB

- Double click the Shop.exe file to run the application, then enter your root password. After that you can enter your application account to start using the application.

Appendix

Terms and abbreviations

Admin – Staff level user's role

Super Admin – Manager level user's role

Some other issues

You have to run VSC and use dotnet run for the application file to load.